

THE SPECTRAL SHORTCUT: CONDITIONAL THEORY AND DIAGNOSTIC LIMITS OF ENCODER STARVATION IN SERIAL SPECTRAL–SPATIAL PIPELINES

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ABSTRACT

Serial pipelines that reduce a per-element signal with a small encoder and then apply a wide downstream model are the default in spectroscopic imaging, and the same structure appears wherever a compact encoder feeds a large head. We separate three ingredients of selective encoder adaptation in such models: block curvature, residual excitation, and optimizer-dependent displacement. Conditional bounds identify when a fast downstream module limits encoder movement: a width-linear curvature disparity under an explicit head hypothesis, a finite-time displacement bound under a residual envelope, a sharp finite-horizon attribution bound, a transparent affine comparison of joint and frozen training, and a scoped nonlinear existence result. An exactly solvable serial cross-entropy instance exposes the role of the induced training metric at a matched fitting threshold; there the displacement bound is within a factor of six of the truth. Measurements on a production spectral–spatial segmentation model show why scalar curvature can mislead: rank-one inputs invert it, its width trend is carried by batch-normalization gains, and a shallow-model trajectory experiment shows the residual leaving the fast directions. Matched retraining comparisons show that a previously reported freezing advantage cannot be attributed to architecture from that protocol.

1 INTRODUCTION

A common pipeline in spectroscopic imaging reduces each pixel’s spectrum with a small encoder and classifies the resulting feature map with a wide spatial network (Berisha et al., 2019; O’Leary et al., 2026). Hyperspectral remote sensing, frozen image encoders under temporal models, and multimodal fusion share the structure: a compact module sits behind a bottleneck, a high-capacity module sits next to the loss, and both are trained jointly with one learning rate. Practitioners often pretrain the small module and freeze it. Whether freezing is necessary, or merely habitual, is exactly what a fair comparison has to decide, and part of this paper is about what makes such a comparison fair.

We ask what the loss geometry says about how the two modules learn, and what a production model does and does not show. We distinguish three ingredients of selective encoder adaptation: the curvature of the two parameter blocks, the excitation of the encoder’s directions by the training residual, and the displacement the optimizer actually applies. Every statement we prove is conditional, and we say on what.

Contributions.

- **Conditional theory.** Under an explicit head hypothesis the module curvature disparity grows linearly in the downstream width (Theorem 1); under a residual envelope the encoder’s displacement by the fitting time is $\mathcal{O}(\varepsilon \log(1/\delta))$ (Theorem 2); the encoder’s share of the loss decrease is at most $a/(a + \kappa)$ at every horizon (Corollary 1); an affine bridge turns kernel coercivity on the visited residual into joint-versus-frozen closeness (Theorem 3), with a scoped nonlinear existence version (Section 5).
- **A solvable instance.** A serial cross-entropy model with replicated readouts has disparity exactly M , fitting rate $(M + 1)/4$, a spectral update suppressed as $1/(1 + Mv_0^2)$, and

a failure under contextual reversal with probability $\Phi(m/2\sigma)$ (Theorem 4, Figure 1); the displacement bound is within a factor of six of the truth.

- **Diagnostic limits.** On a production spectral-spatial model the head hypothesis is structurally unmet, rank-one inputs invert the scalar disparity, the informative quantity is directional anisotropy, the observed width trend of downstream curvature is carried by batch-normalization gains, and on our own shallow instance the residual leaves the coercive subspace early (Section 7, Figure 2).
- **Matched protocol.** A freezing advantage observed under an earlier protocol is absent once clipping scope, normalization-parameter treatment and checkpoint policy are equalized (Section 8, Figure 3, Table 1); the original contrast cannot identify an architectural shortcut.

What the measurements teach. The theory is a set of sufficient conditions, and the production model evades most of them instructively: its classifier head has a fixed input dimension, so the width-linear floor has no premise; its inputs are effectively rank one, so the top-eigenvalue ratio is uninformative and a directional picture must replace it; the width growth of its downstream curvature is a normalization gain that a function-preserving rescaling moves freely. On the shallow instance where the geometry is proved, the residual leaves the coercive part of the downstream kernel within a hundred steps, so initialization curvature predicts only an early phase. The paper’s practical content is the set of measurements one must make before reading a shortcut into a curvature ratio or a training curve, not a recipe.

2 RELATED WORK

Modality competition and laziness. Joint training of multi-branch models is known to under-train the branch that is slower to fit: Wang et al. (2020) and Peng et al. (2022) document the effect and propose gradient blending and modulation, Du et al. (2023) name the under-used branch lazy, and Huang et al. (2022) prove competition for explicit late-fusion networks. Our setting is a single input passed serially through two modules; the asymmetry is positional rather than between parallel branches, and our bounds concern module displacement rather than feature selection.

Shortcut and starvation theory. Pezeshki et al. (2021) prove, in a linearized regime with coupled feature dynamics, that a fast feature starves a slow one of gradient. We bound displacement of a whole module in a compositional pipeline and take the residual envelope as a hypothesis; neither result subsumes the other. The residual-excitation statement of Section 5 makes precise why curvature anisotropy alone does not imply starvation.

Curvature, width and normalization. The width-linear growth of the top Gauss-Newton eigenvalue in proportional-width networks is the mean-field law of Karakida et al. (2019); block heterogeneity of the Hessian is the optimizer-side antecedent of Zhang et al. (2024), and the top-eigenvector structure $u \otimes \mathbb{E}[x]$ appears in Pappas et al. (2020). The scale invariance of layers feeding batch normalization, and its effect on effective learning rates, is established in van Laarhoven (2017); we use it as a diagnosis of one model, not as a new mechanism.

Lazy training and implicit bias in serial models. Our nonlinear existence result is an explicit-constant instance of lazy training in the standard parameterization (Chizat et al., 2019; Sohl-Dickstein et al., 2020; Yang & Hu, 2021), proved with the gate-flip argument of Du et al. (2019) and the kernel-tracking estimates of Lee et al. (2019); Arora et al. (2019). The solvable serial instance rests on the layer-balance invariant of Du et al. (2018) and sits beside the exact dynamics of Saxe et al. (2014) and the architecture- and initialization-dependent implicit biases of Yun et al. (2021); Moroshko et al. (2020); Berthier (2023); it adds a matched-margin encoder-update bound, a frozen comparator and a reversal test in one system, not a new mechanism of implicit bias.

Frozen features in practice. Last-layer retraining on frozen features recovers robustness to spurious correlations (Kirichenko et al., 2023); some layers barely move during training (Zhang et al., 2022); and Coil & Cheney (2025) hypothesize that freezing acts as regularization. Our matched-

protocol study is a caution for this literature: the apparent benefit of freezing in our production model was a property of an unmatched protocol.

3 SETUP

Model. An input $X \in \mathbb{R}^{H \times W \times S}$ has a spectrum $x_{h,w} \in \mathbb{R}^S$ at every pixel. A *spectral encoder* $f_\theta : \mathbb{R}^S \rightarrow \mathbb{R}^K$ is applied per pixel, producing $Z \in \mathbb{R}^{K \times H \times W}$; a *spatial model* g_ϕ of characteristic width M (channels, embedding dimension or hidden width) maps Z to per-pixel logits $\hat{y} \in \mathbb{R}^{N_{\text{cls}} \times H \times W}$. We write $F_{\theta,\phi} = g_\phi \circ f_\theta$, $C_f = \dim \theta$ and $C_g = \dim \phi$. The theorems are informative when $C_g \gg C_f$; the production model of Section 7 has $C_g/C_f \approx 299$.

Loss, residual and Jacobians. With per-pixel cross-entropy averaged over the $N = N_{\text{data}}HW$ pixels, the gradient with respect to the logits at pixel n is the residual $r_n = \text{softmax}(\hat{y}_n) - y_{n,\text{onehot}}$. We stack with $N^{-1/2}$ so that no constant depends on the dataset size:

$$r := N^{-1/2}(r_1, \dots, r_N), \quad J_b := N^{-1/2} \frac{\partial(\hat{y}_1, \dots, \hat{y}_N)}{\partial b}, \quad \nabla_b \mathcal{L} = J_b^\top r, \quad b \in \{\theta, \phi\}. \quad (1)$$

Two objects must be kept apart. The Gauss–Newton matrix and its diagonal blocks live in parameter space,

$$G = J^\top H_\tau J, \quad J = [J_\theta, J_\phi], \quad G_{bb} = J_b^\top H_\tau J_b, \quad (2)$$

with $H_\tau = \text{diag}(p) - pp^\top$ the softmax Jacobian, blockwise over pixels (Botev et al., 2017); they measure *curvature*. The output-space kernels $K_b = J_b J_b^\top$ measure *gradient gain*: for cross-entropy, $r^\top K_b r = \|\nabla_b \mathcal{L}\|^2$ holds exactly, with no squared-loss approximation. Section 4 concerns curvature; Section 5 concerns gain along the realised residual.

Definition 1 (Module curvature disparity). $\mathcal{D}_{\text{curv}}(G) := \lambda_{\max}(G_{\phi\phi})/\lambda_{\max}(G_{\theta\theta})$, evaluated at random initialization unless stated otherwise, with $\mathcal{D}_{\text{curv}} := +\infty$ if the spectral block vanishes.

Assumptions.

Assumption 1. (i) *Linear encoder*: $f_\theta(x) = W^\top x$ with $\theta = \text{vec}(W)$, $W \in \mathbb{R}^{S \times K}$.

Assumption 2. (ii) *Bounded input Gram*: the mean per-pixel second moment $\Sigma_X := N^{-1} \sum_n x_n x_n^\top$ satisfies $\lambda_{\max}(\Sigma_X) \leq \Lambda_X < \infty$.

Assumption 3. (iii) *Width-uniform input Jacobian*: there is $\tilde{L}$, independent of M and C_g , with $\sup \|\partial g_\phi(Z)/\partial Z\|_{\text{op}} \leq \tilde{L}$ over a parameter regime Φ_{reg} and feature domain $\mathcal{Z}_{\text{reg}}$ (compact when g_ϕ uses attention). This is a modelling assumption: it is proved in expectation for a two-layer ReLU head at initialization (Appendix A), where the naive product of layer norms would give $\Theta(M/K)$ instead.

Assumption 4. (iv) *Regularity*: g_ϕ is C^2 jointly in (ϕ, Z) , or piecewise smooth (ReLU), in which case gradient flow is read almost everywhere with a consistent gate selection (Appendix B).

4 CONDITIONAL GEOMETRY: CAPACITY-INDUCED CURVATURE DISPARITY

Theorem 1 (Capacity-induced module curvature disparity). *Consider $F_{\theta,\phi} = g_\phi \circ f_\theta$ with a fixed-dimensional linear encoder (Assumption 1) and a family of spatial networks $g_\phi^{(M)}$ indexed by width M . Suppose Assumptions 2 and 3 hold, and suppose g_ϕ ends in a dense classifier head over penultimate features $h_n \in \mathbb{R}^{M_h}$ with $M_h = \Theta(M)$ whose softmax-weighted mean Gram $S_h^p := N^{-1} \sum_n p_{\min}^{(n)} h_n h_n^\top$ satisfies $\lambda_{\max}(S_h^p) \geq q_h M_h$ with probability at least $1 - \delta_h$ over the initialization, for constants $q_h > 0$, $\delta_h < 1$ independent of M (the head hypothesis: the features carry a common direction of $\Theta(M)$ energy across inputs on pixels whose initial predictions are non-degenerate). Then there are constants $c_1, c_2 > 0$, depending on the data, the initialization scheme, K , S and the architecture class but not on M or C_g , such that in expectation over Kaiming initialization*

$$\mathcal{D}_{\text{curv}}^{(M)} = \frac{\lambda_{\max}(G_{\phi\phi}^{(M)})}{\lambda_{\max}(G_{\theta\theta})} \geq c_1 M - c_2.$$

For families with $C_g = \Theta(M^2)$, equivalently $\mathcal{D}_{\text{curv}}^{(M)} \geq c'_1 \sqrt{C_g/C_f} - c_2$.

Lemma 1 (Spectral block cap). *Under Assumptions 1–3, $\lambda_{\max}(G_{\theta\theta}) \leq \|J_{\theta}\|_{\text{op}}^2 \leq \tilde{L}^2 \lambda_{\max}(\Sigma_X) =: C_{\theta}$, a constant independent of M , C_g and N .*

The proof (Appendix A) is elementary. The rank-one head-weight perturbation $V = cu^{\top}$, with c a unit class contrast and u the dominant feature direction, gives the deterministic inequality $\lambda_{\max}(G_{\phi\phi}) \geq \lambda_{\max}(S_h^n)$, and the head hypothesis supplies $\Omega(M)$; the cap follows from the chain rule $J_{\theta} = (\partial\hat{y}/\partial Z)(\partial Z/\partial\theta)$, whose second factor has squared operator norm exactly $\lambda_{\max}(\Sigma_X)$ under linearity; the rate matches Karakida et al. (2019) where that law applies.

The hypothesis boundary. The theorem measures the downstream module’s *readiness to move*, not the usefulness of its features: the width-linear spike exists before any feature is learned. It concerns the top of the spatial spectrum only, is not a dynamical claim, and is stated for the standard parameterization (Yang & Hu, 2021). The head hypothesis is a cross-input correlation condition on $\Theta(M)$ penultimate features: met by dense per-pixel heads on CNN and ViT backbones, and structurally *unmet* by the production model of Section 7, whose classifier input has a fixed dimension at every width, so that the theorem has no premise there rather than a counterexample.

Empirical check (Experiment 1.1). We measure both blocks of G by power iteration on a synthetic spectral–spatial problem ($S = 64$, $K = 16$, two classes; five seeds per width; Appendix D). For the shallow verified instance (linear encoder, two-layer ReLU head) under default fan-in initialization, $\lambda_{\max}(G_{\phi\phi})$ rises from 0.62 to 46.0 over $M = 16 \rightarrow 8192$ (log–log slope 0.71) while $\lambda_{\max}(G_{\theta\theta}) \approx 0.02$ is flat (slope -0.02), so $\mathcal{D}_{\text{curv}}$ grows from 40 to 2405. Under the theorem’s own initialization (He weights with a width-independent bias variance) the slope is 0.90 ($2.1 \rightarrow 442$) with $\lambda_{\max}(G_{\theta\theta}) \approx 0.6$ flat. A deeper head separates the two regimes sharply: over $M = 16 \rightarrow 1024$ its spatial slope is 0.105 under default initialization and 0.989 under the theorem’s, and isolation arms (fixed bias variance alone 0.924; He weights with scaled or zero bias 0.696, 0.606) show that the floor survives when either variance term of the head hypothesis is width-independent, while default fan-in initialization makes both vanish with width. Slopes are finite-range fits quoted with their levels; $\mathcal{D}_{\text{curv}}$ is not used to attribute differences across arms, whose denominators differ.

5 DYNAMICS: FINITE-TIME STARVATION, ATTRIBUTION, AND THE BRIDGE

We analyze the continuous-time limit of joint training with a shared learning rate, $\dot{\theta} = -\nabla_{\theta}\mathcal{L}$, $\dot{\phi} = -\nabla_{\phi}\mathcal{L}$, along absolutely continuous trajectories (almost everywhere for ReLU). For unregularized cross-entropy on separable data there is no finite minimizer (Soudry et al., 2018), so the honest statement is a finite-time bound on how far the encoder can move before the loss is fitted. By (1), $\|\dot{\theta}\| \leq \|J_{\theta}\|_{\text{op}} \|r\|$. Fix a horizon T and suppose the trajectory stays in $\Phi_{\text{reg}} \times \mathcal{Z}_{\text{reg}}$ (*containment*);

Lemma 1 then caps $B_T := \sup_{t \leq T} \|J_{\theta}(t)\|_{\text{op}} \leq \tilde{L} \sqrt{\lambda_{\max}(\Sigma_X)}$, a constant independent of M .

Assumption 5 (Shortcut residual decay). There are $C \geq 1$ and $\mu > 0$ with $\|r(t)\| \leq C/(1 + \mu t)$ along the joint trajectory on $[0, T]$; μ is the *shortcut fitting rate*, and $T_{\delta} := \inf\{t : \|r(t)\| \leq \delta\} \leq (C/\delta - 1)/\mu$.

Theorem 2 (Finite-time spectral starvation). *Under Assumptions 1–3 and 5, with containment up to T ,*

$$\|\theta(T) - \theta_0\| \leq \frac{B_T C}{\mu} \log(1 + \mu T), \quad \|\theta(T_{\delta}) - \theta_0\| \leq \frac{B_T C}{\mu} \log \frac{C}{\delta} \quad (T \geq T_{\delta}).$$

With $\varepsilon := B_T/\mu$, the encoder’s displacement by the time the residual reaches δ is $\mathcal{O}(\varepsilon \log(1/\delta))$.

Proof. $\|\dot{\theta}\| \leq B_T \|r\| \leq B_T C/(1 + \mu t)$ almost everywhere; integrate, and use $1 + \mu T_{\delta} \leq C/\delta$. $\square$

B_T is capped independently of C_g , while the growth of μ with spatial capacity is a hypothesis, not a consequence of Theorem 1. Discrete versions hold for gradient descent, momentum with clipping before accumulation (our trainer’s ordering), velocity clipping and Adam, with $\mu_{\text{step}} = \eta\mu$ and an extra term for decoupled encoder decay (Lemmas B.1–B.6). The hypothesis constrains the loss, not the encoder, so it is not circular; read contrapositively, substantial spectral learning by the fitting time requires slow fitting.

Corollary 1 (Sharp finite-horizon attribution). *Let both blocks follow unit-rate gradient flow with the almost-everywhere chain rule, and suppose that along the realised residual, for almost every $t \leq T$, $r^\top K_\theta r \leq a \|r\|^2$ and $r^\top K_\phi r \geq \kappa \|r\|^2$, with $\mathcal{L}(0) > \mathcal{L}(T)$. Then*

$$\frac{\int_0^T \|\nabla_\theta \mathcal{L}\|^2 dt}{\mathcal{L}(0) - \mathcal{L}(T)} \leq \frac{a}{a + \kappa},$$

and the constant is attained by scalar kernels. No invariant subspace is required (Appendix B, Corollary B.2).

Theorem 3 (Affine bridge). *Let $z = z_0 + A(\theta - \theta_0) + B(\phi - \phi_0)$ with $\mathcal{L} = \frac{1}{2} \|z - y\|^2$, and train the joint model and the model with $\theta \equiv \theta_0$ by unit-rate gradient flow from the same initialization. If a common invariant subspace contains $e_0 = z_0 - y$ and on it $K_\phi = BB^\top \succeq \kappa I$ and $0 \preceq K_\theta = AA^\top \preceq aI$, then $\sup_t \|z_J(t) - z_F(t)\| \leq a \|e_0\| / (\epsilon \kappa)$, $\|\theta_J(t) - \theta_0\| \leq (\sqrt{a}/\kappa) \|e_0\|$, and $\int_0^\infty \|\nabla_\theta \mathcal{L}\|^2 dt \leq a \|e_0\|^2 / (2\kappa)$ (Appendix B, Theorem B.2). For a shallow biased-ReLU head with M hidden units on distinct inputs, coercivity holds with $\kappa = \kappa_* M/2$ and M -independent a above a width threshold $M_0 \geq 72N^2/\rho$ (Corollary B.3); the threshold is an existence statement and numerically vacuous.*

The coercivity must hold on the residual actually visited: a large top eigenvalue of K_ϕ certifies nothing when the residual lives elsewhere (Appendix B, Proposition B.2), which is what Section 7 measures.

A scoped nonlinear consequence. For the fully trained shallow biased-ReLU model under squared loss in the standard parameterization, for $M \geq M_*$ and with probability at least $1 - \rho$ over the initialization, every joint and every frozen solution fits at rate $\Omega(M)$, the encoder moves $\mathcal{O}(1/M)$ while the head moves $\mathcal{O}(M^{-1/2})$, both paths track one common linear reference to $\mathcal{O}(1/M)$, and the encoder’s share of the loss decrease is at most $A^2/(A^2 + \kappa M/2)$ (Appendix B, Theorem B.6). This is lazy training in the standard parameterization (Chizat et al., 2019; Sohl-Dickstein et al., 2020), proved with the gate-flip argument of Du et al. (2019); its content here is the two-block displacement split. The disparity disappears under a readout $M^{-1/2}Uh + \beta$ with $\mathcal{O}(1)$ entries trained at unit rate, and M_* is existence-only and numerically vacuous.

Residual excitation. Curvature anisotropy does not imply cumulative starvation: the gradient energy an encoder direction receives is $\int (r^\top J_u)^2 dt$, which vanishes whenever the residual has no component along that direction’s image, however large the curvature elsewhere (Appendix B, Theorem B.4). Attribution is therefore phase dependent. If on a phase E the residual keeps a fraction α of its mass in the top- k eigenspace of K_ϕ whose k th eigenvalue is at least ℓ , the encoder’s share of the decrease accrued on E is at most $a/(a + \alpha\ell)$, and the total share is at most $w_E a/(a + \alpha\ell) + (1 - w_E)$ with w_E the phase’s fraction of the decrease; a cumulative share of order $1/M$ then requires the complementary phase to carry order $1/M$ of the energy (Proposition B.12, Corollary B.4).

6 AN EXACTLY SOLVABLE SERIAL INSTANCE

The hypotheses above can all be checked in one model. A scalar spectral coefficient a and a context coefficient v feed M replicated readouts β_j , and the margin is $q = a + bv$ with $b = \sum_j \beta_j$; the loss is logistic cross-entropy on a training distribution in which spectrum, context and label coincide, trained by unit-rate gradient flow until the margin first reaches m (Appendix B, Theorem B.5).

Theorem 4 (Serial instance). *For this model, $\mathcal{D}_{\text{curv}}(0) = M$ exactly and the residual envelope of Assumption 5 holds with $\mu = (M + 1)/4$. For any $a_0 < m$, $v_0 \neq 0$ and zero-sum readout initialization, the spectral update at the fitting time satisfies*

$$0 < \frac{a(T_m) - a_0}{m - a_0} \leq \frac{1}{1 + Mv_0^2}, \quad 0 \leq q_J(t) - q_F(t) \leq \frac{C_0}{p_0 M v_0^2} \quad (t \leq T_m),$$

where q_F is the margin of the frozen-encoder comparator and $C_0 = 1 + 2(m - a_0)^2/v_0^2$, $p_0 = e^{a_0}/(1 + e^{a_0})$. Under contextual reversal at test time the trained model’s error tends to $\Phi(m/2\sigma)$ for isotropic Gaussian $(a_0, v_0) \sim \mathcal{N}(0, \sigma^2 I)$ while a spectral-only comparator has zero error, and for $a_0 > m/2$ the classifier is correct at every width. With the readout normalized as $M^{-1/2}Uh$, with $\mathcal{O}(1)$ entries trained at unit rate, the M -dependence disappears.

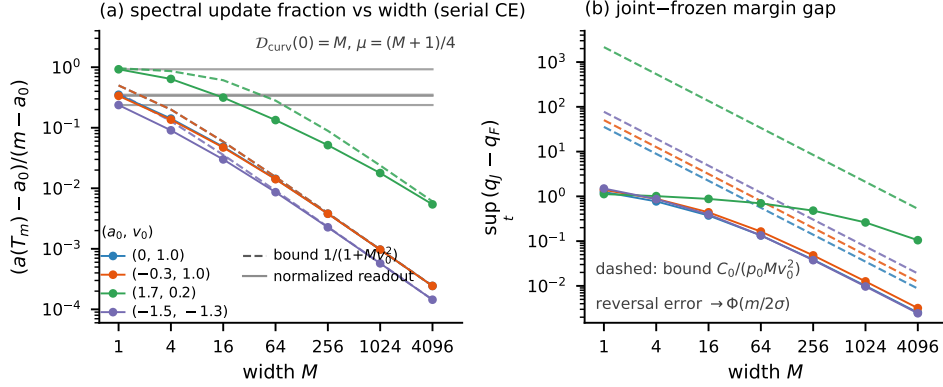


Figure 1: The serial instance. (a) Spectral update fraction $(a(T_m) - a_0)/(m - a_0)$ against width M for several (a_0, v_0) , with the exact bound $1/(1 + Mv_0^2)$ (dashed) and the normalized-readout control (flat). (b) Joint-minus-frozen margin gap $\sup_t(q_J - q_F)$ with its bound $C_0/(p_0 M v_0^2)$. Closed form and full integration agree to 10^{-11} .

The mechanism is a replication-induced effective learning rate, made exact by the layer-balance invariant of gradient flow (Du et al., 2018). What is suppressed is the *update* of the spectral coefficient as a fraction of the fitting task, not the coefficient itself, which can be informative at initialization and remain so. Theorem 2’s displacement bound is within a factor 4.5–6.3 of the realised displacement here, and every displayed quantity agrees with the closed form to 10^{-11} against a full $(2 + M)$ -parameter integration over 579 cases (Appendix D). We use a solvable serial CE example to connect a known optimization-metric imbalance to a finite-margin encoder-update bound, a joint-versus-frozen trajectory comparison, and an explicit failure under contextual reversal. Replication changes the effective learning rate, and its normalization removes the effect. The example supplies an exact consistency check and a scoped existence result; it does not establish a new general mechanism of implicit bias.

7 DIAGNOSTIC LIMITS ON A PRODUCTION MODEL AND A SHALLOW INSTANCE

7.1 PRODUCTION MODEL

Setup. BlockViT-v2 is a spectral-spatial segmentation model from a companion tokenization study (anonymous, under review): a linear per-pixel reduction $942 \rightarrow K = 64$ ($C_f = 61,108$ parameters) followed by a 12-layer ViT of embedding width M with a convolutional decoder ($C_g = 18,271,540$ at $M = 192$; $C_g/C_f \approx 299$; a $\Theta(M^2)$ family). Data are breast-tissue FTIR micro-array cores (fold 0 of a five-fold patient-level split) with a prostate QCL replication (Appendix E). The decoder ends in $\text{Conv}_{3 \times 3}(M + K \rightarrow 96)$ –BN–ReLU–Conv($96 \rightarrow 48$)–BN–ReLU–Conv($48 \rightarrow 4$), so the classifier’s input has dimension 432 at every width: Theorem 1’s head hypothesis is structurally unmet, and the width-linear floor has no premise here.

Geometry at initialization. The cap holds: the log-log slope of $\lambda_{\max}(G_{\theta\theta})$ against $M \in \{48, 96, 192, 384\}$ is -0.003 (breast) and -0.012 (prostate). The scalar disparity is *inverted*: $\mathcal{D}_{\text{curv}} = 0.46 \pm 0.21$ at $M = 192$, rising from 0.33 ± 0.12 to 0.70 ± 0.19 over $48 \rightarrow 384$ on breast and reaching 1.09 ± 0.52 at 384 on prostate. The reason is the input: Σ_X has effective rank 1.07 (breast) and 1.02 (prostate) out of 942, and 1.21 after the module’s input normalization, so the cap $C_\theta \propto \lambda_{\max}(\Sigma_X)$ is enormous and the top-ratio is uninformative. The informative statement is directional. With v_1 the leading input direction, which is the mean spectrum ($|\cos| > 0.995$), the spectral block’s top eigenvector places 0.954 ± 0.017 of its norm in the K -dimensional subspace induced by v_1 , the top 40 of 61,108 directions hold 0.83 ± 0.12 of the trace, and along the clinically decisive CancerEpi–CAS contrast the curvature is $3.7 \times$ (2.6–4.8) smaller than along v_1 , or 12 – $28 \times$ smaller along the contrast’s v_1 -orthogonal part, which retains 76–84% of its squared norm (nine

measurements; Figure 2a). The measurement is anisotropy; starvation along those directions is a hypothesis it motivates.

What carries the width trend. The spatial block’s top eigenvalue does grow, sublinearly (slope 0.38 breast, 0.45 prostate). The only decoder layer whose input dimension grows with M is the first head convolution, and the second-moment matrix of the patch-unfolded features entering it is flat: its top eigenvalue is unchanged to five significant figures across $M \in \{48, 192, 384\}$ (three seeds), with its leading direction inside the width-independent $K = 64$ skip channels. The trend is instead carried by the two head BatchNorm layers. Using fixed initial statistics in both removes it while preserving the incoming features (first-convolution block, slope of the seed-averaged curvature $+0.61$ with batch statistics, per seed $+0.35$ to $+0.79$, versus $+0.006$ with both fixed; whole spatial block $+0.40 \rightarrow -0.33$), and either BatchNorm alone carries it ($+0.56, +0.57$), which argues against attributing it to one layer. The pre-BN per-channel variance of the first head BN falls as $1/(M+K)$ (0.15, 0.064, 0.035), because the fan-in grows with M while the incoming energy sits in the width-independent skip channels, and the BN’s gain grows accordingly (2.6, 4.0, 5.4); a forward-mode propagation of a unit weight perturbation through the actual head confirms the stage at which the energy grows, and a spatially constant bias perturbation is annihilated exactly. A function-preserving rescaling of that convolution (weights and bias by c , stabilizer by c^2) leaves every logit unchanged and scales the block’s top curvature by exactly c^{-2} , so this curvature is a parameterization quantity (van Laarhoven, 2017); the conditional gain law (Proposition B.10) is not a new mechanism. This sweep therefore does not validate the width-linear feature-Gram mechanism of our shallow theory. Three-width slopes are descriptive, and fixing a BatchNorm’s statistics also changes its forward output and downstream gates, so the interventions establish sensitivity, not a unique cause.

7.2 RESIDUAL SUBSPACE ALONG TRAINING OF THE SHALLOW INSTANCE

On the instance where the geometry is proved (linear $S = 64 \rightarrow K = 16$ encoder, theorem-initialized ReLU head of width D), we train jointly with Adam for 2400 steps ($D \in \{128, 512, 2048\}$, three seeds) and, on a fixed probe of $N = 256$ sites ($C = 2$), export at eleven checkpoints the residual’s Rayleigh quotients under K_θ and K_ϕ and its mass in the top- k eigenspaces of K_ϕ ; on the probe, $r^\top K_\theta r = \|\nabla_b \mathcal{L}\|^2$ exactly. We report probe results as probe results and cumulative quantities as *gradient-energy fractions*, the share of summed squared minibatch gradient norms over the run, which is not a decomposition of Adam’s loss decrease.

The cumulative encoder fraction falls with width, with medians 0.29, 0.062 and 0.012. At $t = 0$ the residual sits in the coercive top of K_ϕ (top-20 mass 0.34–0.47), its Rayleigh quotient under K_ϕ grows with D (0.95, 7.1, 64), and the instantaneous share is 0.43, 0.061, 0.0077, close to the top-eigenvalue proxy at the largest width. After about 75 steps the residual has left that subspace: its mass in the top-1, top-20 and top-100 eigenspaces is 8.6×10^{-5} , 0.030 and 0.16 (medians), its Rayleigh quotient is about $0.0025 \lambda_{\max}(K_\phi)$ while the kernel’s condition number is of order 10^6 , and the instantaneous share returns to about 0.4 at every width (Figure 2c). The top-eigenvalue proxy $\lambda_{\max}(K_\theta)/(\lambda_{\max}(K_\theta) + \lambda_{\max}(K_\phi))$ understates the share in 85% of checkpoints (median $2.7\times$) and overstates it by up to $11\times$ in the rest; the $\lambda_{\min}$ proxy is 0.9998–1.0000 throughout. The diagnostic limitation illustrated by Proposition B.2 is observed: the coercivity Theorem 3 needs is available here only as the measured quotient along the visited residual, and by Proposition B.12 a cumulative law of order $1/D$ would require the late phase’s energy fraction to be of that order, which three widths do not establish. With CNN and ViT heads, joint training against a frozen random encoder gives endpoint gaps of at most 0.04 in test accuracy, unresolved at three seeds, with indistinguishable peaks (Appendix D).

8 MATCHED-PROTOCOL EVIDENCE

Original protocol. We retrained the production model at reduced depth (six layers, width 48, AdamW, sixty epochs, three seeds) in a pre-registered design comparing a jointly trained linear encoder with encoders frozen at random or PCA initialization, plus momentum-SGD controls. The joint arm finished 0.110 below the frozen-random arm on the pre-specified final-five-epoch validation macro-F1, a difference that was itself unresolved at three seeds. The comparison was confounded by three asymmetries: the joint arms trained the reduction-stage BatchNorm affine parame-

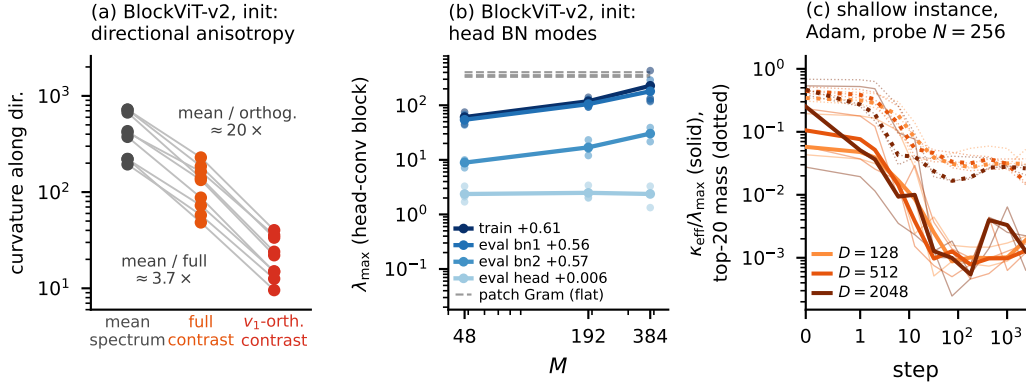


Figure 2: Diagnostic limits. (a) BlockViT-v2 at initialization, no optimizer: curvature along the mean-spectrum direction, the full CancerEpi-CAS contrast and its v_1 -orthogonal part (nine measurements). (b) BlockViT-v2 at initialization: first-head-convolution block curvature against width under four BatchNorm modes, with the flat incoming Gram (dashed). (c) Shallow verified instance under Adam, probe $N = 256$: the residual’s Rayleigh quotient relative to $\lambda_{\max}(K_\phi)$ (solid) and its top-20 mass (dotted) along training.

Table 1: Matched protocol at width 48, three seeds, mean $\pm$ sd of validation macro-F1 (best checkpoint, exploratory; final five epochs, pre-specified) and relative encoder drift $\|\theta_T - \theta_0\|/\|\theta_0\|$. Validation-selected; no test set; two matched pairs.

arm	best-val	final-5	θ drift
frozen random encoder	0.814 ± 0.040	0.684 ± 0.037	0
joint linear encoder	0.837 ± 0.029	0.694 ± 0.056	0.57
frozen pretrained encoder	0.861 ± 0.022	0.690 ± 0.033	0
fine-tuned pretrained	0.857 ± 0.009	0.710 ± 0.052	0.02
joint, spectral lr $\times 0.1$	0.841 ± 0.007	0.626 ± 0.055	0.08
joint, spectral lr $\times 10$	0.886 ± 0.037	0.561 ± 0.114	4.91
joint, cosine schedule	0.825 ± 0.033	0.660 ± 0.020	0.36

ters, gradient clipping acted on θ and ϕ together in joint arms and on ϕ alone in frozen arms, and no best-validation checkpoints were saved. Momentum-SGD controls and a BatchNorm recalibration that recovers 7–17% of the joint arm’s peak-to-final decline are reported in Appendix E.

Matched protocol. We reran seven arms at width 48 with the same clipping rule and scope, identical treatment of every BatchNorm affine parameter, and saved best-validation checkpoints (Table 1). The arms form two matched pairs, not a single-factor design: joint-linear and frozen-random share a linear reduction with BatchNorm; frozen-pretrained and fine-tuned share a pretrained MLP encoder without it; comparisons across pairs change architecture, training history and validation selection. The pre-registered predictions were a fine-tuning deficit relative to the frozen pretrained encoder and a harmful ordering of spectral learning-rate multipliers.

Verdicts. Paired by seed with 90% t -intervals ($n = 3$), fine-tuning minus the frozen pretrained encoder is $+0.021$ $[-0.022, +0.063]$ on final-five and -0.004 $[-0.048, +0.039]$ on best-validation: the predicted deficit is not supported, with verified nonzero applied encoder updates under AdamW. Joint minus frozen-random is $+0.011$ $[-0.142, +0.163]$ and $+0.023$ $[-0.066, +0.112]$; the original -0.110 deficit is absent, and the paired change in the final-five gap between protocols is $+0.120$ $[-0.230, +0.470]$ (best-validation -0.010 $[-0.105, +0.085]$), descriptive seed summaries only. The $\times 10$ multiplier raised best-validation by $+0.048$ $[-0.016, +0.112]$ in all three seeds while its endpoint fell to 0.561 with a fivefold encoder drift; three multiplier means do not establish a relation between displacement and peak score, and this particular cosine schedule did not improve either mean

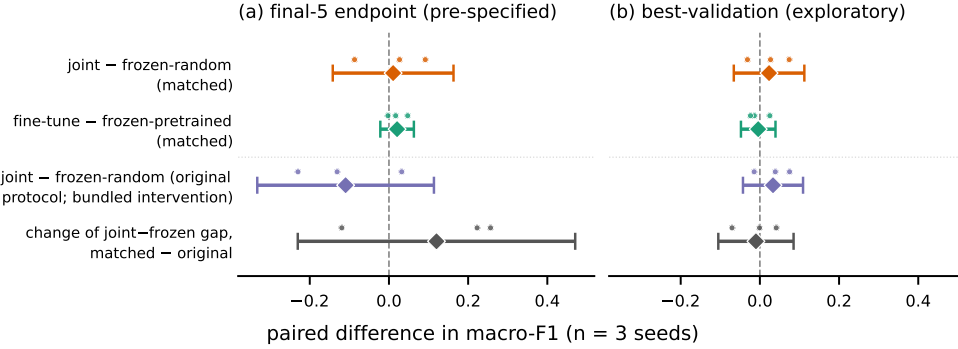


Figure 3: Paired arm differences in validation macro-F1 with 90% paired- t intervals ($n = 3$; seed points shown). (a) Final five epochs, pre-specified. (b) Best validation checkpoint, exploratory. The original joint-minus-frozen contrast is a bundled protocol intervention; the last row is the change of that gap between protocols.

endpoint. The bound of Theorem 2 is not instantiated by these runs, whose optimizer, clipping and momentum lie outside its hypotheses. Disclosures: one width; three seeds; a single validation fold; no test set; best-validation exploratory, final-five pre-specified; pretrained encoders were selected on the same validation cores and pretrained with weight decay 0.01 under an unaugmented centre-crop protocol, whereas subsequent training used random crops with augmentation, which affects cross-pair comparisons but not the frozen versus fine-tuned pair (Figure 3).

Reading. The original protocol confounded freezing with clipping scope and normalization-parameter treatment. Its nominal final-window freezing advantage was absent after these differences were removed, while checkpoint selection reversed the ordering within the original runs. At width 48 and three seeds, neither the within-protocol freezing contrasts nor the change in their gap establishes a population effect. These results demonstrate that the original contrast cannot identify an architectural shortcut and that checkpoint policy changes its descriptive conclusion. They support a protocol-confounded apparent effect; they do not show that architecture has no effect, nor that the named changes alone caused the earlier difference.

9 LIMITATIONS AND CONCLUSION

We have separated three ingredients of selective encoder adaptation in serial models and given conditional bounds for each: curvature disparity under a head hypothesis, displacement under a residual envelope, and a finite-horizon attribution that needs coercivity only on the residual actually visited. The solvable instance shows the bounds biting; the production model shows the diagnostics failing in instructive ways; the matched protocol shows an apparent training effect dissolving.

The limits are these. The theorems assume a linear encoder, and the strongest constructive guarantees hold for deliberately scoped models: an affine system, a shallow ReLU head with a numerically vacuous width threshold, and a scalar serial instance. The production study does not establish their mechanism; it establishes that the scalar disparity is the wrong lens on rank-one inputs, that a width trend can be a normalization gain, and that the head hypothesis is unmet, with descriptive three-width slopes. The residual experiment is a probe-batch diagnostic under Adam: eleven checkpoints cannot certify an almost-everywhere condition, and its comparison with an isotropic residual awaits a reference computed in the class-contrast subspace of dimension $N(C - 1)$. The matched study is one width, three seeds, one validation fold, no test set and two matched pairs; it rules out reading the earlier gap as architectural, not the existence of an effect. Mitigations (per-module learning rates, gradient modulation, freezing) are not evaluated here, and the matched study is a reason to distrust untuned comparisons of them. What the paper adds beyond lazy-training, optimization-metric and feature-competition results is the distinction itself, with measurements showing where each ingredient fails to transfer. The open problem is a same-sequence phase certificate: a run on which the phase membership, its energy fraction w_E and the bound of Proposition B.12 are evaluated on one sequence, or a full-batch flow instance where the energy identity is exact.

AI USE STATEMENT

Generative AI tools (large language model assistants) were used throughout this work: for literature search; for drafting and editing text; for writing and checking code, including the numerical verification scripts of Appendix F; and for adversarial review of the proofs and of the claims made about the measurements. Every proof was independently re-derived by the authors, and every number reported in the paper was recomputed by the authors from the recorded artifacts. All AI-generated text and code were reviewed by the authors, who take full responsibility for the content of this paper, including any remaining errors.

REPRODUCIBILITY STATEMENT

Code, configurations and audit scripts, with seeds and exact commands, are provided in the anonymized repository accompanying this submission. Every number in the paper traces to a named results file; Appendix F lists the verification scripts and their outcomes, including the two that do not exit cleanly. Complete hypotheses and proofs are in Appendices A and B, the diagnostic of Appendix C included. Generators, architectures, data, splits and training protocols are described in Appendices D and E.

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A PROOFS FOR SECTION 4

Theorem 1 bounds a ratio of two top eigenvalues, and its proof separates cleanly into a lower bound on the numerator and an upper bound on the denominator. Appendix A.1 proves the first by an explicit witness direction in the classifier head; Appendix A.2 proves the second, which is Lemma 1 of the main text; Appendix A.3 exhibits a concrete architecture for which *both* structural hypotheses are theorems rather than assumptions, so that the disparity law itself is proved outright for at least one member of each side of the pipeline; Appendix A.4 assembles the two bounds. Appendix A.5 records how the disparity relates to the classical causes of ill-conditioning, and Appendix A.6 states what remains open around Theorem 2. Throughout, Grams are normalized as empirical means over the $N := N_{\text{data}}HW$ pixel positions; the unnormalized (summed) Grams differ by the fixed factor N , which does not affect any statement in M .

A.1 THE SPATIAL BLOCK: AN EXPLICIT WITNESS

Lemma A.1 (Spatial block scaling). *Let M denote the characteristic width of g_ϕ (channel count for a CNN, embedding dimension for a ViT, hidden layer width for an MLP), let $N_{\text{cls}} \geq 2$, and suppose g_ϕ ends in a dense classifier head over penultimate features $h_n \in \mathbb{R}^{M_h}$ with $M_h = \Theta(M)$ whose softmax-weighted mean feature Gram*

$$S_h^p := \frac{1}{N} \sum_{n=1}^N p_{\min}^{(n)} h_n h_n^\top, \quad p_{\min}^{(n)} := \min_i p_i^{(n)}, \quad (3)$$

satisfies $\lambda_{\max}(S_h^p) \geq q_h M_h$ with probability at least $1 - \delta_h$ over the random initialization, for constants $q_h > 0$ and $\delta_h < 1$ independent of M (the head hypothesis: the features carry a common direction of $\Theta(M)$ energy across inputs, on pixels whose initial predictions are non-degenerate). Then the spatial Gauss–Newton block of (2) satisfies

$$\lambda_{\max}(G_{\phi\phi}) = \Omega(M)$$

in expectation over random Kaiming (variance-scaled Gaussian) initialization. The logit Gram $J_{\phi}^{\top} J_{\phi} \succeq G_{\phi\phi}$ inherits the bound unconditionally; the residual Gram $J_{\phi}^{\text{res}\top} J_{\phi}^{\text{res}}$ satisfies it under the analogous hypothesis on the p^2 -weighted Gram $S_h^{p^2}$. For architecture families with $C_g = \Theta(M^2)$ (fixed depth with at least two consecutive width-proportional layers), equivalently $\lambda_{\max}(G_{\phi\phi}) = \Omega(\sqrt{C_g})$.

The proof has three steps: (1) context—the exact width-linear law of Karakida et al. (Karakida et al., 2019) for the top Fisher eigenvalue in their model class, which we use as the asymptotic benchmark but do *not* invoke directly; (2) an exact spectral lower bound on the softmax inner matrix at initialization; (3) an elementary explicit witness direction in the head weights that yields $\Omega(M)$ self-containedly.

Proof. Step 1: The width-linear law of Karakida et al. (context). Karakida, Akaho, and Amari (Karakida et al., 2019) analyze fully connected networks of depth $L \geq 2$ with layer widths $M_{\ell} = \alpha_{\ell} M$, linear outputs, and variance-scaled Gaussian (Kaiming-type) initialization $W_{ij}^{\ell} \sim \mathcal{N}(0, \sigma_w^2/M_{\ell-1})$, $b_i^{\ell} \sim \mathcal{N}(0, \sigma_b^2)$. For the empirical Fisher information of the squared-loss model, $F = \frac{1}{T} \sum_{t=1}^T \sum_k \nabla f_k(x_t) \nabla f_k(x_t)^{\top}$ (the empirical-mean Gram of the output Jacobian), their Theorem 4 gives, in the limit $M \gg 1$,

$$\lambda_{\max}(F) = \alpha \left(\frac{T-1}{T} \kappa_2 + \frac{1}{T} \kappa_1 \right) M = \Theta(M),$$

where $\kappa_1, \kappa_2 = \Theta(1)$ are macroscopic constants computed from the layer-wise signal-propagation recursions and $\alpha = \sum_{\ell} \alpha_{\ell} \alpha_{\ell-1}$. Their Theorem 1 simultaneously shows the *mean* eigenvalue is $m_{\lambda} = C\kappa_1/M = \mathcal{O}(1/M)$: as the network widens, the spectrum becomes extremely skewed—almost all directions flatten while the top eigenvalue grows linearly in width. We record two cautions. First, Theorem 4 fixes the *output* layer at $C = \mathcal{O}(1)$ but requires the input and hidden widths to scale proportionally with M ; a classifier reading a fixed bottleneck K violates the input-side requirement (its coefficient α degenerates there), so the theorem cannot be applied verbatim to the spatial modules considered here. Second, both statements are asymptotic mean-field equalities with no finite- M rate. Steps 2–3 therefore prove the $\Omega(M)$ bound self-containedly; Karakida’s law serves as the sharp benchmark within its model class, validated numerically in their Figure 1.

Step 2: The softmax inner matrix at initialization. The loss is softmax cross-entropy, not squared loss, so the spatial Gauss–Newton block carries the softmax inner matrix

$$G_{\phi\phi} = \frac{1}{N} \sum_{n=1}^N J^{(n)\top} H_{\tau}^{(n)} J^{(n)}, \quad H_{\tau} := \text{diag}(p) - pp^{\top},$$

where $J^{(n)} := \partial \hat{y}^{(n)} / \partial \phi$ is the per-pixel logit Jacobian and $p = \text{softmax}(\hat{y})$. For any $v \in \mathbb{R}^{N_{\text{cls}}}$, the quadratic form is a variance: $v^{\top} H_{\tau} v = \sum_i p_i v_i^2 - (\sum_i p_i v_i)^2 = \text{Var}_{i \sim p}(v_i) \geq p_{\min} \sum_i (v_i - \bar{v}_p)^2 \geq p_{\min} \|P_{\perp} v\|^2$, where $p_{\min} := \min_i p_i$, $\bar{v}_p := \sum_i p_i v_i$, the last inequality uses that the unweighted mean minimizes the sum of squared deviations, and $P_{\perp} := I - \frac{1}{N_{\text{cls}}} \mathbf{1}\mathbf{1}^{\top}$ projects onto the mean-zero subspace $\mathbf{1}^{\perp}$. Hence

$$H_{\tau} \succeq p_{\min} P_{\perp}.$$

The bound is used per pixel in Step 3; no minimum over pixels is taken, so no uniform-over-pixels lower bound on $p_{\min}$ is needed anywhere in the proof. (For orientation: at Kaiming initialization the logits are $\mathcal{O}(1)$ uniformly in M , so $p_{\min}^{(n)}$ is of order $1/N_{\text{cls}}$ for typical n .)

Step 3: An explicit witness direction. Write the classifier head of g_{ϕ} as $\hat{y}_n = Wh_n + b$ with penultimate features $h_n = h(x_n) \in \mathbb{R}^{M_h}$, $M_h = \Theta(M)$. A perturbation of the head weights is a matrix $V \in \mathbb{R}^{N_{\text{cls}} \times M_h}$; it is a unit vector in parameter space precisely when $\|V\|_F = 1$, under any vectorization convention, and the logit perturbation it induces at pixel n is Vh_n . Let u be a top unit

eigenvector of S_h^p in (3), and fix a unit contrast $c \in \mathbb{R}^{N_{\text{cls}}}$ with $c \perp \mathbf{1}$ (possible since $N_{\text{cls}} \geq 2$); by Step 2, at every pixel

$$c^\top H_\tau^{(n)} c = \text{Var}_{i \sim p^{(n)}}(c_i) \geq p_{\min}^{(n)} \|P_\perp c\|^2 = p_{\min}^{(n)}.$$

Take the rank-one witness $V := c u^\top$, for which $\|V\|_F = \|c\| \|u\| = 1$ and $V h_n = (h_n^\top u) c$. Then

$$\begin{aligned} \lambda_{\max}(G_{\phi\phi}) &\geq \frac{1}{N} \sum_{n=1}^N (V h_n)^\top H_\tau^{(n)} (V h_n) = \frac{1}{N} \sum_{n=1}^N (h_n^\top u)^2 c^\top H_\tau^{(n)} c \\ &\geq u^\top S_h^p u = \lambda_{\max}(S_h^p). \end{aligned} \quad (4)$$

Note what (4) is: a deterministic, hypothesis-free inequality—one identity and one application of Step 2—valid at every parameter value. All N pixels contribute, and no minimum over pixels is taken, so no union bound and no factor of N enters.

The head hypothesis is exactly $\lambda_{\max}(S_h^p) \geq q_h M_h$ with probability at least $1 - \delta_h$. On that event $\lambda_{\max}(G_{\phi\phi}) \geq q_h M_h = \Omega(M)$; since $\lambda_{\max} \geq 0$ always,

$$\mathbb{E}[\lambda_{\max}(G_{\phi\phi})] \geq (1 - \delta_h) q_h M_h = \Omega(M),$$

which is the claim.

The other two conventions. The bound just proved is for the Gauss–Newton block $G_{\phi\phi} = J_\phi^\top H_\tau J_\phi$ of (2), the object appearing in Definition 1. Two neighbouring matrices inherit it. For the *logit* Gram, $H_\tau \preceq I$ gives $J_\phi^\top J_\phi \succeq G_{\phi\phi}$, so $\lambda_{\max}(J_\phi^\top J_\phi) \geq \lambda_{\max}(G_{\phi\phi}) = \Omega(M)$ immediately. For the *residual* Gram $J_\phi^{\text{res}\top} J_\phi^{\text{res}} = J_\phi^\top H_\tau^2 J_\phi$, note that H_τ is symmetric with $H_\tau \mathbf{1} = 0$, so $\mathbf{1}^\perp$ is invariant and, by Step 2, $H_\tau^{(n)} \succeq p_{\min}^{(n)} I$ there; squaring the eigenvalues on that subspace gives $(H_\tau^{(n)})^2 \succeq (p_{\min}^{(n)})^2 P_\perp$, so the identical witness argument runs with the weights $(p_{\min}^{(n)})^2$ in place of $p_{\min}^{(n)}$ and yields $\lambda_{\max}(J_\phi^{\text{res}\top} J_\phi^{\text{res}}) \geq \lambda_{\max}(S_h^{p^2}) = \Omega(M)$ under the corresponding head hypothesis. $\square$

On the head hypothesis, and its counterpart in Karakida et al. Write $S_h := N^{-1} \sum_n h_n h_n^\top$ for the unweighted mean feature Gram. The hypothesis $\lambda_{\max}(S_h^p) = \Omega(M_h)$ asks two things at once: that the penultimate features carry a common direction of $\Theta(M_h)$ energy across inputs ($\lambda_{\max}(S_h) = \Omega(M_h)$), and that the initial predictions are non-degenerate on the pixels supplying that energy. The second is mild at initialization—the logits are $\mathcal{O}(1)$ uniformly in M , since the head weights have variance $\Theta(1/M_h)$ against $\|h_n\|^2 = \mathcal{O}(M_h)$, so $p_{\min}^{(n)}$ is bounded below by a constant of order $1/N_{\text{cls}}$ on all but a small fraction of pixels, and S_h^p dominates that constant times the unweighted Gram restricted to those pixels. (Proposition A.1 below turns this sketch into a proof for a concrete head.) The first is the substantive condition. It is implied by $\|\bar{h}\|^2 \geq q M_h$ on the *mean* feature vector $\bar{h} := N^{-1} \sum_n h_n$: for $\bar{u} = \bar{h} / \|\bar{h}\|$,

$$\lambda_{\max}(S_h) \geq \bar{u}^\top S_h \bar{u} = \frac{1}{\|\bar{h}\|^2} \cdot \frac{1}{N} \sum_n (h_n^\top \bar{h})^2 \geq \frac{1}{\|\bar{h}\|^2} \left(\frac{1}{N} \sum_n h_n^\top \bar{h} \right)^2 = \frac{\|\bar{h}\|^4}{\|\bar{h}\|^2} = \|\bar{h}\|^2,$$

the middle step being Jensen’s inequality (the mean of squares dominates the square of the mean). At variance-scaled initialization each entry of h_n has $\Theta(1)$ second moment (the signal-propagation recursions of Karakida et al. (Karakida et al., 2019), Eq. (9)), so the condition holds whenever the activations are correlated across inputs—in particular for any nonnegative or nonzero-mean activation (ReLU and its variants), and for odd activations whenever the bias variance $\sigma_b^2 > 0$ or the inputs are themselves correlated.

Per-entry second moments alone are *not* enough: $\|h_n\|^2 = \Theta(M_h)$ for every n gives only $\text{tr}(S_h) = \Theta(M_h)$, hence $\lambda_{\max}(S_h) \geq \Theta(M_h) / \min(N, M_h)$, which is weaker by a factor of N . Cross-input correlation, not per-vector norm, is what makes the top eigenvalue grow. The same quantity carries the width-linear term in Karakida et al.’s Theorem 4: their κ_2 is built from the cross-input activation correlations $\hat{q}_{st}$. Their footnote notes that odd activations with $\sigma_b = 0$ give $\hat{q}_{st} = 0$ and $\kappa_2 = 0$, and is explicit that this exceptional case falls outside their analysis: “In such exceptional cases, we need to evaluate the lower order terms of s_λ and $\lambda_{\max}$ (outside the scope of this study).” We therefore

attribute no finite- T claim to their theorem in the degenerate case. Our hypothesis has the parallel structure, and its degenerate behaviour *can* be evaluated directly: for odd activations with $\sigma_b = 0$ and uncorrelated inputs the features lose their common direction, and the per-vector norms alone give only $\lambda_{\max}(S_h) \geq \text{tr}(S_h)/\min(N, M_h)$. In our regime $N \gg M_h$, so this floor is $\Theta(1)$ —no width scaling. Width-linear growth in the degenerate case would require a separate argument, which we do not attempt.

Remark A.1 (Scope, and the role of the mean-field law). The witness argument uses only: (a) a dense classifier head over penultimate features of dimension $\Theta(M)$; (b) the head hypothesis $\lambda_{\max}(S_h^p) \geq q_h M_h$ at initialization—a *cross-input correlation* condition, which variance-scaled initialization does not by itself deliver (see the per-vector-norm counterexample above) but which holds for the standard activation choices discussed there; and (c) softmax cross-entropy with $N_{\text{cls}} \geq 2$. Given (b), the inequality $\lambda_{\max}(G_{\phi\phi}) \geq \lambda_{\max}(S_h^p)$ is deterministic and architecture-agnostic, so the argument applies to per-pixel dense heads on MLP, CNN (channel count M), and ViT (embedding dimension M) spatial modules alike. For fully connected modules (b) follows from standard signal propagation at Kaiming initialization; for CNN and ViT heads it is an explicit assumption, to be checked. No i.i.d. Gaussian input assumption is required. Karakida et al.’s Theorem 4 plays the role of the sharp asymptotic benchmark: within its model class (fully connected, input and hidden widths proportional to M , output fixed at $C = \mathcal{O}(1)$, squared loss), it upgrades the witness lower bound to an exact width-linear law with explicit constant, and their Theorem 1 supplies the complementary fact that the *mean* eigenvalue shrinks as $\mathcal{O}(1/M)$, so curvature concentrates in a vanishing fraction of directions as the module widens. The loss transfer from squared loss to cross-entropy is handled exactly at initialization by Step 2, at the cost of the constant $p_{\min}$.

Remark A.2 (Empirical status, and width versus parameter count). Karakida et al.’s own numerical validation (their Figure 1, right column) confirms $\lambda_{\max}$ growing linearly in M , matching the exact constant of their Theorem 4, for tanh, ReLU, and linear fully connected networks. The theorem’s native variable is width, not parameter count: for architecture families with $C_g = \Theta(M^2)$ the equivalent parameter-count statement is $\lambda_{\max} = \Omega(\sqrt{C_g})$ (log–log slope 0.5 against C_g), whereas the shallow family measured in Section 4 has $C_g = \Theta(M)$, for which the width-linear law predicts slope 1.0 against C_g as well as against M . The two parameter-count forms must not be conflated. The measured slopes, their dependence on the initialization scheme—which controls whether either variance term of the head hypothesis is width-independent—and the levels that accompany them are reported in Section 4 and Appendix D; we quote slope and level together, and read finite-range fits as finite-range fits.

A.2 THE SPECTRAL BLOCK: AN OPERATOR-NORM CAP

The proof of Lemma 1 factors the spectral Jacobian via the chain rule and bounds each factor’s operator norm using Assumptions 1, 2, and 3. We bound the *logit* Jacobian $J_{\theta} = \partial \hat{y} / \partial \theta$ of (1); the Gauss–Newton block and the residual Jacobian then inherit the bound, since both insert factors of H_{τ} with $0 \preceq H_{\tau} \preceq I$.

Proof of Lemma 1. Step 1: Chain-rule factorisation. With $Z = f_{\theta}(X)$ and $\hat{y} = g_{\phi}(Z)$, the chain rule gives, in the normalization of (1) (the $N^{-1/2}$ riding on the right-hand factor),

$$J_{\theta} = J_Z \cdot \frac{\partial Z}{\partial \theta}, \quad J_Z := \partial g_{\phi} / \partial Z,$$

where J_Z is unnormalized—the object Assumption 3 bounds. Since g_{ϕ} acts on one image at a time, the dataset-stacked J_Z is block diagonal across images, so its operator norm is the maximum over per-image blocks and Assumption 3 applies to the stack with the same constant $\tilde{L}$. (For the residual map $r(\theta, \phi) = \tau(g_{\phi}(Z))$ with $\tau(\hat{y}) = \text{softmax}(\hat{y}) - y_{\text{onehot}}$, the same computation carries the extra left factor $J_{\tau} := \partial \tau / \partial \hat{y} = \text{diag}(p) - pp^{\top} = H_{\tau}$, the softmax Jacobian, which satisfies $\|H_{\tau}\|_{\text{op}} \leq 1$; see Botev, Ritter, and Barber (Botev et al., 2017).)

Step 2: Operator-norm inequality. Sub-multiplicativity of the operator norm gives

$$\|J_{\theta}\|_{\text{op}} \leq \|J_Z\|_{\text{op}} \cdot \left\| \frac{\partial Z}{\partial \theta} \right\|_{\text{op}} \leq \tilde{L} \cdot \left\| \frac{\partial Z}{\partial \theta} \right\|_{\text{op}},$$

using Assumption 3 on the first factor.

Step 3: The input-Jacobian factor, exactly. Under Assumption 1, $Z_n = W^\top x_n$ with $\theta = \text{vec}(W)$, $W \in \mathbb{R}^{S \times K}$. The per-pixel Jacobian is $\partial Z_n / \partial \text{vec}(W) = I_K \otimes x_n^\top$ (Magnus and Neudecker (Magnus & Neudecker, 1988), Ch. 9). Stacking over the N pixel positions with the $N^{-1/2}$ normalization of (1) gives $\partial Z / \partial \theta \in \mathbb{R}^{NK \times SK}$ with

$$\left(\frac{\partial Z}{\partial \theta} \right)^\top \frac{\partial Z}{\partial \theta} = \frac{1}{N} \sum_{n=1}^N (I_K \otimes x_n) (I_K \otimes x_n^\top) = I_K \otimes \Sigma_X,$$

hence, *exactly*,

$$\left\| \frac{\partial Z}{\partial \theta} \right\|_{\text{op}}^2 = \lambda_{\max}(I_K \otimes \Sigma_X) = \lambda_{\max}(\Sigma_X),$$

with $\Sigma_X := N^{-1} \sum_{n=1}^N x_n x_n^\top$ the mean per-pixel input Gram of the main text; Assumption 2 bounds $\lambda_{\max}(\Sigma_X) \leq \Lambda_X$ uniformly in the dataset size N . No factor of K appears: the Kronecker structure replicates the same spectrum K times rather than accumulating it.

Step 4: Assembling. Combining Steps 2 and 3,

$$\|J_\theta\|_{\text{op}}^2 \leq \tilde{L}^2 \cdot \lambda_{\max}(\Sigma_X) =: C_\theta.$$

For the Gauss–Newton block, $G_{\theta\theta} = J_\theta^\top H_\tau J_\theta \preceq J_\theta^\top J_\theta$ since $H_\tau \preceq I$, so $\lambda_{\max}(G_{\theta\theta}) \leq C_\theta$; for the residual Jacobian, $\|H_\tau J_\theta\|_{\text{op}} \leq \|J_\theta\|_{\text{op}}$ gives the same cap. The constant C_θ depends on the classifier’s input Lipschitz constant $\tilde{L}$ (which may grow with the depth of g_ϕ but not with its width M) and the top eigenvalue of the mean input Gram $\lambda_{\max}(\Sigma_X)$. It does not depend on M , on the spatial parameter count C_g , or on the dataset size N . $\square$

Remark A.3 (Why an operator-norm cap suffices). An alternative route via the smallest positive eigenvalue $\lambda_{\min}^+(J_\theta^\top J_\theta)$ requires either an exact Kronecker factorisation (valid only for per-pixel classifiers) or a trace-over-rank argument via the softmax residual dimension—both of which need additional structural assumptions on g_ϕ . The operator-norm bound above is significantly simpler, applies to general nonlinear spatial classifiers (CNN, ViT), and is exactly what Theorem 1 needs: a C_g -independent cap on the spectral block’s *top* eigenvalue, so that $\mathcal{D}_{\text{curv}} = \lambda_{\max}(G_{\phi\phi}) / \lambda_{\max}(G_{\theta\theta}) \geq \Omega(M) / C_\theta$.

Remark A.4 (Softmax Jacobian bound). The bound $\|\text{diag}(p) - pp^\top\|_{\text{op}} \leq 1$ is standard: $\text{diag}(p) - pp^\top$ is the covariance matrix of the categorical distribution with parameter p (equivalently, the Jacobian of the softmax map), hence positive semi-definite with $\lambda_{\max} \leq \max_i p_i \leq 1$. See Botev, Ritter, and Barber (Botev et al., 2017) for its role in Gauss–Newton computations.

A.3 A FULLY PROVED INSTANCE

For general heads, the head hypothesis of Lemma A.1 and the uniform input-Jacobian cap of Assumption 3 are hypotheses. For one concrete architecture they are both theorems, and the disparity law follows outright.

Proposition A.1 (A fully proved instance). *Fix bottleneck features $z_1, \dots, z_N \in \mathbb{R}^K$ with $\|z_n\| \leq R$, and let the spatial module’s head be the two-layer ReLU network*

$$h_n = \text{ReLU}(W_1 z_n + b_1), \quad \hat{y}_n = W_2 h_n + b_2,$$

with Kaiming Gaussian initialization: entries of $W_1 \in \mathbb{R}^{M_h \times K}$ i.i.d. $\mathcal{N}(0, \sigma_w^2/K)$, entries of $W_2 \in \mathbb{R}^{N_{\text{cls}} \times M_h}$ i.i.d. $\mathcal{N}(0, \sigma_w^2/M_h)$, biases i.i.d. $\mathcal{N}(0, \sigma_b^2)$ with $\sigma_b > 0$, and $N_{\text{cls}} \geq 2$. Let the spectral module be the linear per-pixel f_θ of Assumption 1 with inputs satisfying Assumption 2. Then there exist constants $q_h > 0$, $M_0 < \infty$, and $C_G < \infty$, depending only on $\sigma_w, \sigma_b, R, K, N_{\text{cls}}$, and $\text{tr}(\Sigma_X)$ —not on M_h or N —such that for all $M_h \geq M_0$, at initialization:

- (i) (head hypothesis) $\lambda_{\max}(S_h^p) \geq q_h M_h$ with probability at least $1/2$; consequently $\mathbb{E}[\lambda_{\max}(G_{\phi\phi})] \geq \frac{1}{2} q_h M_h$;
- (ii) (spectral-block cap, trace form) $\mathbb{E}[\lambda_{\max}(G_{\theta\theta})] \leq \mathbb{E}[\text{tr}(G_{\theta\theta})] \leq C_G := N_{\text{cls}} \sigma_w^4 \text{tr}(\Sigma_X)$, uniformly in M_h and N ;

(iii) (instance-level disparity law) with probability at least $3/8$, $\mathcal{D}_{\text{curv}} \geq (q_h/(8C_G)) M_h$; consequently $\mathbb{E}[\mathcal{D}_{\text{curv}}] \geq \frac{3}{64}(q_h/C_G) M_h = \Omega(M)$.

For this architecture, the disparity law of Theorem 1—not merely its hypotheses—is proved outright at initialization. What remains assumption-level in general is the uniform-over-inputs, during-training form of Assumption 3: a per-input Jacobian cap at initialization follows from the computation in Step D below, but a supremum over all inputs by the natural union route would cost a $\sqrt{\log N}$ factor, and training-time persistence is exactly what Assumption 3 postulates. The trace-form cap (ii) bypasses both issues for the instance.

Proof. Step A (mean feature vector). Write $g_{n,i} = w_i^\top z_n + b_{1,i}$ for the pre-activations, so $g_{n,i} \sim \mathcal{N}(0, s_n^2)$ with $s_n^2 = \sigma_w^2 \|z_n\|^2 / K + \sigma_b^2 \in [\sigma_b^2, s_{\max}^2]$, $s_{\max}^2 := \sigma_w^2 R^2 / K + \sigma_b^2$. Since $\mathbb{E} \text{ReLU}(\mathcal{N}(0, s^2)) = s/\sqrt{2\pi}$, each coordinate of the mean feature vector $\bar{h} = N^{-1} \sum_n h_n$ has expectation

$$m := \mathbb{E}[\bar{h}_i] = \frac{1}{N} \sum_n \frac{s_n}{\sqrt{2\pi}} \geq \frac{\sigma_b}{\sqrt{2\pi}} > 0.$$

To obtain constants free of the data and of N , we use throughout only the uniform floor $m \geq m_0 := \sigma_b/\sqrt{2\pi}$. The rows $(w_i, b_{1,i})$ are i.i.d. across i , so the coordinates $\bar{h}_i$ are i.i.d. with $\mathbb{E}[\bar{h}_i^2] \leq N^{-1} \sum_n \mathbb{E}[\text{ReLU}(g_{n,i})^2] \leq s_{\max}^2$ and (being averages of sub-Gaussian variables) finite fourth moment bounded by a constant $C_4(s_{\max})$. By Chebyshev’s inequality applied to the i.i.d. sum $\|\bar{h}\|^2 = \sum_i \bar{h}_i^2$, whose mean is at least $m^2 M_h \geq m_0^2 M_h$,

$$\Pr\left[\|\bar{h}\|^2 < \frac{1}{2} m_0^2 M_h\right] \leq \frac{4C_4(s_{\max})}{m_0^4 M_h} \xrightarrow{M_h \rightarrow \infty} 0.$$

On the complementary event E_A , the Jensen step above gives $\lambda_{\max}(S_h) \geq \|\bar{h}\|^2 \geq \frac{1}{2} m_0^2 M_h$, and moreover the witness direction $\bar{u} = \bar{h} / \|\bar{h}\|$ certifies $A := N^{-1} \sum_n (h_n^\top \bar{u})^2 \geq \frac{1}{2} m_0^2 M_h$.

Step B (softmax weights). Condition on $\mathcal{F}_1 = \sigma(W_1, b_1)$, which fixes the features h_n , the direction $\bar{u}$, and the energies $a_n := (h_n^\top \bar{u})^2$; the head randomness (W_2, b_2) is independent of $\mathcal{F}_1$. Call pixel n tame if $\|h_n\|^2 \leq 2s_{\max}^2 M_h$. For a tame pixel, the logits $\hat{y}_n | \mathcal{F}_1$ have i.i.d. Gaussian entries with variance $\sigma_w^2 \|h_n\|^2 / M_h + \sigma_b^2 \leq 2\sigma_w^2 s_{\max}^2 + \sigma_b^2 =: s_*^2$, so by a Gaussian tail bound on the logit range, there are constants $c_p(s_*, N_{\text{cls}}) > 0$ and $\delta_0 \leq 1/16$ with $\Pr[p_{\min}^{(n)} < c_p | \mathcal{F}_1] \leq \delta_0$ for every tame pixel. Untame pixels are exponentially rare: $\|h_n\|^2 / M_h$ is an average of M_h i.i.d. sub-exponential variables with mean $\leq s_{\max}^2/2$, so $\Pr[n \text{ untame}] \leq e^{-cM_h}$ (Bernstein’s inequality; Vershynin (Vershynin, 2018), Thm. 2.8.1), and by Cauchy–Schwarz with $\mathbb{E}\|h_n\|^4 \leq 3s_{\max}^4 M_h^2$ (i.i.d. coordinates), the $\mathcal{F}_1$ -measurable untame energy $U := N^{-1} \sum_n a_n \mathbf{1}[n \text{ untame}]$ satisfies

$$\mathbb{E}[U] \leq \max_n \sqrt{\mathbb{E}\|h_n\|^4} \sqrt{\Pr[n \text{ untame}]} \leq \sqrt{3} s_{\max}^2 M_h e^{-cM_h/2} =: \varepsilon(M_h) M_h,$$

with $\varepsilon(M_h) \rightarrow 0$ exponentially (we used $a_n \leq \|h_n\|^2$, valid since $\bar{u}$ is a unit vector).

Step C (assembly). We control the two bad masses by separate arguments, respecting which randomness each depends on. The untame energy U is $\mathcal{F}_1$ -measurable; by (unconditional) Markov, the event $E_U := \{U \leq 16\varepsilon(M_h) M_h\}$ has $\Pr[E_U] \geq 15/16$, and for $M_h \geq M_0$ large enough that $16\varepsilon(M_h) \leq m_0^2/8$, on $E_A \cap E_U$ we have $U \leq \frac{1}{8} m_0^2 M_h \leq \frac{1}{4} A$. The p -degenerate mass over tame pixels, $P := N^{-1} \sum_{n \text{ tame}} a_n \mathbf{1}[p_{\min}^{(n)} < c_p]$, is (W_2, b_2) -random given $\mathcal{F}_1$, with $\mathbb{E}[P | \mathcal{F}_1] \leq \delta_0 A$ by Step B; conditional Markov with $\delta_0 \leq 1/16$ gives $\Pr[P > \frac{1}{4} A | \mathcal{F}_1] \leq 4\delta_0 \leq 1/4$. On the intersection of the three events—probability at least $1 - 4C_4/(m_0^4 M_h) - 1/16 - 1/4 > 1/2$ for $M_h \geq M_0$ —the total bad mass is $P + U \leq \frac{1}{2} A$, and

$$\lambda_{\max}(S_h^p) \geq \bar{u}^\top S_h^p \bar{u} \geq c_p (A - P - U) \geq \frac{c_p}{2} A \geq \frac{c_p m_0^2}{4} M_h.$$

Take $q_h := c_p m_0^2/4 = c_p \sigma_b^2/(8\pi)$, manifestly free of the data and of N . The in-expectation clause follows from $\lambda_{\max} \geq 0$ and (4). This proves (i).

Step D (spectral-block cap, trace form). At any input z , the head’s input Jacobian is $\partial \hat{y} / \partial z = W_2 D_z W_1$ with $D_z = \text{diag}(\mathbf{1}[W_1 z + b_1 > 0])$. Conditioning on (W_1, b_1) (which fixes D_z) and using that the rows of W_2 are i.i.d. $\mathcal{N}(0, (\sigma_w^2/M_h)I)$,

$$\mathbb{E} \|W_2 D_z W_1\|_F^2 = N_{\text{cls}} \cdot \frac{\sigma_w^2}{M_h} \cdot \mathbb{E} \|D_z W_1\|_F^2 \leq N_{\text{cls}} \cdot \frac{\sigma_w^2}{M_h} \cdot \mathbb{E} \|W_1\|_F^2 = N_{\text{cls}} \sigma_w^4,$$

using $\mathbb{E} \|W_1\|_F^2 = M_h K \cdot \sigma_w^2 / K = M_h \sigma_w^2$ —uniformly over inputs z . (The naive product bound $\|W_2\|_{\text{op}} \|W_1\|_{\text{op}}$ would grow like $\sqrt{M_h/K}$; the composition is $O(1)$ because the top directions of the two random factors are generically misaligned—which the Frobenius computation captures without any alignment analysis.) Now bound the spectral block. With $J_{\theta}^{(n)} = (W_2 D_{z_n} W_1)(I_K \otimes x_n^\top)$ the per-pixel spectral logit Jacobian (Appendix A.2, Step 3) and $H_\tau \preceq I$,

$$\begin{aligned} \mathbb{E} \text{tr}(G_{\theta\theta}) &\leq \frac{1}{N} \sum_{n=1}^N \mathbb{E} \|J_{\theta}^{(n)}\|_F^2 \leq \frac{1}{N} \sum_{n=1}^N \mathbb{E} \|W_2 D_{z_n} W_1\|_F^2 \cdot \|x_n\|^2 \\ &\leq N_{\text{cls}} \sigma_w^4 \cdot \frac{1}{N} \sum_n \|x_n\|^2 = N_{\text{cls}} \sigma_w^4 \text{tr}(\Sigma_X), \end{aligned}$$

using $\|AB\|_F \leq \|A\|_F \|B\|_{\text{op}}$ and $\|I_K \otimes x_n^\top\|_{\text{op}} = \|x_n\|$. Since $\lambda_{\max} \leq \text{tr}$ for positive semidefinite matrices, (ii) follows with $C_G = N_{\text{cls}} \sigma_w^4 \text{tr}(\Sigma_X)$, free of M_h and N .

Step E (combining). If $C_G = 0$ then $\Sigma_X = 0$, the spectral block vanishes almost surely, and $\mathcal{D}_{\text{curv}} = +\infty$ by the convention of Definition 1; assume $C_G > 0$ below. Let E_1 be the event of (i) ($\Pr \geq 1/2$) and $E_2 := \{\lambda_{\max}(G_{\theta\theta}) \leq 8C_G\}$, which by (ii) and Markov has $\Pr[E_2] \geq 7/8$. On $E_1 \cap E_2$ (probability at least $1 - 1/2 - 1/8 = 3/8$), the witness bound (4) gives

$$\mathcal{D}_{\text{curv}} = \frac{\lambda_{\max}(G_{\phi\phi})}{\lambda_{\max}(G_{\theta\theta})} \geq \frac{q_h M_h}{8C_G},$$

and since $\mathcal{D}_{\text{curv}} \geq 0$, $\mathbb{E}[\mathcal{D}_{\text{curv}}] \geq \frac{3}{8} \cdot q_h M_h / (8C_G) = \frac{3}{64} (q_h / C_G) M_h$. This proves (iii). $\square$

Remark A.5 (Minimality). Proposition A.1 is deliberately minimal: one hidden layer, ReLU, bias variance $\sigma_b^2 > 0$ (which guarantees activation correlation across inputs; cf. the κ_2 discussion above). Its role is to certify that the two structural hypotheses of Theorem 1 are *theorems*, not merely assumptions, for at least one concrete member of each side of the pipeline—so the disparity law has a fully verified instance. For deeper heads and attention-based modules the hypotheses revert to assumption status, argued by signal propagation and checked empirically.

Remark A.6 (What the instance shows—and what it does not). The mechanism of Proposition A.1 deserves a candid reading. The width-linear spike is generated by the *mean feature direction*: with $\sigma_b > 0$, every hidden unit has positive mean activation, so the features of all inputs share a common direction of $\Theta(M_h)$ energy. This requires no information in the features at all. Take every bottleneck feature $z_n = 0$, so that all inputs are indistinguishable: the features $h_n = \text{ReLU}(b_1)$ are identical across pixels, carry no sample-specific structure whatsoever, and the proposition applies verbatim. Width-linear downstream curvature is therefore *not* evidence that the spatial module has found, or will find, useful structure: $\mathcal{D}_{\text{curv}}$ measures the downstream module’s readiness to move—the loss curvature its parameters command at initialization—not the value of what it computes. It is the counterpart, on the spatial side, of an ambiguity on the spectral side: small spectral curvature cannot by itself distinguish a direction that is irrelevant to the task from one whose learning signal is buried behind the bottleneck; here, large spatial curvature cannot distinguish signal from capacity. It is also why we read the geometry as a hazard rather than as a mechanism: the capacity advantage is in place before the data has spoken, so whatever the spatial module can most easily fit is what the geometry favours. Whether it in fact fits it first is the separate dynamical question of Theorem 2 and of the residual-excitation results of Appendix B.

Remark A.7 (Smoothness scope). ReLU is not twice differentiable, but no smoothness enters Proposition A.1: the Gauss–Newton blocks involve only first-order Jacobians, which at Gaussian initialization exist almost surely (every preactivation is a.s. nonzero). Assumption 4 plays no role here—its function is the well-posedness of the gradient flow of Theorem 2, whose statements are formulated for absolutely continuous trajectories satisfying the flow almost everywhere and therefore accommodate piecewise-smooth models.

A.4 PROOF OF THEOREM 1

Proof of Theorem 1. By Lemma A.1, in expectation over random Kaiming (variance-scaled Gaussian) initialization,

$$\lambda_{\max}(G_{\phi\phi}^{(M)}) \geq c_{\phi} \cdot M$$

for a constant $c_{\phi} > 0$ bounded below independently of M : the witness bound (4) gives $\mathbb{E}[\lambda_{\max}] \geq (1 - \delta_h) q_h M_h \geq (1 - \delta_h) q_h \beta M$, where $\beta > 0$ is the width-ratio floor $M_h \geq \beta M$ supplied by the head hypothesis, so one may take $c_{\phi} := (1 - \delta_h) q_h \beta$. The witness route delivers this with no additive correction; within Karakida et al.’s (Karakida et al., 2019) proportional-width model class, their Theorem 4 sharpens the width-linear rate to an exact asymptotic law. By Lemma 1,

$$\lambda_{\max}(G_{\theta\theta}) \leq \|J_{\theta}\|_{\text{op}}^2 \leq C_{\theta}$$

with C_{θ} independent of M and C_g . (If $C_{\theta} = 0$ the spectral block vanishes identically and the bound holds trivially under the $+\infty$ convention of Definition 1; assume $C_{\theta} > 0$ below.) The cap is deterministic—valid for every realization of the initialization—so it divides through inside the expectation:

$$\mathbb{E}[\mathcal{D}_{\text{curv}}^{(M)}] = \mathbb{E}\left[\frac{\lambda_{\max}(G_{\phi\phi}^{(M)})}{\lambda_{\max}(G_{\theta\theta})}\right] \geq \frac{\mathbb{E}[\lambda_{\max}(G_{\phi\phi}^{(M)})]}{C_{\theta}} \geq \frac{c_{\phi} \cdot M}{C_{\theta}} = c_1 \cdot M, \quad (5)$$

where $c_1 := c_{\phi}/C_{\theta}$ depends only on the initialization scheme, the architecture class of g_{ϕ} , the classifier’s input Lipschitz constant $\tilde{L}$, the bottleneck dimension K , the input spectral dimension S , and the data distribution—exactly the dependencies claimed in the theorem statement. The witness route needs no additive correction ($c_2 = 0$ suffices); the subtracted constant is retained in the statement so that sharper asymptotic constants, which hold only for M large, may also be substituted. Under $C_g = \Theta(M^2)$, $M = \Theta(\sqrt{C_g})$ and the ratio form $\mathcal{D}_{\text{curv}} \geq c'_1 \sqrt{C_g}/C_f - c_2$ follows on multiplying and dividing by $\sqrt{C_f}$. $\square$

Remark A.8 (What Theorem 1 does and does not claim). Theorem 1 is a scaling law for the curvature ratio between two Gauss–Newton blocks; it does not claim that the full joint Gauss–Newton matrix has a large condition number in the strict $\lambda_{\max}/\lambda_{\min}$ sense (which is trivially infinite under rank-deficiency), nor that joint gradient descent produces disparate effective learning rates (the separate content of Theorem 2, under its own hypotheses). The scaling $\mathcal{D}_{\text{curv}} = \Omega(M)$ (equivalently $\Omega(\sqrt{C_g/C_f})$ for $\Theta(M^2)$ -parameter families) is a purely geometric statement about how curvature concentrates in the downstream module as the spatial network widens. In particular $\mathcal{D}_{\text{curv}}$ concerns the *top* of the spatial block’s spectrum, while the fitting rate μ of Assumption 5 is governed by how fast the residual can be driven down—a property of other parts of the spatial spectrum. The two are linked only under spectral-shape assumptions we do not make.

A.5 RELATION TO CLASSICAL ILL-CONDITIONING

Ill-conditioned optimization has classical causes with classical remedies, and it is natural to ask whether they remove the width scaling of Theorem 1. Each acts on a different lever, and none acts on the one the theorem uses.

Feature normalization. Scale asymmetry or collinearity among input features degrades the conditioning of Σ_X and is classically cured by standardizing (e.g. mean-centering) the inputs. Here the cure has a sharper, quantitative role: Lemma 1 places $\lambda_{\max}(\Sigma_X)$ in the spectral cap, so input transformations that deflate the top of the input spectrum *raise* the bound on the disparity at every width—normalization moves the constants. What no input transformation can touch is the width scaling of the spatial block (Lemma A.1): capacity moves the rate. On raw nonnegative spectra, whose second moment is dominated by the shared mean shape, the cap is large and the disparity the bound permits correspondingly small. Whether an input transformation moves the *realized* disparity, however, depends on the architecture between f_{θ} and the loss: internal normalization layers can absorb input shifts entirely (their Jacobians annihilate constant directions), in which case the effective regressor is already centered and the lever acts through the normalization choice, not the preprocessing. Section 7 measures exactly this on the production architecture.

Weight decay. Adding $\alpha \|w\|^2$ to the loss shifts the Hessian by $2\alpha I$, lifting every eigenvalue by 2α and famously repairing classical condition numbers. But the shift acts on *both* blocks equally: the disparity becomes $(\lambda_{\max}(G_{\phi\phi}) + 2\alpha)/(\lambda_{\max}(G_{\theta\theta}) + 2\alpha)$, which remains $\Theta(M)$ for any fixed α as M grows. Removing the disparity this way would require $2\alpha \sim \lambda_{\max}(G_{\phi\phi})$, a regularization strength that would dominate the data term and prevent fitting. The same arithmetic applies to any penalty that shifts the two blocks by a common multiple of the identity, including logit penalties of the Spectral Decoupling type (Pezeshki et al., 2021).

Adaptive optimizers. Adam and RMSProp precondition per-coordinate gradient magnitudes with a diagonal rescaling. The raw block spectra are optimizer-invariant, but the *dynamics* under a diagonal preconditioner P are governed by the preconditioned spectra of $P^{-1/2}GP^{-1/2}$ and need not respect the disparity: under Adam, per-coordinate steps are near-constant in gradient magnitude, so small spectral gradients do not imply small spectral movement. The gradient-flow statements around Theorem 2 are accordingly scoped to (stochastic) gradient descent, with the discrete pathwise versions of Appendix B; joint-versus-frozen gaps observed under Adam are evidence about the phenomenon, not applications of the theorem. The synthetic dynamics experiment reported in Section 7 and Appendix D is run under Adam, and at three seeds its arm ordering is not resolved; we treat the stronger claim—that no shared-learning-rate adaptive method can remove the disparity’s dynamical consequence—as an open question rather than a theorem. Optimizers with explicitly different learning rates per module (two-timescale update rules; Heusel et al. (Heusel et al., 2017); Borkar (Borkar, 2008), Ch. 6) are the natural remedy at the optimizer level; we do not evaluate them here, and Section 8 is our reason for not recommending untuned comparisons.

Remark A.9 (Parameterization scope). $\mathcal{D}_{\text{curv}}$ is not invariant to per-module reparameterization, and Theorem 1 is a statement about the standard variance-scaled (Kaiming) parameterization—the one used in the pipelines we analyze and in all our experiments. Under alternative parameterizations that rescale layers with width, such as μP (Yang & Hu, 2021), both the initialization magnitudes and the effective per-layer learning rates change, and the width scaling of the blocks would need to be re-derived; we do not claim the disparity law transfers. Our uses of μP elsewhere are limited to a different role: as one route to keeping the input-Jacobian bound (Assumption 3) uniform in width during training. Whether principled reparameterization can neutralize the disparity—rather than merely re-expressing it—is open. A closely related caution applies to normalization layers, whose gain can move measured curvature with width at fixed function (Appendix B and Section 7; cf. van Laarhoven (van Laarhoven, 2017)).

Remark A.10 (Defense against linear triviality). A natural objection is that, since f_{θ} is linear, the analysis must reduce to principal component analysis (PCA) of the input. This is not the case. PCA finds directions of maximum input variance *without reference to labels*. At initialization, however, the distinction is limited, and we state this candidly: the Gauss–Newton blocks carry no label information there (the softmax weights are near-uniform), and $G_{\theta\theta}$ is, by Lemma 1, an input-second-moment object propagated through a label-free random network. The defense against PCA-triviality is therefore a statement about *training*: along the trajectory the supervised residual shapes $\partial\hat{y}/\partial Z$ and hence the curvature, and the conditional pathology we identify is precisely that this supervised signal may fail to reach θ —not that f_{θ} is incapable of representing useful features.

A.6 SCOPE AND OPEN EXTENSIONS FOR THEOREM 2

Remark A.11 (Open extensions). The proof of Theorem 2 is complete for the gradient-flow model: the displacement bound follows by direct integration of the residual envelope, with the operator-norm cap supplied by Lemma 1. Two extensions are deliberately left open rather than sketched, and one is supplied elsewhere. (i) *Discrete updates*: pathwise discrete counterparts—gradient descent, momentum with clipping in either order, Adam with a positive stabilizer, and decoupled weight decay—are proved in Appendix B with $\mu_s = \eta\mu_f$, under the same containment and gradient-bound hypotheses imposed at *each* iterate; a stochastic-approximation route that instead tracks the flow under gradient noise (Borkar (Borkar, 2008), Ch. 2) would relax that requirement, and we have not carried it out. (ii) *Deriving the envelope*: obtaining Assumption 5—in particular the growth of μ with C_g —from the architecture, for instance by a kernel analysis of the spatial pathway at large width adapting Pezeshki et al. (Pezeshki et al., 2021), is the missing link that would chain Theorems 1 and 2 into a single causal statement; we test its content empirically rather than claiming it, and the residual-excitation and phase-restricted results of Appendix B record why curvature anisotropy

alone does not supply it. (iii) *Envelope verification*: fitting $C/(1 + \mu t)$ envelopes to measured residual trajectories, with the fitted $\hat{\mu}$ increasing in capacity, is an empirical programme, reported for the shallow model in Appendix D.

B ADDITIONAL THEORY

This appendix collects the mathematical results that the main text cites but does not prove. In order: pathwise discrete-time counterparts of the displacement bound (gradient descent, momentum under either clipping order, velocity clipping, Adam with a positive stabilizer, decoupled decay); monotone and affine joint-versus-frozen comparisons, the finite-horizon attribution bound that Corollary 1 states in the main text, the counterexample showing that a large *top* kernel eigenvalue is not a rate certificate, and a finite-data initialization certificate; interior curvature witnesses and the normalization propositions, including the conditional gain law under which a batch-normalized block’s measured curvature moves with width at fixed function; the residual-excitation theorem and the phase-restricted attribution bound with its width corollary; the solvable serial cross-entropy instance with its isotropic extension, its displacement and margin-gap bounds and its normalized-readout control; and a scoped nonlinear existence result for the shallow biased-ReLU model under squared loss. It is reproduced verbatim from a self-contained mathematical note, so its internal labels are namespaced and its normalization is restated in its first paragraph; in particular $K_b = J_b J_b^\top$ there is an output-space kernel and is *not* the parameter-space block G_{bb} , and the two must not be substituted for one another. Its section headings are typeset as subsections of this appendix.

B.1 MATHEMATICAL SCOPE AND NORMALIZATION

We use the parameter blocks θ, ϕ of the main text and $z = N^{-1/2} \text{stack}(\hat{y}_n)$, $r = \nabla_z \ell(z)$, $J_b = \partial z / \partial b$, $\mathcal{L} = \ell \circ z$. Write

$$K_b = J_b J_b^\top, \quad G_{bb} = J_b^\top H_\tau J_b, \quad \nabla_b \mathcal{L} = J_b^\top r.$$

Thus K_b is an output-space logit kernel; it is not the parameter-space GGN block G_{bb} . For averaged softmax CE, $r = N^{-1/2} \text{stack}(p_n - y_n)$ and H_τ is the block-diagonal softmax Hessian. For normalized squared loss, $r = z - y$ and $H_\tau = I$. All parameter norms are Euclidean (Frobenius for matrix parameters); matrix-map bounds are operator norms unless explicitly marked otherwise. Here $C = N_{\text{cls}}$ denotes output dimension; envelope amplitudes are written C_r to avoid a collision. A normalized convention removes extraneous powers of N , but does not make data-dependent constants such as kernel minimum eigenvalues independent of N .

Theorem 1 and Lemma A.1 are geometric statements. Theorem 2 additionally requires a trajectory Jacobian cap and a residual envelope. Neither a large $\lambda_{\max}(G_{\phi\phi})$ nor parameter count alone supplies residual-subspace coercivity or a comparison with a separately trained frozen model. For unit gradient flow, $\dot{z} = -(K_\theta + K_\phi)r$; under CE, $\dot{r} = -H_\tau(K_\theta + K_\phi)r$, not the squared-loss constant-kernel residual equation.

B.2 DISCRETE DISPLACEMENT, CLIPPING, AND OPTIMIZER CONVENTIONS

The following seven statements collect the bounds numbered (4.1)–(4.7) in the review notes and distinguish the two clipping algorithms. They are pathwise: no convergence, stochastic independence, or smoothness assumption is needed beyond the stated gradient bound at every iterate. There is no encoder weight decay unless explicitly included. A uniform bound $\|g_k\| \leq B_k R_k \leq B R_k$ derived from the main theorem requires its containment/Jacobian hypothesis at *each discrete iterate*. A full-data or smoothed residual envelope does not establish a sampled-gradient envelope. An observed residual floor limits a fitted envelope; a positive floor is not a universal property of mini-batch training.

Lemma B.1 (Pathwise GD and clipping before a GD update). *Let $K \geq 0$, $\eta_k \geq 0$, $a_k \in [0, 1]$, and $\|g_k\| \leq B_k R_k$, where $B_k, R_k \geq 0$. If $\theta_{k+1} = \theta_k - \eta_k a_k g_k$, then*

$$\|\theta_K - \theta_0\| \leq \sum_{k=0}^{K-1} \eta_k a_k B_k R_k \leq \sum_{k=0}^{K-1} \eta_k B_k R_k. \quad (6)$$

The clipping scalar may depend on all parameter groups and the history.

Proof. Telescope the updates and apply the triangle inequality. $\square$

Corollary B.1 (Polynomial envelope in step units). *In Lemma B.1, suppose $\eta_k = \eta \geq 0$, $B_k \leq B$, and $R_k \leq C_r/(1 + \mu_s k)$, with $C_r, \mu_s > 0$. For $K \geq 1$,*

$$\|\theta_K - \theta_0\| \leq \eta B C_r \left[1 + \frac{\log(1 + \mu_s(K-1))}{\mu_s} \right]. \quad (7)$$

For $0 < \delta < C_r$, suppose the envelope holds through $K_ = \lceil (C_r/\delta - 1)/\mu_s \rceil$. Then the first index $K_\delta = \min\{k : R_k \leq \delta\}$ satisfies $K_\delta \leq K_*$ and*

$$\|\theta_{K_\delta} - \theta_0\| \leq \eta B C_r [1 + \mu_s^{-1} \log(C_r/\delta)]. \quad (8)$$

The displacement is zero if $K_\delta = 0$.

Proof. For decreasing $f(x) = (1 + \mu_s x)^{-1}$, $\sum_{k=1}^{K-1} f(k) \leq \int_0^{K-1} f(x) dx$. The envelope gives $R_{K_*} \leq \delta$. If $K_\delta \geq 1$, $K_\delta - 1 \leq K_* - 1 < (C_r/\delta - 1)/\mu_s$; substitute in (7). $\square$

Remark B.1 (Flow-to-step dictionary). For constant positive η , $t_k = \eta k$ and $\mu_s = \eta \mu_f$. Consequently $\eta B C_r / \mu_s = B C_r / \mu_f$. Omitting the leading η in (7) is a units error. The extra $\eta B C_r$ is a left-sum allowance, not an additional mechanism.

Lemma B.2 (Variable-step envelope). *Under Lemma B.1, let $t_k = \sum_{i < k} \eta_i$, $B_k \leq B$, $R_k \leq C_r/(1 + \mu_f t_k)$ with $C_r, \mu_f > 0$. For $K \geq 1$ and $\eta_{\max} = \max_{k < K} \eta_k$,*

$$\|\theta_K - \theta_0\| \leq B C_r \left[\frac{\log(1 + \mu_f t_K)}{\mu_f} + \eta_{\max} \left(1 - \frac{1}{1 + \mu_f t_K} \right) \right]. \quad (9)$$

Proof. The excess of the left sum for $f(t) = (1 + \mu_f t)^{-1}$ over its integral on $[t_k, t_{k+1}]$ is at most $\eta_k(f(t_k) - f(t_{k+1}))$. Bound η_k by $\eta_{\max}$ and telescope. Zero step sizes cause no difficulty. $\square$

Lemma B.3 (Momentum, including gradient clipping before accumulation). *Let $0 \leq \beta < 1$, $\eta_k \geq 0$, $a_k \in [0, 1]$, and $\|g_k\| \leq B_k R_k$. For $v_{k+1} = \beta v_k + a_k g_k$ and $\theta_{k+1} = \theta_k - \eta_k v_{k+1}$,*

$$\theta_K - \theta_0 = -v_0 \sum_{k < K} \eta_k \beta^{k+1} - \sum_{i < K} a_i g_i \sum_{k=i}^{K-1} \eta_k \beta^{k-i}. \quad (10)$$

For constant η this implies

$$\|\theta_K - \theta_0\| \leq \frac{\eta \beta (1 - \beta^K)}{1 - \beta} \|v_0\| + \eta \sum_{i < K} \frac{1 - \beta^{K-i}}{1 - \beta} a_i B_i R_i. \quad (11)$$

For $v_0 = 0$, (7)–(8) therefore remain valid with an additional factor $1/(1 - \beta)$.

Proof. Induction gives $v_{k+1} = \beta^{k+1} v_0 + \sum_{i=0}^k \beta^{k-i} a_i g_i$. Substitute into the parameter update, interchange the finite sums, and sum the geometric series. For variable steps the inner coefficient is at most $\eta_{\max}/(1 - \beta)$. $\square$

Remark B.2. This is zero-dampening, non-Nesterov momentum. It includes global *gradient* clipping through its actual a_k ; the gradients and later iterates belong to the clipped run itself. It does not compare a clipped run's displacement with a separately evolved unclipped run. The factor $1/(1 - \beta)$ is approached by long sequences of aligned gradients.

Lemma B.4 (Velocity clipping: sharper norm induction). *Let $c_k \geq 0$, $0 \leq \beta < 1$, and let $\text{clip}_c(u) = u \min\{1, c/\|u\|\}$, with $\text{clip}_c(0) = 0$. Suppose*

$$v_{k+1} = \text{clip}_{c_k}(\beta v_k + g_k), \quad \theta_{k+1} = \theta_k - \eta_k v_{k+1}, \quad \eta_k \geq 0.$$

Define the comparison coefficient $a_k = \min\{1, c_k/\|g_k\|\}$ for $g_k \neq 0$ and $a_k = 1$ otherwise. Then

$$\|v_{k+1}\| \leq \beta^{k+1} \|v_0\| + \sum_{i=0}^k \beta^{k-i} a_i \|g_i\|.$$

Consequently (11) holds when $\|g_i\| \leq B_i R_i$. For variable steps the corresponding norm bound uses the same finite weights as (10). The vector identity (10) is not asserted.

Proof. Write $x = \beta\|v_k\|$, $y = \|g_k\|$. Then

$$\|v_{k+1}\| \leq \min(c_k, x + y) \leq x + \min(c_k, y) = \beta\|v_k\| + a_k\|g_k\|.$$

If $y \leq c_k$ use the second argument of the minimum; if $y > c_k$ use its first. Induct and sum the parameter-step norms. $\square$

Remark B.3 (Two valid coefficient conventions). The comparison a_k above differs from the actual multiplier $t_k = \min\{1, c_k/\|\beta v_k + g_k\|\}$, defined as one at a zero argument. The same norm bound is also valid with t_i in place of a_i , by the separate induction $\|v_{k+1}\| \leq \beta\|v_k\| + t_k\|g_k\|$. In fact the exact unrolling is

$$v_{k+1} = \beta^{k+1} \left(\prod_{j=0}^k t_j \right) v_0 + \sum_{i=0}^k \beta^{k-i} \left(\prod_{j=i}^k t_j \right) g_i.$$

Dropping subsequent multipliers gives the claimed t_i bound. The two coefficient conventions need not be ordered, owing to gradient cancellation. State which one is used. For radial implementations with $c/(\|u\| + \varepsilon_{\text{clip}})$ and clipping multiplier in $[0, 1]$, this actual-multiplier argument still applies; the ideal-clipping comparison a_i is not obtained by silently equating the two formulas.

Lemma B.5 (Positive-stabilizer Adam). *Suppose the first moment is initialized at zero, $0 \leq \beta_1 < 1$, $0 \leq \eta_k \leq \eta_{\max}$, and*

$$\hat{m}_k = \sum_{i=0}^k \frac{(1 - \beta_1)\beta_1^{k-i}}{1 - \beta_1^{k+1}} g_i, \quad \theta_{k+1} - \theta_k = -\eta_k D_k \hat{m}_k, \quad \|D_k\| \leq \zeta^{-1}, \quad \zeta > 0.$$

Then

$$\|\theta_K - \theta_0\| \leq \frac{\eta_{\max}}{\zeta(1 - \beta_1)} \sum_{i < K} \|g_i\|. \quad (12)$$

Standard Adam denominators $\sqrt{\hat{v}_k} + \zeta$ satisfy the preconditioner bound when $\hat{v}_k \geq 0$ coordinate-wise. If clipping occurs before moment accumulation, g_i here is the gradient after that clipping.

Proof. Bound the update by $\eta_{\max}\zeta^{-1}\|\hat{m}_k\|$ and interchange sums. For each i , the sum of its non-negative weights is at most $\sum_{k \geq i} \beta_1^{k-i} \leq (1 - \beta_1)^{-1}$, since $(1 - \beta_1)/(1 - \beta_1^{k+1}) \leq 1$. $\square$

Remark B.4 (Numerically weak optimizer constant). The factor $1/\zeta$ makes this certificate numerically vacuous at standard small stabilizers unless a much smaller preconditioner bound is independently certified. At $\zeta = 0$, the first scalar Adam step has magnitude η for every nonzero g_0 , so no constant uniform in residual scale bounds it by a multiple of $\eta\|g_0\|$.

Lemma B.6 (Decoupled encoder decay must be included). *If an update is $\theta_{k+1} - \theta_k = -u_k - \eta_k \lambda_k \theta_k$ with $\eta_k, \lambda_k \geq 0$, then $\|\theta_K - \theta_0\| \leq \sum_{k < K} \|u_k\| + \sum_{k < K} \eta_k \lambda_k \|\theta_k\|$. Even with $u_k = 0$, constant decay λ gives $\theta_K = \prod_{k < K} (1 - \eta_k \lambda) \theta_0$.*

Proof. Telescope for the bound and iterate the scalar multiplication for the identity. A nonzero decay displacement can occur at identically zero residual, so a residual-only bound cannot omit this term. $\square$

B.3 JOINT AND FROZEN TRAJECTORY COMPARISONS

Theorem B.1 (Monotone frozen flow; review Theorem 1.1). *On $[0, T]$ let the joint and frozen paths be absolutely continuous, with common spatial initialization and $\dot{\phi}_J = -\nabla_{\phi} \mathcal{L}(\theta_J, \phi_J)$, $\dot{\phi}_F = -\nabla_{\phi} \mathcal{L}(\theta_0, \phi_F)$ a.e. Let $d \geq 0$ be locally integrable with $\|\theta_J(t) - \theta_0\| \leq d(t)$. Assume the differentiable frozen objective has an α -monotone gradient on a region containing both spatial paths, $\alpha \geq 0$:*

$$\langle v - w, \nabla_{\phi} \mathcal{L}(\theta_0, v) - \nabla_{\phi} \mathcal{L}(\theta_0, w) \rangle \geq \alpha \|v - w\|^2.$$

Assume finite nonnegative constants $K_{\theta\phi}, L_{\theta}, L_{\phi}$ satisfy, on the parameter pairs used below,

$$\begin{aligned} \|\nabla_{\phi} \mathcal{L}(\theta_J, \phi_J) - \nabla_{\phi} \mathcal{L}(\theta_0, \phi_J)\| &\leq K_{\theta\phi} d(t), \\ \|z(\theta_J, \phi_J) - z(\theta_0, \phi_J)\| &\leq L_{\theta} d(t), \\ \|z(\theta_0, \phi_J) - z(\theta_0, \phi_F)\| &\leq L_{\phi} \|\phi_J - \phi_F\|. \end{aligned}$$

Then

$$\|\phi_J(t) - \phi_F(t)\| \leq K_{\theta\phi} \int_0^t e^{-\alpha(t-s)} d(s) ds, \quad (13)$$

$$\|z_J(t) - z_F(t)\| \leq L_\theta d(t) + L_\phi K_{\theta\phi} \int_0^t e^{-\alpha(t-s)} d(s) ds. \quad (14)$$

If d is nondecreasing, the last bound is $[L_\theta + L_\phi K_{\theta\phi} \psi_\alpha(t)]d(t)$, where $\psi_\alpha(t) = (1 - e^{-\alpha t})/\alpha$ for $\alpha > 0$ and $\psi_0(t) = t$.

Proof. For $\Delta = \phi_J - \phi_F$, split the derivative into a frozen-gradient difference and an encoder-induced forcing. Then $\frac{1}{2}(d/dt)\|\Delta\|^2 \leq -\alpha\|\Delta\|^2 + K_{\theta\phi}d\|\Delta\|$. The scalar norm comparison (or the regularization $\sqrt{\|\Delta\|^2 + \epsilon^2}$ followed by $\epsilon \downarrow 0$) proves (13). Add and subtract $z(\theta_0, \phi_J)$ for the output bound; monotonicity of d gives the final expression. $\square$

Remark B.5. Under Theorem 2, one may use $d(t) = BC_r \log(1 + \mu t)/\mu$ on its valid horizon. If the first fitting time lies in that horizon, replace this logarithm by $\log(C_r/\delta)$ only up to that fitting time. The constants $L_\theta, L_\phi, K_{\theta\phi}$ and α are additional path-dependent hypotheses; all must be controlled across widths before inferring a width rate. A PL inequality alone supplies none of this monotonicity.

Proposition B.1 (Counterexample 1.2: PL does not imply contraction). *For $a, b > 0$, $f(x, y) = \frac{1}{2}(x + ay^2)^2$ satisfies $\frac{1}{2}\|\nabla f\|^2 \geq f - \inf f$ globally, but nearby gradient-flow trajectories can expand by an arbitrarily large factor at time one.*

Proof. $\inf f = 0$ and $\|\nabla f\|^2 = (x + ay^2)^2(1 + 4a^2y^2) \geq 2f$. Along $(x(t), y(t)) = (-be^{-t}, 0)$, transverse linearized variation satisfies $\dot{\xi} = 2abe^{-t}\xi$, hence $\xi(t) = \xi(0) \exp(2ab(1 - e^{-t}))$. Vary a with the PL constant fixed. This uses the PL convention of Karimi et al. (2016). $\square$

Theorem B.2 (Affine squared-loss bridge; review Theorem 1.3). *Let $z = z_0 + A(\theta - \theta_0) + B(\phi - \phi_0)$ and $\mathcal{L} = \frac{1}{2}\|z - y\|^2$. Train the joint model and the model with $\theta = \theta_0$ by unit Euclidean gradient flow from the same initialization. Let $K_\theta = AA^\top$, $K_\phi = BB^\top$. Suppose a common invariant linear subspace $\mathcal{V}$ contains $e_0 = z_0 - y$ and, on $\mathcal{V}$, $K_\phi \succeq \kappa I$, $0 \preceq K_\theta \preceq aI$ with $\kappa > 0$, $a \geq 0$. For $t \geq 0$,*

$$\|z_J(t) - z_F(t)\| \leq ate^{-\kappa t}\|e_0\|, \quad \sup_{t \geq 0} \|z_J(t) - z_F(t)\| \leq \frac{a\|e_0\|}{e\kappa}, \quad (15)$$

$$\|\theta_J(t) - \theta_0\| \leq \frac{\sqrt{a}}{\kappa}(1 - e^{-\kappa t})\|e_0\|, \quad \int_0^\infty \|\nabla_\theta \mathcal{L}\|^2 dt \leq \frac{a\|e_0\|^2}{2\kappa}. \quad (16)$$

At every finite horizon with positive loss decrease the encoder share is at most $a/(a + \kappa)$, by Corollary B.2.

Proof. On $\mathcal{V}$, both semigroups have norm at most $e^{-\kappa t}$. Their difference obeys the noncommutative Duhamel identity

$$e_J(t) - e_F(t) = - \int_0^t e^{-K_\phi(t-s)} K_\theta e^{-(K_\phi + K_\theta)s} e_0 ds.$$

This gives $ate^{-\kappa t}\|e_0\|$, maximized at $t = 1/\kappa$. Since $\|A^\top v\| \leq \sqrt{a}\|v\|$ for $v \in \mathcal{V}$, integrate $\dot{\theta} = -A^\top e_J$ and its squared norm to get (16). Both residuals converge to zero. $\square$

Corollary B.2 (Sharp finite-horizon attribution; main Corollary 1). *Explicitly: $\mathcal{L} = \ell \circ z$ is differentiable, the parameter path is absolutely continuous, the residual is $r = \nabla_z \ell$ with $K_\theta = J_\theta J_\theta^\top$, $K_\phi = J_\phi J_\phi^\top$, both blocks follow unit Euclidean gradient flow, and the chain rule holds a.e. Suppose for constants $a \geq 0$, $\kappa > 0$ that the Rayleigh inequalities*

$$r^\top K_\theta r \leq a\|r\|^2, \quad r^\top K_\phi r \geq \kappa\|r\|^2$$

hold at almost every time along the realized trajectory on $[0, T]$, and $\mathcal{L}(0) - \mathcal{L}(T) > 0$. Then

$$\frac{\int_0^T \|\nabla_\theta \mathcal{L}\|^2 dt}{\mathcal{L}(0) - \mathcal{L}(T)} \leq \frac{a}{a + \kappa}. \quad (17)$$

No common invariant subspace is required: that hypothesis of Theorem B.2 is a device for guaranteeing the Rayleigh bounds along the path, and here they are assumed directly.

Proof. Put $q_b = r^\top K_b r = \|\nabla_b \mathcal{L}\|^2$. Pointwise $\kappa_{q\theta} \leq a q_\phi$. Integrate and use $\mathcal{L}(0) - \mathcal{L}(T) = \int_0^T (q_\theta + q_\phi) dt$. Scalar kernels $K_\theta = a$, $K_\phi = \kappa$ attain equality. $\square$

Remark B.6. The same result holds for the nonsmooth flow convention defined below whenever its a.e. energy identity and Rayleigh inequalities hold. The unit rate is not cosmetic: with per-block rates $\eta_\theta \neq \eta_\phi$, the numerator $\int \|\nabla_\theta \mathcal{L}\|^2 dt$ is no longer the encoder share. For positive constant rates, the actual share uses η_θ in the numerator and is bounded by $\eta_\theta a / (\eta_\theta a + \eta_\phi \kappa)$.

Proposition B.2 (Counterexample 1.4: top eigenvalues are insufficient). *Let $K_\phi = \text{diag}(M, 0)$, $K_\theta = \text{diag}(0, 1)$, $e_0 = (0, 1)^\top$, $M > 0$. In the affine squared-loss model, the joint/frozen logit gap is $1 - e^{-t}$ although $\lambda_{\max}(K_\phi) = M$.*

Proof. $e_F(t) = (0, 1)^\top$ and $e_J(t) = (0, e^{-t})^\top$. Each kernel is realized by its PSD square root as an affine Jacobian. The fast spatial direction contains none of the residual. $\square$

Proposition B.3 (Robustness of the affine comparison). *Under the kernel/subspace assumptions of Theorem B.2, suppose instead the residuals solve $\dot{e}_J = -(K_\phi + K_\theta)e_J + \xi_J$ and $\dot{e}_F = -K_\phi e_F + \xi_F$, with common e_0 and locally integrable $\xi_J, \xi_F \in \mathcal{V}$. Then add*

$$\int_0^t e^{-\kappa(t-s)} (\|\xi_J(s)\| + \|\xi_F(s)\|) ds$$

to the first bound in (15). If $\|\xi_J\| \leq \nu_J$ and $\|\xi_F\| \leq \nu_F$, the additional uniform term is at most $(\nu_J + \nu_F)/\kappa$.

Proof. Compare each perturbed solution with its own constant-kernel unperturbed solution by variation of constants, then use the triangle inequality. $\square$

Proposition B.4 (Logit margins imply a disagreement bound). *If normalized logits satisfy $\|z_J - z_F\| \leq h$ and at most a fraction b_γ of the N samples have frozen top-two unnormalized logit margin at most $\gamma > 0$, the disagreement fraction is at most $b_\gamma + 2h^2/\gamma^2$ (and at most one).*

Proof. On a remaining disagreement, for the frozen winner c_* and the joint winner c , the logit perturbation d_n satisfies $d_{n,c} - d_{n,c_*} > \gamma$. Thus $\|d_n\|^2 \geq (d_{n,c} - d_{n,c_*})^2/2 > \gamma^2/2$. Average and apply Markov. Macro-F1 needs further class-specific confusion and denominator control. $\square$

B.3.1 A FINITE-DATA INITIALIZATION CERTIFICATE

Corollary B.3 (Biased-ReLU initialization; review Corollary 1.5). *Fix $N < \infty$, $x_n \in \mathbb{R}^S$, and a deterministic encoder matrix Θ_0 for which $u_n = \Theta_0 x_n \in \mathbb{R}^K$ are pairwise distinct and $\|u_n\| \leq R$. Let*

$$h_i(u) = \text{ReLU}(w_i^\top u + b_i), \quad \hat{y} = Vh + b_2,$$

with independent $w_i \sim \mathcal{N}(0, s_w^2 I_K/K)$, $b_i \sim \mathcal{N}(0, s_b^2)$, $V_{ci} \sim \mathcal{N}(0, s_w^2/M)$, $b_{2,c} \sim \mathcal{N}(0, s_b^2)$, and $s_w, s_b > 0$, $1 \leq c \leq C$. Let A, B be the normalized encoder/spatial logit Jacobians at this initialization, including all hidden and readout parameters in the spatial block. Train the resulting exactly affine model with squared loss as in Theorem B.2. Define

$$\psi = (h_1(u_n))_{n=1}^N, \quad K_* = N^{-1} \mathbb{E} \psi \psi^\top, \quad \kappa_* = \lambda_{\min}(K_*), \quad s_{\max}^2 = s_w^2 R^2/K + s_b^2.$$

Then $\kappa_ > 0$. For $0 < \rho < 1$, put*

$$\begin{aligned} M_0 &= \left\lceil \frac{18s_{\max}^4}{\rho \kappa_*^2} \right\rceil, & A_\rho &= \frac{3C_G}{\rho}, & R_\rho &= \sqrt{3C_r/\rho}, \\ C_G &= C s_w^4 \text{tr}(\Sigma_X), & C_r &= 2C(s_w^2 s_{\max}^2/2 + s_b^2) + 2\|y\|^2, & \Sigma_X &= N^{-1} \sum_n x_n x_n^\top. \end{aligned}$$

For each $M \geq M_0$, with probability at least $1 - \rho$,

$$K_\phi \succeq \kappa_* M I / 2, \quad \|K_\theta\| \leq A_\rho, \quad \|e_0\| \leq R_\rho.$$

Consequently

$$\sup_t \|z_J - z_F\| \leq \frac{2A_\rho R_\rho}{e\kappa_* M}, \quad \sup_t \|\theta_J - \theta_0\| \leq \frac{2\sqrt{A_\rho R_\rho}}{\kappa_* M}.$$

The displayed sufficient threshold necessarily satisfies $M_0 \geq 72N^2/\rho$; it is an existence certificate, not a practical onset width or a lower bound on widths at which coercivity actually holds.

Proof. If $c^\top K_* c = 0$, the continuous function $\sum_n c_n \text{ReLU}(w^\top u_n + b)$ vanishes almost surely under a Gaussian with full support, hence everywhere. The hyperplanes $w^\top u_n + b = 0$ have pairwise nonproportional normals $(u_n, 1)$. At a point of the n th hyperplane on none of the others, crossing the hyperplane produces the gradient jump $c_n(u_n, 1)$; all other terms are affine locally. The jump must vanish, so each $c_n = 0$.

Let $H_{ni} = h_i(u_n)$ be the unnormalized feature matrix. The readout kernel is $(HH^\top/N) \otimes I_C$. Gaussian ReLU fourth moments and Cauchy–Schwarz give $\mathbb{E}\|\psi\|^4 \leq 3N^2 s_{\max}^4/2$. Independence of columns yields

$$\mathbb{E} \left\| \frac{HH^\top}{NM} - K_* \right\|_F^2 \leq \frac{3s_{\max}^4}{2M}, \quad \Pr \left(\left\| \frac{HH^\top}{NM} - K_* \right\| > \kappa_*/2 \right) \leq \frac{6s_{\max}^4}{M\kappa_*^2}.$$

For the encoder Jacobian, conditioning on the hidden weights kills the cross terms in the independent centered readouts. Bounding each gate by one gives $\mathbb{E}\|J_\theta\|_F^2 \leq C_G$. Also $\mathbb{E}\|e_0\|^2 \leq C_r$ by the activation second moment and $\|z_0 - y\|^2 \leq 2\|z_0\|^2 + 2\|y\|^2$. Markov and a three-event union bound prove the asserted event. Apply Theorem B.2 on the full output space. Finally $\text{tr}(K_*) \leq s_{\max}^2/2$ implies $\kappa_* \leq s_{\max}^2/(2N)$ and the threshold inequality. $\square$

Remark B.7. This proves a frozen-linearization instance only. Constants depend on the fixed dataset and encoder; nearly duplicate bottleneck inputs can make κ_* arbitrarily small. Exact duplicates require a compatible restricted-space formulation. The displayed constants are numerically vacuous for the reported large datasets and are not empirical width thresholds.

Proposition B.5 (Truncated-Chernoff alternative). *For the same Gaussian/ReLU feature law, let $X_i = \psi_i \psi_i^\top / N$. Choose $R_c > 0$ such that*

$$2N(R_c + 2s_{\max}^2)e^{-R_c/(2s_{\max}^2)} \leq \kappa_*/2.$$

Then

$$\Pr \left\{ \lambda_{\min} \left(\sum_{i=1}^M X_i \right) < M\kappa_*/4 \right\} \leq N \exp[-M\kappa_*/(16R_c)]. \quad (18)$$

In particular $M \geq (16R_c/\kappa_*) \log(N/\eta)$ suffices for failure probability $\eta \in (0, 1)$. If used in the previous corollary or the nonlinear theorem below, replace the head floor $\kappa_* M/2$ by $\kappa_* M/4$ and propagate this changed constant.

Proof. Let Z be the maximum of the N squared Gaussian preactivations. Then $\|X_i\| \leq Z$ and $\Pr(Z > u) \leq 2Ne^{-u/(2s_{\max}^2)}$. Integrating the tail gives $\mathbb{E}[\|X_i\|1_{\{\|X_i\| > R_c\}}] \leq 2N(R_c + 2s_{\max}^2)e^{-R_c/(2s_{\max}^2)}$. Thus $Y_i = X_i 1_{\{\|X_i\| \leq R_c\}}$ satisfies $0 \preceq Y_i \preceq R_c I$ and $\mathbb{E}Y_i \succeq \kappa_* I/2$. Since each X_i is positive semidefinite, $\sum_i X_i \succeq \sum_i Y_i 1_{\{\|X_i\| \leq R_c\}}$ and therefore $\lambda_{\min}(\sum_i X_i) \geq \lambda_{\min}(\sum_i Y_i)$; the lower-tail bound for the truncated sum applies verbatim to the untruncated sum. Apply matrix Chernoff (Tropp, 2012) with $\delta = 1/2$ and $\mu_{\min} \geq M\kappa_*/2$. $\square$

Remark B.8. The constant is Tropp’s $\delta = 1/2$ lower tail after the standard simplification $[e^{-\delta}(1 - \delta) - (1 - \delta)]\mu_{\min}/R_c \leq e^{-\delta^2\mu_{\min}/(2R_c)}$. Keeping the exact constant $-\log(e^{-1/2}\sqrt{2}) = 0.15343\dots$ replaces $1/16$ by approximately $1/13.04$; the difference is immaterial at the widths involved. The one-sided ReLU tail also permits N in place of $2N$ in the truncation estimate. This certificate removes a power of κ_*^{-1} at the cost of logarithmic truncation factors; it can still be numerically vacuous. Neither its threshold nor the Frobenius threshold is a necessity theorem.

B.4 INTERIOR CURVATURE WITNESSES AND NORMALIZATION

Lemma B.7 (Pointwise interior witness; review Lemma 3.1). *At N sites let $h_n \in \mathbb{R}^d$, $a_n = Wh_n + b \in \mathbb{R}^q$, and $\hat{y}_n = \Phi_n(a_n) \in \mathbb{R}^C$, $C \geq 2$. Perturbing a_n must not change logits at other sites in this lemma. Fix a parameter point with finite logits and existing downstream Jacobians $D_n = \partial\Phi_n/\partial a_n$. Let $\Pi = I - \mathbf{1}\mathbf{1}^\top/C$, $p_{\min,n} = \min_c p_{n,c}$. For a unit $a \in \mathbb{R}^q$ define*

$$S_{h,a} = N^{-1} \sum_n p_{\min,n} \|\Pi D_n a\|^2 h_n h_n^\top, \quad S_h^p = N^{-1} \sum_n p_{\min,n} h_n h_n^\top.$$

If W belongs to ϕ , then

$$\lambda_{\max}(G_{\phi\phi}) \geq \lambda_{\max}(G_{WW}) \geq \lambda_{\max}(S_{h,a}).$$

If additionally $\|\Pi D_n a\| \geq s$ for every n with $h_n \neq 0$, then the final lower bound is at least $s^2 \lambda_{\max}(S_h^p)$.

Proof. For each unit $u \in \mathbb{R}^d$, $\Delta W = au^\top$ has unit Frobenius norm. The CE Hessian satisfies $H_n \succeq p_{\min,n} \Pi$: indeed $v^\top H_n v = \min_c \sum_j p_{n,j} (v_j - c)^2 \geq p_{\min,n} \min_c \sum_j (v_j - c)^2$. Consequently the witness Rayleigh quotient is at least $N^{-1} \sum_n (h_n^\top u)^2 p_{\min,n} \|\Pi D_n a\|^2 = u^\top S_{h,a} u$. Maximize over u . Zero-padding the W perturbation proves the full-block inequality. The additional hypothesis gives $S_{h,a} \succeq s^2 S_h^p$. $\square$

Remark B.9 (Mixing, masks, and nonsmooth points). For a general interior perturbation, let $\mathcal{R}_h \text{vec}(\Delta W)$ be its full unnormalized preactivation stack, including all sites that can affect the loss. If D_ϕ is the full downstream Jacobian and P_{out} selects the supervised logits, the valid replacement is

$$\lambda_{\max}(G_{\phi\phi}) \geq \frac{1}{N_{\text{out}}} \|H_\tau^{1/2} P_{\text{out}} D_\phi \mathcal{R}_h \text{vec}(\Delta W)\|^2, \quad \|\Delta W\|_F = 1. \quad (19)$$

This follows by the chain rule and the Rayleigh principle and includes BN, convolutions, and ignored sites. A product lower bound requires a restricted downstream gain on this same witness, with matching site normalization; a smallest nonzero singular value does not protect a witness in a nullspace. At ReLU kinks one must specify a gate selection; for nondegenerate Gaussian preactivations the ordinary Jacobian exists a.s. Sites with zero feature and zero bias contribute zero to the incoming lower bound, but a Jacobian convention is still necessary.

Theorem B.3 (Fixed bottleneck, growing interior width; review Theorem 3.2). *Fix integers $q \geq 1$, $C \geq 2$, and constants $R, a_0, s_w, s_v > 0$, $s_b \geq 0$. For each M let deterministic features $h_n \in \mathbb{R}^M$ satisfy $\|h_n\|^2 \leq R^2 M$ and $\lambda_{\max}(S_h) \geq a_0 M$, where $S_h = N^{-1} \sum_n h_n h_n^\top$. Consider $\hat{y}_n = V \text{ReLU}(Wh_n + b)$ with independent $W_{ij} \sim \mathcal{N}(0, s_w^2/M)$, $b_i \sim \mathcal{N}(0, s_b^2)$, $V_{ci} \sim \mathcal{N}(0, s_v^2/q)$, independent of the features. At any identically zero preactivation fix a gate in $[0, 1]$ consistently in all Jacobian blocks; such sites have $h_n = 0$ and cannot contribute to the W witness below. Under averaged CE in these Euclidean coordinates there exist $c_0, p_0 > 0$ independent of M, N with*

$$\Pr\{\lambda_{\max}(G_{WW}) \geq c_0 a_0 M\} \geq p_0, \quad \mathbb{E} \lambda_{\max}(G_{\phi\phi}) \geq p_0 c_0 a_0 M$$

whenever W belongs to ϕ . One explicit choice is

$$s_{\max}^2 = s_w^2 R^2 + s_b^2, \quad L_0 = \sqrt{48 q s_{\max}^2}, \quad c_0 = e^{-4L_0}/(32C), \quad p_0 = p_V/4,$$

where $p_V = \Pr\{\|V\|_F \leq 2, \|\Pi V e_1\| \geq 1/2\} > 0$. These constants are existence-only and numerically vacuous; this is a positive-probability/expectation statement, not a probability approaching one as M grows.

Proof. Fix a top unit eigenvector u of S_h , write $a_n = (h_n^\top u)^2$, $A = N^{-1} \sum_n a_n \geq a_0 M$, and $\chi_n = 1_{\{(Wh_n+b)_1 > 0\}}$. At sites with $a_n > 0$ this preactivation is a nondegenerate centered Gaussian. For $Q = N^{-1} \sum_n a_n \chi_n$, $\mathbb{E}Q = A/2$ and $0 \leq Q \leq A$, hence $\Pr(Q \geq A/4) \geq 1/3$ by splitting the expectation. The exact moment bound $\mathbb{E}\|\text{ReLU}(Wh_n+b)\|^2 \leq q s_{\max}^2/2$ gives $\Pr(\|\text{ReLU}(Wh_n+b)\| > L_0) \leq 1/96$. Thus the weighted untame energy U has $\mathbb{E}U \leq A/96$ and $\Pr(U > A/8) \leq 1/12$. With probability at least $1/4$, active tame sites therefore carry energy at least $A/8$. No independence across sites was used.

Independently, the V event has positive probability: a small open ball around a matrix with first column a unit class contrast and other columns zero is contained in it. At tame sites on that event, $\|\hat{y}_n\|_\infty \leq 2L_0$ and $p_{\min,n} \geq e^{-4L_0}/C$. The unit witness $\Delta W = e_1 u^\top$ has logit perturbation $(h_n^\top u)\chi_n V e_1$, hence curvature at least $(e^{-4L_0}/C)(1/4)(A/8) = c_0 A$. Multiply the independent event probabilities. Nonnegativity gives the expectation bound. $\square$

Remark B.10 (Scope and a squared-loss counterpart). A fixed channel direction may be inactive on every site, making its bound zero on some draws. Adaptive directions can improve the witness, but are not required for validity. For random upstream features, condition on an event where both feature bounds hold; its probability multiplies p_0 . The theorem covers a pointwise ReLU bottleneck followed by a linear readout, not an arbitrary convolutional BN decoder. It supplies no residual-subspace coercivity for Theorem B.2. For scalar squared loss, independent readout entries $v_i \sim \mathcal{N}(0, s_v^2/q)$ give the simpler bound

$$\mathbb{E}\lambda_{\max}(G_{WW}) \geq \frac{s_v^2}{2q} \lambda_{\max}(S_h).$$

Indeed, the same witness has curvature $v_1^2 Q$, and independence gives $\mathbb{E}(v_1^2 Q) = (s_v^2/q)(A/2)$.

Proposition B.6 (Counterexamples 3.3 and 3.4). *A fixed ReLU bottleneck does not in general admit a positive interior floor with probability $1 - o_M(1)$. Moreover, train-mode BN can erase an entire width-linear incoming witness even with a positive stabilizer.*

Proof. For the first claim take all $h_n = h \neq 0$ with $\|h\|^2 = M$ and q independent centered nondegenerate Gaussian preactivations. With probability 2^{-q} all q gates are inactive at every site, and $J_W = 0$. This probability is independent of M ; centered Gaussian bias does not remove it.

For the second take $h_n = h$ throughout the whole normalization group, $\|h\|^2 = M$, and place train-mode BN immediately after Wh_n . For every positive stabilizer $\varepsilon > 0$, its output is the BN affine offset, independent of W . Thus every weight perturbation is annihilated and $G_{WW} = 0$, whereas $S_h^p = p_{\min} h h^\top$ has eigenvalue $p_{\min} M$ if the downstream logits have fixed finite nondegenerate probabilities. At $\varepsilon = 0$ this zero-variance forward pass is undefined; it is not that counterexample. Constant inputs with a 1×1 convolution or periodic boundaries give literal convolutional realizations. $\square$

Proposition B.7 (BN derivative: an upper bound, not a lower witness). *For one normalization group of size L , let $P = I - \mathbf{1}\mathbf{1}^\top/L$, $t = Pu$, $s^2 = \|t\|^2/L$, and $\tilde{\sigma} = \sqrt{s^2 + \varepsilon} > 0$, with $\varepsilon \geq 0$. With fixed affine scale γ and offset,*

$$D_{\text{BN}} = \frac{\gamma}{\tilde{\sigma}} \left(P - \frac{tt^\top}{L(s^2 + \varepsilon)} \right), \quad \|D_{\text{BN}} d\|^2 \leq \frac{\gamma^2}{s^2 + \varepsilon} \|Pd\|^2. \quad (20)$$

If $s^2 > 0$, decompose $Pd = d_\parallel + d_\perp$ parallel and orthogonal to t . Then exactly

$$\|D_{\text{BN}} d\|^2 = \frac{\gamma^2}{s^2 + \varepsilon} \left[\|d_\perp\|^2 + \left(\frac{\varepsilon}{s^2 + \varepsilon} \right)^2 \|d_\parallel\|^2 \right].$$

Proof. Differentiate $\gamma Pu / \sqrt{\|Pu\|^2/L + \varepsilon}$. The middle matrix annihilates constants, has eigenvalue one on centered vectors orthogonal to t , and eigenvalue $\varepsilon/(s^2 + \varepsilon)$ in the t direction. This proves both identities and the contraction upper bound. If $s^2 = 0$ and $\varepsilon > 0$, the derivative is simply $\gamma P / \sqrt{\varepsilon}$. $\square$

Proposition B.8 (Function-preserving BN scale law). *Suppose a linear or convolutional block W is used downstream only through its immediately following train-mode BN. Scale W and its bias by $c > 0$, keep the other parameters fixed, and co-scale the stabilizer $\varepsilon \mapsto c^2 \varepsilon$. On batches with defined forward passes, logits are unchanged and*

$$J_W(cW) = c^{-1} J_W(W), \quad G_{WW}(cW) = c^{-2} G_{WW}(W), \quad (21)$$

where the left side uses the co-scaled bias and stabilizer. For $\varepsilon = 0$ the same identities hold without a stabilizer change provided every channel has positive variance.

Proof. Centering is linear and variance scales by c^2 , so $\text{BN}_{c^2\varepsilon}(cu) = \text{BN}_\varepsilon(u)$. This identity holds as a function of W in a neighborhood where defined; differentiate it and use unchanged output probabilities in the GGN. The incoming bias Jacobian also scales by c^{-1} . Jacobians with respect to parameters that are not co-scaled remain unchanged, by differentiating the function identity with respect to those parameters. $\square$

Remark B.11 (Known parameterization sensitivity). The effective-learning-rate effect of normalization is established (van Laarhoven, 2017). The scalar $\|W\|_F^2 \lambda_{\max}(G_{WW})$ is invariant under this uniform rescaling, not under arbitrary channelwise rescaling. For positive diagonal D , channelwise scaling W to DW with channelwise stabilizers co-scaled is also function preserving, but the GGN transforms by a nonuniform inverse congruence; the scalar generally changes. The audited four-channel example with scales $(0.2, 1, 3, 8)$ changed it by a factor 235 at identical logits. This is an example, not a universal factor. At fixed positive ε , report each s_i^2/ε and the scale-control departure: neither a universal accuracy threshold such as $s_i^2 \geq 10^3\varepsilon$ nor a final-logit error bound follows without bounds on c and on downstream sensitivity. Empirical tolerances from a particular block are diagnostics only.

Proposition B.9 (A constant centered Gram can coexist with growing curvature). *For $M \geq 2$, take three feature rows $(1, 0)$, $(0, 1)$, $(-1, -1)$ padded with zeros to dimension M , and $w_M = e_1/\sqrt{M}$. Put a single train-mode BN with $\gamma = 1$, $\varepsilon = 0$ after $w_M^\top h$, followed by binary logits $(\text{BN}(w_M^\top h), 0)$. Then the centered incoming Gram has top eigenvalue one, while $\lambda_{\max}(G_{ww}) = Mc_*$ for a constant $c_* > 0$.*

Proof. The nonzero Gram block is $\begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{pmatrix}$. At $w = e_1$ the centered preactivation is $(1, 0, -1)$, with positive variance, and the perturbation e_2 induces $(0, 1, -1)$ with a nonzero component transverse to it. The BN derivative and finite-logit CE curvature give a positive Rayleigh quotient. Apply Proposition B.8 with $c = M^{-1/2}$. This growth is precisely that function-preserving rescaling: the normalization $\|w_M\| = M^{-1/2}$ is the whole mechanism. It refutes an *unqualified* necessity claim for a width-linear centered Gram, not a claim that assumes width-uniform downstream gain. It is not a normalization-free mechanism. $\square$

Proposition B.10 (Conditional normalization-gain width law). *Let input dimension be d (for a fixed-size convolution, d is a fixed multiple of $M + K$). On a fixed BN group let the centered feature covariance be S_c , and initialize a row w independently with $\mathbb{E}w = 0$, $\mathbb{E}ww^\top = \sigma_w^2 I_d/d$. Its pre-BN variance satisfies*

$$\mathbb{E}_w s^2 = \sigma_w^2 \text{tr}(S_c)/d. \quad (22)$$

For a realized row assume $c_-/d \leq s^2 \leq c_+/d$, with $0 < c_- \leq c_+ < \infty$, and $0 < \gamma_- \leq |\gamma| \leq \gamma_+$. Then

$$\frac{\gamma_-^2 d}{c_+ + \varepsilon d} \leq \frac{\gamma^2}{s^2 + \varepsilon} \leq \frac{\gamma_+^2 d}{c_- + \varepsilon d}. \quad (23)$$

For a given unit parameter witness whose pre-BN perturbation d' has $\|d'_\perp\|^2 \geq \tau \|d'\|^2$ with $\tau > 0$, its post-BN energy is at least $\tau \gamma_-^2 d \|d'\|^2 / (c_+ + \varepsilon d)$. A final GGN lower bound further requires a positive lower gain of the actual downstream softmax-weighted, masked map on that post-BN perturbation.

Proof. $s^2 = w^\top S_c w$, so taking a trace proves (22). Invert the two realized variance bounds to obtain (23). The transverse energy identity in Proposition B.7 proves the witness assertion; then apply (19) with the stated downstream gain. $\square$

Remark B.12 (Diagnostic model and its premises). If $s^2 = c_0/(M + K)$, its squared BN gain is exactly

$$\gamma^2(M + K)/(c_0 + \varepsilon(M + K)).$$

The premise is fixed positive *total centered* incoming energy as channels are added, together with control of the realized row variance. Standard fan-in initialization with per-channel centered energy fixed instead gives width-independent *expected* pre-BN variance. A fixed total uncentered second moment is insufficient (constant features have zero centered variance). An expectation of variance cannot be inverted into a concentration statement about gain. Even with fixed pre-BN witness energy, a radial witness is annihilated at $\varepsilon = 0$ and need not inherit the gain. These qualifications make this

a conditional diagnostic identity, not a new general BN theorem or an unconditional replacement for Lemma A.1.

B.5 RESIDUAL EXCITATION AND PHASE-RESTRICTED ATTRIBUTION

Lemma B.8 (Directional gradient gain is not directional GGN curvature). *Let R_u isometrically embed a directional parameter subspace and $J_u = J_\theta R_u$. Then*

$$\|J_u^\top r\|^2 \leq \lambda_{\max}(J_u^\top J_u) \|r\|^2.$$

If $H \succeq 0$ and $r \in \text{range}(H)$, also

$$\|J_u^\top r\|^2 \leq \lambda_{\max}(J_u^\top H J_u) r^\top H^\dagger r. \quad (24)$$

For finite-logit softmax CE the range condition holds. A bound $H \succeq h_0 \Pi_{\text{cls}}$ on the class-contrast space, $h_0 > 0$, permits $r^\top H^\dagger r \leq \|r\|^2 / h_0$.

Proof. The first claim is the operator-norm inequality. For the second, write

$$J_u^\top r = (H^{1/2} J_u)^\top H^{\dagger/2} r.$$

Apply the operator-norm inequality again. The last assertion follows by diagonalizing H on its range. Here the Hessian floor is understood for the softmax Hessian, whose range is exactly the class-contrast space at finite logits. $\square$

Remark B.13 (CE counterexample and saturation). For one example with logits $(\theta, 0)$, $\theta = 0$, and label one, the residual and Jacobian are

$$r = (-1/2, 1/2), \quad J = (1, 0)^\top.$$

Then $|\nabla_\theta \mathcal{L}|^2 = 1/4$, while $G_{\theta\theta} \|r\|^2 = (1/4)(1/2) = 1/8$. Thus replacing the logit Gram by the GGN without the factor in (24) is false. At general first-class probability p the ratio is $1/(2p(1-p))$, unbounded near zero or one. A uniform positive h_0 is a further assumption, not supplied by saturation. For a linear encoder and unit input direction u , $R_u b = \text{vec}(bu^\top)$, so $J_u^\top r = (\nabla_\theta \mathcal{L})u$.

Theorem B.4 (Cumulative energy and excitation; review Theorem 2.1). *Let A_u, A_v be constant directional logit Jacobians along a path, $r(t) = \nabla_z \ell(z(t))$ square integrable on $[0, T]$, and $C_u = A_u A_u^\top$, $\lambda_u = \|C_u\|$, with analogous v notation. Define $Q_T = \int_0^T r(t) r(t)^\top dt$ and $E_u = \int_0^T \|A_u^\top r(t)\|^2 dt$. Then*

$$E_u = \text{tr}(C_u Q_T) \leq \lambda_u \text{tr}(Q_T).$$

Suppose $\lambda_v > 0$, $\lambda_u > 0$, $\text{tr}(Q_T) > 0$, and a top unit eigenvector q_v of C_v satisfies $q_v^\top Q_T q_v \geq \rho \text{tr}(Q_T)$ with $\rho > 0$. Then $E_v \geq \rho(\lambda_v/\lambda_u) E_u$; division by E_u requires $E_u > 0$. More generally, if $C_v \succeq c \lambda_v P$ and $\text{tr}(P Q_T) \geq \rho \text{tr}(Q_T)$ for a projector P , replace ρ by $c\rho$.

Proof. Integrate $\|A_u^\top r\|^2 = r^\top C_u r$ and use $C_u \preceq \lambda_u I$. Since $C_v \succeq \lambda_v q_v q_v^\top$, $E_v \geq \lambda_v q_v^\top Q_T q_v \geq \rho \lambda_v \text{tr}(Q_T)$. The projector version is identical. $\square$

Remark B.14. This theorem is loss-agnostic under its constant-Jacobian and integrability hypotheses; it includes affine CE. An envelope $\|r(t)\| \leq C_r/(1 + \mu t)$ gives $E_u \leq \lambda_u C_r^2/\mu$. Here λ_u is a logit-kernel eigenvalue. The exponential residual formulas in the next counterexamples specifically use squared loss. Excitation is an output-space condition on the visited residual; an input principal component alone does not establish it.

Proposition B.11 (Counterexamples 2.2 and 2.3 and the decoupled ratio). *For affine squared loss take $C_v = L e_1 e_1^\top$, $C_u = \ell e_2 e_2^\top$, $L, \ell > 0$, with total kernel $\text{diag}(L, \ell)$. If $r(0) = e_2$, then $E_v(T) = 0 < E_u(T)$ for $T > 0$. If $r(0) = e_1 + e_2$, then*

$$E_v(T) = (1 - e^{-2LT})/2, \quad E_u(T) = (1 - e^{-2\ell T})/2,$$

so their ratio tends to one even for arbitrarily large L/ℓ . More generally assume e_1, e_2 are orthonormal eigenvectors of the total constant kernel with positive eigenvalues γ_v, γ_u , $C_v = \lambda_v e_1 e_1^\top$, $C_u = \lambda_u e_2 e_2^\top$, and the initial residual components are a_v, a_u with $a_u \neq 0$, $\lambda_u > 0$, $T > 0$. Then

$$\frac{E_v(T)}{E_u(T)} = \frac{\lambda_v a_v^2 \gamma_u}{\lambda_u a_u^2 \gamma_v} \frac{1 - e^{-2\gamma_v T}}{1 - e^{-2\gamma_u T}}. \quad (25)$$

Proof. Each residual component is $a_j e^{-\gamma_j t}$; integrate $\lambda_j a_j^2 e^{-2\gamma_j t}$. The first two constructions are realized by the affine map $(\sqrt{\ell}\theta_1, \sqrt{\ell}\theta_2)$. Without the decoupling hypothesis, these scalar exponentials do not represent mixed residual components. $\square$

Proposition B.12 (Attribution during a residual-excited phase). *Assume the differentiability, absolute continuity, unit-flow and energy-identity conventions of Corollary B.2, but not its global Rayleigh bounds. Let $q_b(t) = r^\top K_b r$, $D(t) = q_\theta(t) + q_\phi(t)$, and $\Delta L = \int_0^T D(t)dt = \mathcal{L}(0) - \mathcal{L}(T) > 0$. For a measurable phase $E \subset [0, T]$, put*

$$\tau_E = |E|, \quad w_E = \frac{\int_E D(t)dt}{\Delta L}.$$

On E suppose a measurable top- k projector $P_k(t)$ of $K_\phi(t)$ (with a fixed measurable tie convention) satisfies

$$\|P_k(t)r(t)\|^2 \geq \alpha \|r(t)\|^2, \quad \lambda_k(K_\phi(t)) \geq \ell_* > 0, \quad q_\theta(t) \leq a \|r(t)\|^2,$$

where $0 < \alpha \leq 1$, $a \geq 0$. Then, when the phase denominator is positive,

$$\frac{\int_E q_\theta}{\int_E D} \leq \frac{a}{a + \alpha \ell_*}, \quad (26)$$

$$\frac{\int_0^T q_\theta}{\Delta L} \leq w_E \frac{a}{a + \alpha \ell_*} + (1 - w_E). \quad (27)$$

A verified complementary share bound b replaces the last term by $(1 - w_E)b$. Time-varying pointwise constants give the sharper $D(t)dt/\Delta L$ -weighted average of their share caps.

Proof. PSD spectral decomposition gives $q_\phi \geq \lambda_k \|P_k r\|^2 \geq \alpha \ell_* \|r\|^2$. Thus $q_\theta \leq [a/(a + \alpha \ell_*)]D$ on E . Integrate on E and its complement, where $q_\theta \leq D$. The phase duration is explicit but imposes no value on w_E . For $E = [0, \tau]$, $\int_E D = \mathcal{L}(0) - \mathcal{L}(\tau)$; under squared loss additionally $\|r(t)\| \leq e^{-\alpha \ell_* t} \|r(0)\|$ through this interval by differentiating its squared norm. $\square$

Corollary B.4 (What a width-dependent cumulative share additionally requires). *For a family satisfying Proposition B.12, suppose its early-phase constants obey $a \leq a_*$ and $\alpha \ell_* \geq cM$, with fixed $a_* \geq 0, c > 0$. On the complement assume $0 < b_- \leq q_\theta/D \leq b_+ \leq 1$ at almost every time with $D > 0$, for constants independent of M . Writing the cumulative share as S_M , one has*

$$b_-(1 - w_E) \leq S_M \leq w_E \frac{a_*}{a_* + cM} + b_+(1 - w_E).$$

Consequently $S_M = O(M^{-1})$ holds if and only if $1 - w_E = O(M^{-1})$. The same statement holds for nonnegative weighted sums on one discrete sequence, interpreting w_E and S_M as gradient-energy fractions.

Proof. The upper bound is Proposition B.12. For the lower bound retain only the complementary numerator and use its share floor. Both implications follow since $b_- > 0$ and $a_*/(a_* + cM) = O(M^{-1})$. This characterizes what the claimed cumulative scaling needs under the stated late-phase floor; early excitation or a short phase alone is insufficient. $\square$

Remark B.15 (Discrete diagnostic and admissible reference space). Replacing integrals by non-negative weighted sums of gradient squares proves the same *gradient-energy* inequalities on any one sampled sequence, regardless of optimizer. This is not Adam loss-decrease attribution. Probe-defined phases and training-minibatch energies are different sequences unless explicitly matched. No phase-share bound alone replaces the stability assumptions needed to compare two trajectories. For CE let $\Pi_{\text{cls}} = I_N \otimes (I_C - \mathbf{1}\mathbf{1}^\top/C)$. The residual belongs to its rank- $N(C-1)$ range. For a uniformly random unit vector in that range, expected energy in a projector P is $\text{tr}(P\Pi_{\text{cls}})/[N(C-1)]$, since its second moment is $\Pi_{\text{cls}}/[N(C-1)]$. The full-space baseline $\text{rank}(P)/(NC)$ is generally inappropriate.

B.6 A SOLVABLE SERIAL CROSS-ENTROPY INSTANCE

Proposition B.13 (Information retained by a fitting serial encoder). *If the predictions depend on X only through $Z = f_\theta(X)$ and a classifier $q_\phi(Y | Z)$ has population CE at most $\eta < \infty$, then $H(Y | Z) \leq \eta$ and $I(Y; Z) \geq H(Y) - \eta$. The same statement holds for the empirical distribution with a fixed model.*

Proof. CE is conditional entropy plus the nonnegative expected conditional KL divergence. Suppression of a selected coordinate or failure under a specified shift is therefore different from absence of all training-label information in the serial representation. $\square$

Theorem B.5 (Serial CE instance; review Theorem 5.1). *Let Y be uniform on $\{-1, 1\}$ and let training inputs be $x_1 = (S, 0)$, $x_2 = (0, C_{\text{ctx}})$ with $S = C_{\text{ctx}} = Y$. The shared linear encoder is $W = (a, v)$, giving $z_1 = aS$, $z_2 = vC_{\text{ctx}}$. For integer $M \geq 1$ define*

$$F = z_1 + \sum_{j=1}^M \beta_j z_2 = aS + bvC_{\text{ctx}}, \quad b = \sum_j \beta_j, \quad \theta = (a, v), \quad \phi = \beta.$$

All coordinates follow unit Euclidean gradient flow on the averaged binary CE $\mathcal{L} = \mathbb{E} \log(1 + e^{-YF}) = \log(1 + e^{-q})$, $q = a + bv$, from $a_0 = 0, v_0 = 1, \beta_0 = 0$, with no clipping, decay or momentum. Fix $0 < \delta < 1/2$, $m = \log((1 - \delta)/\delta)$, and let T_m be the first time q reaches m . Write $\varrho = (1 + e^q)^{-1}$ and $\Psi(q) = q + e^q - 1$. Then:

$$\dot{a} = \varrho, \quad \dot{v} = b\varrho, \quad \dot{b} = Mv\varrho, \quad (28)$$

$$v = \cosh(\sqrt{M}a), \quad b = \sqrt{M} \sinh(\sqrt{M}a), \quad (29)$$

$$q = a + \frac{\sqrt{M}}{2} \sinh(2\sqrt{M}a), \quad \dot{q} = (M + 1 + 2b^2)\varrho. \quad (30)$$

The solution is global and reaches m in finite time, with

$$\frac{\Psi(m)}{M + 1 + 2m^2} \leq T_m \leq \frac{\Psi(m)}{M + 1}. \quad (31)$$

At that time, writing $a_M = a(T_m)$,

$$0 < a_M \leq \frac{m}{M + 1}, \quad b(T_m)v(T_m) \geq \frac{Mm}{M + 1}, \quad Ma_M \longrightarrow m, \quad (32)$$

$$0 \leq v(T_m) - 1 \leq \frac{m^2}{2M}, \quad \|W(T_m) - W_0\| \leq \sqrt{\frac{m^2}{(M + 1)^2} + \frac{m^4}{4M^2}}. \quad (33)$$

For fixed m , a_M decreases strictly with M ; removing the contextual path entirely requires $a = m$ at the same fitting margin. Moreover,

$$\varrho(t) \leq \frac{1/2}{1 + (M + 1)t/4}, \quad t \geq 0. \quad (34)$$

In the stated parameter blocks and loss convention,

$$\lambda_{\max}(G_{\theta\theta}) = \varrho(1 - \varrho)(1 + b^2), \quad \lambda_{\max}(G_{\phi\phi}) = M\varrho(1 - \varrho)v^2,$$

so at initialization they equal $1/4, M/4$ and the disparity is M . For the separately trained frozen encoder $W = (0, 1)$, $\beta_F(0) = 0$, its margin obeys $\Psi(q_F(t)) = Mt$ and

$$0 \leq q(t) - q_F(t) \leq \frac{(1 + 2m^2)t}{2 + Mt/2} \leq \frac{2(1 + 2m^2)}{M}, \quad 0 \leq t \leq T_m. \quad (35)$$

Under the specified reversal $S = Y, C_{\text{ctx}} = -Y$, the joint model at T_m has error one; the fitted spectral-only comparator has error zero.

Proof. The training margin is the same on each example, so direct differentiation gives (28). Divide by $\dot{a} = \varrho > 0$ to get $dv/da = b$, $db/da = Mv$, which proves (29). It implies $b^2 = M(v^2 - 1)$ and (30). Since $0 < \dot{a} \leq 1/2$, a stays finite on every finite time interval; the phase formula then gives global existence. The strictly increasing map $Q_M(a) = a + (\sqrt{M}/2) \sinh(2\sqrt{M}a)$ maps $[0, \infty)$ onto itself and has a finite inverse at m . On this compact interval, $\dot{a} \geq \delta > 0$, so the hitting time is finite.

$\sinh x \geq x$ gives $Q_M(a) \geq (M+1)a$, strictly for $a > 0$. Before fitting, $v \geq 1$, $0 \leq bv = q - a \leq m$, hence $b \leq m$. The invariant gives $v - 1 = b^2/[M(v+1)] \leq m^2/(2M)$. For fixed positive a the contextual term increases strictly with M ; invert to get strict decrease of a_M . Its upper bound implies $\sqrt{M}a_M \rightarrow 0$, and the expansion $m = (M+1)a_M + O(M^2a_M^3)$ gives $Ma_M \rightarrow m$.

$\dot{\varrho} = -(M+1+2b^2)\varrho^2(1-\varrho)$ and $\varrho \leq 1/2$ imply $(d/dt)(1/\varrho) \geq (M+1)/2$. Integrate from $\varrho(0) = 1/2$ for the envelope. Since $\Psi'(q) = 1/\varrho$, before fitting $\frac{d}{dt}\Psi(q) = M+1+2b^2 \in [M+1, M+1+2m^2]$; integration proves (31). The margin gradients are $Y(1, b)$ and $Yv\mathbf{1}_M$; the scalar logistic Hessian is $\varrho(1-\varrho)$, giving the GGN.

For the frozen system $\dot{q}_F = M/(1+e^{q_F})$. Thus $0 \leq \Psi(q) - \Psi(q_F) = \int_0^t (1+2b^2)ds \leq (1+2m^2)t$. Convexity gives a lower bound $\Psi'(q_F)(q-q_F)$ for this difference. As $q_F \leq e^{q_F} - 1$, $Mt = \Psi(q_F) \leq 2(e^{q_F} - 1)$, so $\Psi'(q_F) \geq 2 + Mt/2$, proving (35). The shifted margin is $a - bv = 2a_M - m < 0$: for $M > 1$ use (32), and for $M = 1$ use the strict inequality $Q_1(a) > 2a$. The spectral-only shifted margin is m . $\square$

Corollary B.5 (Analytic instantiation of the displacement bound). *For Theorem B.5, embed the scalar prediction in symmetric logits $(F/2, -F/2)$. The packet residual norm is $R = \sqrt{2}\varrho$, and on $[0, T_m]$*

$$\|J_\theta\| = \sqrt{(1+b^2)/2} \leq B_m := \sqrt{(1+m^2)/2}.$$

With envelope amplitude $C_r = 1$ and $\mu = (M+1)/4$, Theorem 2 gives

$$\|W(T_m) - W_0\| \leq \frac{4B_m}{M+1} \log \frac{1}{\sqrt{2}\delta}. \quad (36)$$

Also $\|W(T_m) - W_0\| \leq \sqrt{1+m^2}a_M \leq \sqrt{1+m^2}m/(M+1)$.

Proof. The symmetric logit derivative has squared norm $(1+b^2)/2$, while the full residual norm is $\sqrt{2}\varrho$, so their product is the exact encoder gradient norm. The envelope has true initial amplitude $1/\sqrt{2}$; $C_r = 1$ is a valid larger amplitude matching the main text's $C_r \geq 1$ convention. Its target is $\sqrt{2}\delta < 1$. All constants apply on the fitting region $|b| \leq m$. Alternatively integrate $\|\dot{W}\| = \varrho\sqrt{1+b^2} \leq \sqrt{1+m^2}\dot{a}$. $\square$

Remark B.16 (Embedding, head, and initialization conventions). For logits $(F, 0)$, loss, residual norm and GGN agree, but the unprojected logit-Jacobian norm is $\sqrt{1+b^2}$; the symmetric embedding makes the norm-product bound exact. A freely trained dense two-row head is not the parameter block $\phi = \beta$: its initial head eigenvalue is $M/2$, not $M/4$, while the encoder eigenvalue remains $1/4$ at the same initial function. Labels need not be balanced for the training ODE; balance is used in the information/noisy-accuracy statement below. The initial encoder orientation in Theorem B.5 is explicitly asymmetric; the next corollary removes that choice for W , not the routing/readout asymmetry.

Proposition B.14 (Readout normalization and representation interpretation). *Replace the contextual readout by $\alpha_M \sum_j \gamma_j z_j$, $\gamma_j(0) = 0$, at unit rates. Its reduced equation for $b = \alpha_M \sum_j \gamma_j$ is $\dot{b} = M\alpha_M^2 v \varrho$. In particular $\alpha_M = M^{-1/2}$ gives exactly the $M = 1$ dynamics for every M . At every finite M in the original theorem, the noiseless spectral coordinate $a_M S$ still determines Y . If it is instead observed as $Z_S = a_M Y + \sigma \xi$, with $\xi \sim \mathcal{N}(0, 1)$ independent, $\sigma > 0$ fixed, then Bayes accuracy is $\Phi(a_M/\sigma)$ and $I(Y; Z_S) \leq a_M^2/(2\sigma^2)$.*

Proof. Differentiate each readout and sum. The Gaussian likelihood-ratio test is thresholding at zero, giving the accuracy. For $Q = \mathcal{N}(0, \sigma^2)$, $\mathbb{E}_Y \text{KL}(P_{Z_S|Y} \| Q) = a_M^2/(2\sigma^2) = I(Y; Z_S) + \text{KL}(P_{Z_S} \| Q)$. This noise is a specified observation perturbation, not training noise. $\square$

Corollary B.6 (General and isotropic encoder initialization). *In the serial model let $m > 0$, $a_0 < m$, $v_0 \neq 0$, and allow any readout initialization with $\sum_j \beta_{j0} = 0$. Set $u = a - a_0$, $\Delta = m - a_0 > 0$. Then*

$$v = v_0 \cosh(\sqrt{M}u), \quad b = \sqrt{M}v_0 \sinh(\sqrt{M}u), \quad (37)$$

$$q = a_0 + u + \frac{\sqrt{M}v_0^2}{2} \sinh(2\sqrt{M}u),$$

$$0 < u_M \leq \frac{\Delta}{1 + Mv_0^2}, \quad Mu_M \rightarrow \frac{\Delta}{v_0^2}, \quad \frac{a(T_m) - a_0}{m - a_0} \rightarrow 0. \quad (38)$$

The fitting root decreases strictly with M and

$$|v(T_m) - v_0| \leq \frac{\Delta^2}{2M|v_0|^3}, \quad \|W(T_m) - W_0\| \leq \sqrt{\frac{\Delta^2}{(1 + Mv_0^2)^2} + \frac{\Delta^4}{4M^2|v_0|^6}}. \quad (39)$$

With $\Psi_{a_0}(q) = q - a_0 + e^q - e^{a_0}$,

$$\frac{\Psi_{a_0}(m)}{1 + Mv_0^2 + 2\Delta^2/v_0^2} \leq T_m \leq \frac{\Psi_{a_0}(m)}{1 + Mv_0^2}.$$

For a frozen encoder at (a_0, v_0) and the same initial readouts, $\Psi_{a_0}(q_F) = Mv_0^2 t$. If $p_0 = e^{a_0}/(1 + e^{a_0})$ and $C_0 = 1 + 2\Delta^2/v_0^2$, then

$$0 \leq q_J - q_F \leq \frac{C_0 t}{1 + e^{a_0} + p_0 Mv_0^2 t} \leq \frac{C_0}{p_0 Mv_0^2}, \quad 0 \leq t \leq T_m. \quad (40)$$

Under contextual reversal the fitted margin is $2a(T_m) - m$. For $a_0 < m/2$, error is one for all sufficiently large M , with sufficient condition $Mv_0^2 > m/(m - 2a_0)$. For $m/2 \leq a_0 < m$, the reversed classifier is correct at every width. If $a_0 \geq m$, define fitting to stop immediately, in which case it is also correct.

If $(a_0, v_0) \sim \mathcal{N}(0, \sigma^2 I_2)$, $\sigma > 0$, then $v_0 \neq 0$ almost surely; use the preceding stopping convention. The expected reversal error satisfies

$$\lim_{M \rightarrow \infty} \mathbb{E}_{W_0} \text{Err}_{\text{reversal}} = \Phi\left(\frac{m}{2\sigma}\right). \quad (41)$$

The same-threshold spectral-only comparator has zero reversal error.

Proof. Dividing the ODE by $\dot{a} > 0$ gives (37); only the initial readout sum enters. Each $\beta_i - \beta_j$ is conserved, so equal initial readouts are not needed for closure. Positivity of the contextual margin $bv = (\sqrt{M}v_0^2/2) \sinh(2\sqrt{M}u)$ and the sinh inequality prove (38); Taylor expansion at $\sqrt{M}u_M \rightarrow 0$ gives the limit. The invariant is $b^2 = M(v^2 - v_0^2)$; v keeps its sign and $|v| \geq |v_0|$. Since $bv \leq \Delta$, $|b| \leq \Delta/|v_0|$ and

$$|v - v_0| = \frac{b^2}{M(|v| + |v_0|)} \leq \frac{\Delta^2}{2M|v_0|^3}.$$

This finite- M inequality proves (39). Now $\dot{q} = (1 + Mv_0^2 + 2b^2)\varrho$ and $b^2 \leq \Delta^2/v_0^2$ on the fitting interval; integrate the transformed margin for the time bounds. Global existence follows as before from $\dot{u} \leq 1$ and the phase formula; every finite m is hit.

The transformed joint/frozen difference lies between zero and $C_0 t$. Also $q_F - a_0 \leq (e^{q_F} - e^{a_0})/e^{a_0}$ gives $\Psi'_{a_0}(q_F) \geq 1 + e^{a_0} + p_0 Mv_0^2 t$. Convexity proves (40). For $a_0 < m/2$, use $a(T_m) \leq a_0 + \Delta/(1 + Mv_0^2)$ to get the finite- M reversal condition. For $a_0 > m/2$ the classifier is correct at every fitted width, since $u_M > 0$ gives $2a(T_m) - m > 2a_0 - m > 0$; equality $a_0 = m/2$ also gives a positive margin. Immediate stopping covers $a_0 \geq m$. Almost surely the limiting error is $1_{\{a_0 < m/2\}}$, apart from the zero-probability boundary and $v_0 = 0$ events. Dominated convergence gives (41). $\square$

Remark B.17 (Confidence dependence and limiting quantity). For $a_0 \neq 0$, $a(T_m)/a_0 \rightarrow 1$ but $a(T_m)/m \rightarrow a_0/m$; the vanishing quantity is the *update* fraction in (38). For $0 < \rho < 1$, a Gaussian tail and its maximum density give, with probability at least $1 - \rho$,

$$|a_0| \leq \sigma \sqrt{2 \log(4/\rho)}, \quad |v_0| \geq \frac{\rho \sigma}{2} \sqrt{\pi/2}.$$

Substitute these bounds in the finite- M inequalities for uniform confidence bounds. The dependence is severe: for fixed $(\Delta, v_0) \in (0, \infty) \times (\mathbb{R} \setminus \{0\})$, expansion of the root yields

$$Mu_M = \frac{\Delta}{v_0^2} \left[1 - \frac{1}{Mv_0^2} - \frac{2\Delta^2}{3Mv_0^4} + O(M^{-2}) \right].$$

Indeed substitute $u = A/M + B/M^2$ in the sinh equation and match its constant and M^{-1} coefficients. The expansion's remainder is pointwise, not already uniform in ρ . Its small- $|v_0|$ crossover requires control of $\Delta^2/(Mv_0^4)$; at the confidence cutoff this has scale $\Delta^2\rho^{-4}\sigma^{-4}/M$, a fourth power of $1/\rho$, not a second. A rigorous uniform remainder can instead be bounded from the sinh series on this controlled argument. Do not present this asymptotic crossover as a universal necessary width.

Provenance and interpretation. Architecture- and initialization-dependent implicit bias is established in linear-network analyses (Yun et al., 2021; Moroshko et al., 2020); trajectory-level incremental learning is studied by Berthier (2023). Exact deep-linear dynamics (Saxe et al., 2014) and conservation of layer-norm differences (Du et al., 2018) supply the underlying tools. CE feature competition is central to Pezeshki et al. (2021). The serial example instantiates a known optimization-metric mechanism with finite-margin formulas, an encoder-update comparison and a specified reversal distribution. Replication is equivalent to a rate M on b , and normalization removes it; the function class does not grow. These particular calculations are not a claim of a new general implicit-bias mechanism, and finite-time or serial composition alone is not a novelty claim.

B.7 A SCOPED NONLINEAR EXISTENCE RESULT

Flow convention and normalization. Here the loss is squared loss, the encoder is linear, the decoder has one hidden ReLU layer, and every parameter uses unit Euclidean gradient flow. There is no BatchNorm, convolution, weight decay, clipping, preconditioning, or stochastic sampling. Write

$$f_n = V\sigma(W\Theta x_n + b) + \beta, \quad \mathcal{L} = \frac{1}{2N} \sum_{n=1}^N \|f_n - y_n\|^2, \quad \theta = \Theta, \quad \phi = (W, b, V, \beta).$$

Parameter norms are Euclidean, with Frobenius norms on matrices. Set $\bar{f} = N^{-1/2}(f_1, \dots, f_N)$ and $e = \bar{f} - \bar{y}$, so that $\mathcal{L} = \|e\|^2/2$. At a ReLU kink choose a measurable gate in $[0, 1]$, using the same gate in every parameter block for that neuron and datum. A solution is an absolutely continuous parameter path satisfying the resulting gradient flow almost everywhere. The theorem concerns every such solution; uniqueness is not asserted. Local existence and the energy identity needed below are proved in the proof. This formulation avoids identifying arbitrary backpropagation selections with the Clarke subdifferential.

Theorem B.6 (Theorem N: nonlinear, unnormalized-readout existence result). *Fix positive integers N, D, K, C , fixed data $x_n \in \mathbb{R}^D, y_n \in \mathbb{R}^C$, and a deterministic $\Theta_0 \in \mathbb{R}^{K \times D}$ such that the $\Theta_0 x_n$ are pairwise distinct. Fix $s_w, s_v, s_b > 0$ and initialize independently across neurons, entries, and blocks:*

$$w_i \sim \mathcal{N}(0, s_w^2 I_K/K), \quad b_i \sim \mathcal{N}(0, s_b^2), \quad V_{ci} \sim \mathcal{N}(0, s_v^2/M).$$

The output bias β_0 is fixed, or independent of the other blocks with an M -independent distribution of finite second moment. In particular β_0 need not be centered. Compare joint training from this initialization with training of all decoder parameters from the same initialization while fixing $\Theta = \Theta_0$. For every $\rho \in (0, 1)$ there are finite positive constants $\kappa, A, B, R_0, C_{\text{gap}}, M_$, depending on the fixed data, initialization scales and ρ , but not on M , with the following property. For each $M \geq M_*$, with probability at least $1 - \rho$, both flows have global solutions, and every pair of solutions obeys, for every $t \geq 0$,*

$$\|e^J(t)\|, \|e^F(t)\| \leq R_0 e^{-\kappa M t/2}, \tag{N1}$$

$$\sup_t \|\Theta^J(t) - \Theta_0\| \leq \frac{2AR_0}{\kappa M}, \quad \sup_t \|\phi^{J,F}(t) - \phi_0\| \leq \frac{2BR_0}{\kappa\sqrt{M}}, \tag{N2}$$

$$\sup_t \|\bar{f}^J(t) - \bar{f}^F(t)\| \leq \frac{C_{\text{gap}}R_0}{\kappa M}. \tag{N3}$$

For every finite $T > 0$ with $\mathcal{L}(0) > \mathcal{L}(T)$ on the joint path,

$$\frac{\int_0^T \|\nabla_{\theta} \mathcal{L}\|^2 dt}{\mathcal{L}(0) - \mathcal{L}(T)} \leq \frac{A^2}{A^2 + \kappa M/2}. \quad (\text{N4})$$

For $0 < \delta < R_0$, residual norm at most δ is reached by $2(\kappa M)^{-1} \log(R_0/\delta)$. If the initial residual is already at most δ , the first hitting time can be zero. The constants and threshold constructed below are existence-only and can be numerically vacuous. This is not a uniform theorem for learned encoders, growing datasets, general deep decoders, or production Adam.

Proof. We give explicit finite constants to make all quantifiers and probability events checkable.

Population kernel and initialization events. Let $X = \max_n \|x_n\|$, $z_n = \Theta_0 x_n$, and

$$H_{ni} = N^{-1/2} \sigma(w_i^T z_n + b_i), \quad K_* = \mathbb{E}[H_{:i} H_{:i}^T], \quad \kappa_* = \lambda_{\min}(K_*), \quad \kappa = \kappa_*/2.$$

Here K_* uses one random neuron with the displayed normalization, so that $\mathbb{E}[H H^T] = M K_*$. The distinct-input kink argument in the proof of Corollary B.3 gives $\kappa_* > 0$: if a linear combination of the ReLU features vanishes almost surely, it vanishes everywhere by continuity and full Gaussian support in (w, b) ; one can cross each distinct affine hyperplane away from the others, forcing its coefficient to vanish. Put

$$s_{\max}^2 = \frac{s_w^2}{K} \max_n \|z_n\|^2 + s_b^2, \quad Z_0 = \sqrt{1 + (\|\Theta_0\| X)^2}, \quad Z = Z_0 + X.$$

Operator norm may be used for $\|\Theta_0\|$ here; its Frobenius norm is also a valid upper bound. For $\|\Theta - \Theta_0\| \leq 1$,

$$\sqrt{1 + \|\Theta x_n\|^2} \leq Z. \quad (42)$$

Indeed, with $a = \|\Theta_0\| X$, $(\sqrt{1 + a^2} + X)^2 - [1 + (a + X)^2] = 2X(\sqrt{1 + a^2} - a) \geq 0$.

Choose positive h_0, C_w, C_v, A_0, R_0 such that each of

$$\|H_0\|_F \leq h_0 \sqrt{M}, \quad \|W_0\|_F \leq C_w \sqrt{M}, \quad \|V_0\|_F \leq C_v, \quad \|J_{\Theta,0}\|_{\text{op}} \leq A_0, \quad \|e_0\| \leq R_0 \quad (43)$$

has failure probability at most $\rho/7$. Here and below J is the Jacobian of $\bar{f}$. For example, Markov's inequality applies with respective second-moment upper bounds

$$\frac{s_{\max}^2}{2}, \quad s_w^2, \quad C_s^2 s_v^2, \quad C_s^2 s_v^2 s_w^2 X^2, \quad C_s^2 s_v^2 s_{\max}^2 + 2\mathbb{E}\|\beta_0\|^2 + 2\|\bar{y}\|^2$$

after dividing the first two squared norms by M . Take each squared constant at least $7/\rho$ times the corresponding bound, enlarging to a positive value if necessary. For $J_{\Theta,0}$, conditional centering and independence of the entries of V_0 eliminate cross terms in $V_0 \Gamma_{n,0} W_0$; its expected squared Frobenius norm is at most $C_s^2 s_v^2 s_w^2$. The factors x_n and $N^{-1/2}$ give the stated bound. The residual bound follows by conditioning on (W_0, b_0) and using the independent centered V_0 ; the output bias itself need not be centered. The variance calculation used in Corollary B.3 gives

$$\Pr\{\lambda_{\min}(H_0 H_0^T) < \kappa M\} \leq \frac{6s_{\max}^4}{M\kappa_*^2}.$$

Thus $M \geq 42s_{\max}^4/(\rho\kappa_*^2)$ makes this sixth failure probability at most $\rho/7$.

Boundary event and constants. Set

$$h = h_0 + 1, \quad \Omega = Z(C_v + 1), \quad B = \sqrt{h^2 + \Omega^2 + 1}, \quad R_\phi = 4BR_0/\kappa, \quad \tau_i = \frac{X\|w_{i,0}\| + ZR_\phi}{\sqrt{M}}.$$

For the seventh event define

$$\mathcal{B} = \sum_{i=1}^M \|V_{:i,0}\| \|w_{i,0}\| \mathbf{1}\{\min_n |w_{i,0}^T z_n + b_{i,0}| \leq \tau_i\}.$$

Conditioning on $w_{i,0}$, the Gaussian density of $b_{i,0}$ is bounded by $(s_b\sqrt{2\pi})^{-1}$. A union bound over n , independence of $V_{i,0}$, and $\mathbb{E}\|V_{i,0}\| \leq s_v\sqrt{C/M}$ give

$$\mathbb{E}\mathcal{B} \leq C_{\text{bd}} := \frac{2Ns_v\sqrt{C}}{s_b\sqrt{2\pi}} \left(X\mathbb{E}\|w\|^2 + ZR_\phi\mathbb{E}\|w\| \right).$$

Here $\mathbb{E}\|w\|^2 = s_w^2$ and $\mathbb{E}\|w\| \leq s_w$; either the actual moment or its bound can define the constant. Put $\mathcal{B}_* = 7C_{\text{bd}}/\rho$. Then $\Pr\{\mathcal{B} > \mathcal{B}_*\} \leq \rho/7$. Define

$$\begin{aligned} A &= A_0 + X\{R_\phi C_w + (C_v + 1) + \mathcal{B}_*\}, & R_\theta &= 4AR_0/\kappa, \\ D_H &= XC_wR_\theta + ZR_\phi, & D_K &= (2h_0 + 1)D_H, \\ C_{\text{gap}} &= 2D_K + 2\Omega^2 + A^2. \end{aligned}$$

It suffices to take the integer ceiling of

$$M_* \geq \max\{1, R_\theta^2, R_\phi^2, D_H, 2D_K/\kappa, 42s_{\text{max}}^4/(\rho\kappa_*)\}. \quad (44)$$

All constants are determined before M is chosen. The intersection of the five events in (43), the head-kernel event, and the boundary event has probability at least $1 - \rho$. Initialization at a kink has probability zero and may be excluded without another probability charge.

Deterministic estimates in the bootstrap tube. Work on this intersection and in the closed tube

$$\|\Theta - \Theta_0\| \leq R_\theta/M, \quad \|\phi - \phi_0\| \leq R_\phi/\sqrt{M}.$$

The width conditions imply $\|\Theta - \Theta_0\| \leq 1$ and $\|\phi - \phi_0\| \leq 1$, as well as $R_\theta/M \leq M^{-1/2}$. Write $\omega = (W, b)$. ReLU is 1-Lipschitz, so

$$\begin{aligned} \|H - H_0\|_F &\leq X\|W_0\|_F\|\Theta - \Theta_0\| + Z\|\omega - \omega_0\| \leq D_H/\sqrt{M}, \\ \|H\|_F &\leq h\sqrt{M}, \quad \|HH^T - H_0H_0^T\|_{\text{op}} \leq D_K. \end{aligned}$$

For the second line use $D_H \leq M$ and expand the product; in particular $2h_0D_H + D_H^2/M \leq D_K$. Consequently

$$HH^T \succeq (\kappa M/2)I_N. \quad (45)$$

The preactivation change for neuron i on any datum is bounded by

$$X\|w_{i,0}\|R_\theta/M + Z\|\omega_i - \omega_{i,0}\| \leq \tau_i.$$

Thus a changed gate belongs to the initial boundary event above. To control the encoder Jacobian use the exact three-term decomposition

$$V\Gamma_n W - V_0\Gamma_{n,0}W_0 = (V - V_0)\Gamma_n W_0 + V\Gamma_n(W - W_0) + V_0(\Gamma_n - \Gamma_{n,0})W_0. \quad (46)$$

The last term has operator norm at most $\mathcal{B}$. The first uses the current gates and initial W_0 ; the second uses the current V ; both distinctions matter. Averaging over n and multiplying by x_n yields

$$\|J_\Theta\|_{\text{op}} \leq A_0 + X\{\|V - V_0\|_F\|W_0\|_F + \|V\|_F\|W - W_0\|_F + \mathcal{B}_*\} \leq A.$$

The separate block bounds are

$$\|J_\omega\|_{\text{op}} \leq Z\|V\|_F \leq \Omega, \quad \|J_V\|_{\text{op}} \leq h\sqrt{M}, \quad \|J_\beta\|_{\text{op}} = 1, \quad \|J_\phi\|_{\text{op}} \leq B\sqrt{M}.$$

Furthermore $J_\phi J_\phi^T \succeq (HH^T) \otimes I_C \succeq (\kappa M/2)I_{NC}$. These estimates also hold for the frozen path in its decoder tube.

Flow existence, chain rule and continuation. For completeness, local solutions in the convention above can be obtained by replacing ReLU by $\sigma_\epsilon(u) = (u + \sqrt{u^2 + \epsilon^2})/2$. For fixed M , on a compact parameter neighborhood its values and first derivatives are uniformly bounded for $0 < \epsilon \leq 1$. The smooth gradient vector fields are therefore uniformly bounded there, giving a common positive interval of existence and uniformly Lipschitz paths on a smaller neighborhood. A subsequence converges uniformly. The finitely many gates have a weak-* convergent subsequence in L^∞ with limits in $[0, 1]$ and the correct values away from zero preactivations. Each block gradient is linear in the corresponding gate, with all remaining coefficients converging uniformly. Passing to the integral equations gives one common limiting gate selection and a local solution. Along any such absolutely

continuous solution, each preactivation has derivative zero almost everywhere on its zero level set. Consequently $d\sigma(g(t))/dt = \gamma(t)\dot{g}(t)$ almost everywhere, for every admissible gate selection. The ordinary product rule then gives $\dot{e} = -JJ^Te$ and $\dot{L} = -\|J^Te\|^2$ almost everywhere. These facts follow directly here; related nonsmooth chain-rule frameworks are discussed by Davis et al. (2020); Bolte & Pauwels (2021). We do not infer a Clarke inclusion solely from the computational graph.

Until a first exit from the tube, $\frac{d}{dt}\|e\|^2 \leq -\kappa M\|e\|^2$, proving (N1). The integrated speeds are bounded by

$$\int_0^\infty \|\dot{\Theta}\| dt \leq \frac{2AR_0}{\kappa M} = \frac{R_\theta}{2M}, \quad \int_0^\infty \|\dot{\phi}\| dt \leq \frac{2BR_0}{\kappa\sqrt{M}} = \frac{R_\phi}{2\sqrt{M}}.$$

At a first exit these bounds contradict the tube boundary. Thus no exit occurs. Bounded velocities in this compact tube preclude finite escape, give a limit at any finite endpoint, and the local existence argument continues the path from that endpoint. This proves global existence and (N1)–(N2) for every selection, without uniqueness.

Common linear reference and attribution. Let

$$K_0 = (H_0H_0^T) \otimes I_C + (\mathbf{1}\mathbf{1}^T/N) \otimes I_C, \quad e^{\text{lin}}(t) = e^{-K_0 t}e_0.$$

The second summand is the constant output-bias kernel. For the full output kernels $K^J(t)$ and $K^F(t)$,

$$\|K^J(t) - K_0\|_{\text{op}} \leq D_K + \Omega^2 + A^2, \quad \|K^F(t) - K_0\|_{\text{op}} \leq D_K + \Omega^2.$$

Since $K_0 \succeq \kappa MI$, variation of constants gives

$$e^a(t) - e^{\text{lin}}(t) = - \int_0^t e^{-K_0(t-s)}(K^a(s) - K_0)e^a(s) ds, \quad a \in \{J, F\}.$$

Use $\|e^a(s)\| \leq R_0$ and $\int_0^t e^{-\kappa M(t-s)} ds \leq 1/(\kappa M)$ and add the two bounds to obtain (N3). Both paths track the same linear reference; (N3) is not a general stability theorem for arbitrary nonlinear joint and frozen flows. Applying Corollary B.2 with $a = A^2$ and decoder floor $\kappa M/2$ gives (N4), completing the proof. $\square$

Remark B.18 (Threshold and scope). The threshold (44) is deliberately conservative. The boundary term contains an explicit factor N/ρ and R_0 can scale as $\rho^{-1/2}$, before its further dependence on R_ϕ , the data, and κ_* is substituted. When these factors are nonzero and the remaining scales are held fixed, already R_θ^2 can contain a contribution of order $N^2/(\rho^3\kappa_*^2)$; the full displayed construction can be substantially worse. This is an illustration of vacuity, not a universal necessary lower bound (degenerate inputs can remove some terms), and it is not an empirical width prediction.

Relation to prior work and a normalization control. Positive limiting kernels, small parameter movement, and control of ReLU gate changes are established tools in wide-network convergence proofs (Du et al., 2019; Arora et al., 2019). Linearized dynamics are studied by Lee et al. (2019); their dependence on parameterization and the distinction between lazy and feature-learning limits are central to Chizat et al. (2019); Yang & Hu (2021). The divergent kernel scale of the naive standard parameterization and its relation to layerwise training rates are also explicit in Sohl-Dickstein et al. (2020). The result above is a scoped instantiation with a coupled finite-width encoder and a joint/frozen comparison. It does not establish a new general lazy-training mechanism. Its gain disparity vanishes under an explicit $1/\sqrt{M}$ readout in the following precise convention: write $f = M^{-1/2}U\sigma(W\Theta x + b) + \beta$, initialize the entries of U with variance s_v^2 , and train U at unit Euclidean rate. This preserves the initial function distribution but changes the optimization metric. The readout kernel is then $HH^T/M = O(1)$, and the encoder kernel remains $O(1)$; there is no forced factor- M readout/encoder separation. This does not assert equality of the two kernels or of their directional gains. Merely inserting the factor while retaining variance s_v^2/M for the trained readout would be a different, shrinking-output initialization.

C THE EFFECTIVE GRADIENT RATIO

The theorems of the main text bound the spectral module’s learning signal but say nothing about how a practitioner would notice its decay inside a single run. The *effective gradient ratio* (EGR) is the

cheapest observable that tracks it. We define it, prove the one statement the theory supports—decay of the numerator—and report what it does and does not predict empirically. It appears here, and not in the main text, because the evidence supports it as a complementary measurement rather than as a decision rule.

Definition C.1 (Effective gradient ratio). At training step t ,

$$\text{EGR}(t) = \frac{\|\nabla_{\theta} \mathcal{L}(\theta_t, \phi_t)\|}{\|\nabla_{\phi} \mathcal{L}(\theta_t, \phi_t)\|}.$$

Both norms are available from the standard backward pass, so the cost is one extra reduction per step. EGR is unitless and invariant to the shared learning rate.

Proposition C.1 (Spectral gradient decay). *Under the hypotheses of Theorem 2—Assumptions 1, 2, 3 and 5, with the trajectory contained in the regular set up to a horizon T —write $B_T := \sup_{t \leq T} \|J_{\theta}(t)\|_{\text{op}}$. Then for every $t \leq T$,*

$$\|\nabla_{\theta} \mathcal{L}(t)\| \leq \frac{B_T C}{1 + \mu t}, \quad B_T \leq \sqrt{C_{\theta}} = \tilde{L} \sqrt{\lambda_{\max}(\Sigma_X)},$$

with C_{θ} the cap of Lemma 1 and C, μ the constants of Assumption 5.

Proof. $\|\nabla_{\theta} \mathcal{L}\| = \|J_{\theta}^{\top} r\| \leq \|J_{\theta}\|_{\text{op}} \|r\| \leq B_T \|r\|$ for $t \leq T$, and Assumption 5 bounds $\|r(t)\| \leq C/(1 + \mu t)$. The cap on B_T is Lemma 1. The horizon is not a technicality: in the unregularized separable regime that Assumption 5 models, a cap uniform over $t \in [0, \infty)$ cannot be assumed; in practice T is the length of the run, which is the horizon the diagnostic operates over. $\square$

Remark C.1 (From gradient decay to the ratio). Proposition C.1 bounds the *numerator* only. The denominator $\|\nabla_{\phi} \mathcal{L}\| = \|J_{\phi}^{\top} r\|$ is also proportional to the residual, so both quantities vanish together as the loss is fitted and the behaviour of their ratio is not determined by residual decay alone: it depends on the relative conditioning of the two Jacobians along the trajectory. The EGR trajectory is therefore an empirical object, and we use the proposition only for the absolute claim that the spectral module’s learning signal decays polynomially while the loss is being fitted.

Descriptive statistics. On the synthetic width sweep of Experiment 1.2 ($N = 24$ runs, widths $\{16, 64, 256, 1024\}$, six seeds each), EGR depth correlates with final test accuracy at Pearson $r = -0.723$ and Spearman $\rho = -0.808$, and with the train–test gap at $r = +0.741$. The pooled correlation is confounded with capacity: EGR depth correlates with log width at $r = +0.903$, and the partial correlation with test accuracy controlling for log width collapses to -0.169 . A fixed-capacity companion sweep isolates the remainder: width fixed at $D = 256$, varying data noise $\{0.05, 0.10, 0.15, 0.20\}$, learning rate $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$, weight decay $\{0, 10^{-3}\}$ and six seeds, $N = 144$ runs. There $r(\text{EGR}_{\min}, \text{acc}) = +0.344$; $\text{EGR}_{\min}$ is nearly orthogonal to the swept hyperparameters (marginal correlations $+0.029, -0.073, +0.007$; $R^2 = 0.015$ for the full set), and the partial correlation given all three, $+0.395$, is *larger* than the pooled value—the opposite signature of a hidden hyperparameter confound. Controlling additionally for final training loss (itself a predictor, $r = +0.47$ for accuracy) leaves it at $+0.335$.

Why we report no aggregate test on the cell analysis. Within the 24 (noise, lr, wd) cells in which only the seed varies, 23 of 24 give a positive $r(\text{EGR}_{\min}, \text{acc})$, with mean $+0.584$. We report this descriptively and attach no p -value to it. The same six seeds populate every cell, so the cells are not independent replicates; the between-cell standard deviation of the Fisher-transformed correlations, 0.233, sits far below the 0.577 that independent cells of this size would produce, and a one-sample test over cells would therefore overstate the evidence by an unknown factor. The $N = 144$ partial correlation above, computed once over runs rather than over cells, is the inferential statement we stand behind.

Effect size, honestly. As a binary alarm, $\text{EGR}_{\min}$ reaches ROC-AUC 0.612 for flagging below-median test accuracy and $\text{EGR}_{\text{depth}}$ only 0.538 for a high overfit gap; the best operating point ($\text{EGR}_{\min} < 0.10$) catches 90% of poor runs while also flagging 57% of healthy ones. At fixed capacity, EGR is a real, capacity-independent signal correlated with generalization, and it carries

information that train-loss monitoring does not; the within-capacity effect size is small enough that it belongs alongside loss monitoring as a complementary measurement and not as a standalone run-quality indicator.

D SYNTHETIC EXPERIMENT DETAILS

This appendix records the settings, artifacts and complete tables behind the synthetic measurements quoted in the main text. Every number below is taken from a named file under `results/`; where a value is a mean, a median or a fitted slope that we computed from the raw rows of such a file rather than being printed in it, we say so at that place.

D.1 GENERATOR, ARCHITECTURES AND COMMON PROTOCOL

The generator produces hyperspectral images $X \in \mathbb{R}^{H \times W \times S}$ with $H = W = 16$ and $S = 64$ and a two-class per-pixel label. A known spectral direction $u \in \mathbb{R}^S$ is placed in each pixel, scaled by a per-pixel latent $z \sim \mathcal{N}(0, 1)$ and modulated by a scalar strength α ; smooth per-class spatial templates, modulated by a scalar strength β , enter the label-generating logits. At $\alpha = \beta = 1$ and noise level 0.1 the Bayes-optimal accuracy is approximately 0.85–0.95.

The spectral module f_θ is a linear per-pixel map $\mathbb{R}^{64} \rightarrow \mathbb{R}^{16}$ without bias ($C_f = 1024$ throughout). Four spatial modules are used across the experiments: *SpatialMLP*, a per-pixel one-hidden-layer network of width M (no spatial receptive field, $C_g = \Theta(M)$); *SpatialDeepMLP*, three hidden layers all $M \times M$ ($C_g = \Theta(M^2)$); *SpatialCNN*, two 3×3 convolutional layers; and *SpatialViT*, a two-layer pre-LN pixel-token transformer. The per-pixel loss is softmax cross-entropy averaged over sites.

Two initializations of the spatial module are distinguished throughout, and the spectral module’s initialization is never altered. *Default* is PyTorch’s `nn.Linear` initialization, whose bias variance is $\Theta(1/\text{fan-in})$. *Theorem* initialization draws weights $\mathcal{N}(0, 2/\text{fan-in})$ and biases $\mathcal{N}(0, \sigma_b^2)$ with $\sigma_b = 0.5$ held fixed in width, which is what the curvature-disparity theorem’s head hypothesis requires. Three further controls decouple the weight law from the bias law: `default_fixedbias` (default weights, fixed-variance biases), `he_scaledbias` (He weights, Gaussian biases at the default variance $1/(3 \text{ fan-in})$) and `he_zerobias` (He weights, zero biases).

D.2 EXPERIMENT 1.1: CURVATURE BLOCKS AT INITIALIZATION

What is measured. For each (architecture, width, seed) we compute $\lambda_{\max}$ of the two diagonal Gauss–Newton blocks $G_{\theta\theta}$ and $G_{\phi\phi}$ of the averaged cross-entropy loss at random initialization, in float64, on one fixed batch of 64 images per seed, by Lanczos with full reorthogonalization applied to a matrix-free GGN block operator (80 Lanczos steps, top-1). Before any sweep the runner executes a validation gate that cross-checks the operator’s $\lambda_{\max}$ against a densely materialized $J^\top H_\tau J$ eigendecomposition on tiny configurations for both architectures and both blocks, and aborts unless the relative error is below 10^{-6} . This closes the loss-Hessian versus Gauss–Newton measurement gap present in the first version of this experiment: the quantity measured here is the Gauss–Newton block the theory is stated for. Five seeds per width. Artifacts: `results/exp1_1_v3_ggn_aggregated.csv`, `results/exp1_1_v3_ggn_theorem_init_aggregated.csv` and `results/exp1_1_v3_ggn_siglab_isolation_aggregated.csv`, each with per-width seed means and standard deviations.

Results. Table 2 lists the two shallow-head arms and the five deep-head arms. Eigenvalue entries are the per-width means over five seeds printed in those files; the log–log slopes are ordinary least-squares fits of $\log \lambda$ on $\log M$ over the full width range, computed by us from those per-width means.

Reading. The spectral cap is flat in width in every arm: the seven λ_θ slopes lie in $[-0.050, +0.030]$, and the level shifts by three orders of magnitude between initializations without acquiring a width trend. On the shallow head under the theorem’s initialization the spatial block

Table 2: Experiment 1.1 (v3): top Gauss–Newton block eigenvalues at initialization, five seeds per width. Each eigenvalue is given at the smallest and the largest width of the arm’s range. Slopes are least-squares log–log fits over that range, computed by us from the per-width seed means in the three aggregated CSVs; they use three orders of magnitude of width for the shallow head and two for the deep head, and are finite-range descriptions rather than asymptotic exponents. The five deep-head rows are the five arms of the initialization isolation.

Initialization	M range	λ_θ			λ_ϕ		
		first	last	slope	first	last	slope
<i>Shallow head</i> (one hidden layer, $C_g = \Theta(M)$)							
default	16–8192	0.0184	0.0220	−0.019	0.62	46.0	0.705
theorem	16–8192	0.691	0.538	−0.040	2.14	441.6	0.895
<i>Deep head</i> (three hidden $M \times M$ layers, $C_g = \Theta(M^2)$)							
default	16–1024	0.00032	0.00037	+0.012	0.68	1.11	0.105
theorem	16–1024	0.371	0.398	−0.006	3.62	243.1	0.989
default, fixed bias	16–1024	0.00055	0.00063	−0.041	1.67	73.8	0.924
He, scaled bias	16–1024	0.310	0.343	+0.030	1.50	27.6	0.696
He, zero bias	16–1024	0.336	0.372	−0.050	1.45	17.6	0.606

grows with slope 0.895 from 2.14 to 441.6; under default fan-in initialization the same architecture gives slope 0.705. The deep head separates the two mechanisms most sharply: slope 0.105 under default initialization against 0.989 under theorem initialization. The isolation arms show that the width-linear behaviour survives if *either* the weight-variance term or the bias-variance term of the head hypothesis is width-independent — fixed-variance biases with default weights reach slope 0.924, He weights with default-variance biases 0.696 and He weights with zero biases 0.606 — while default fan-in initialization, which makes both terms vanish with width, does not produce it. The corresponding curvature ratios $\mathcal{D}_{\text{curv}}$, first and last width, are $39.8 \rightarrow 2405$ and $3.4 \rightarrow 869$ for the two shallow arms and $3167 \rightarrow 3060$, $18.1 \rightarrow 668$, $5552 \rightarrow 1.17 \times 10^5$, $5.7 \rightarrow 84.5$ and $4.6 \rightarrow 46.7$ for the five deep arms, in table order. They illustrate why the slope and the level must be quoted together, and why we do not use $\mathcal{D}_{\text{curv}}$ to attribute differences across arms: the deep default arm has the largest ratio of any arm and the smallest λ_ϕ slope, because its λ_θ is 10^{-4} rather than 10^{-1} .

D.3 EXPERIMENT 1.2 (V4): JOINT AND FROZEN TRAJECTORIES

Data as in Appendix D.1 with $\alpha = \beta = 1.0$ and noise 0.10; 512 training and 128 test images; Adam at learning rate 10^{-3} , batch size 32, 150 epochs (2400 steps); three seeds (42,43,44). The frozen arm sets `requires_grad` to false on the spectral parameters, so the spectral gradient is identically zero there by construction. Widths are architecture-specific: $D \in \{16, 64, 256, 1024\}$ for the CNN head and $D \in \{64, 128, 256\}$ for the ViT head. Table 3 gives the full grid; means and standard deviations over the three seeds were computed by us from the per-run rows of `results/exp1_2v4_summary.csv`.

Three observations, all descriptive at $n = 3$. The frozen arm’s endpoint accuracy is ahead of the joint arm’s at every width and both heads, by +0.014 to +0.038; every one of those gaps is smaller than the sum of the two seed standard deviations, so none is resolvable here. Peak accuracies are indistinguishable between arms (the largest difference is 0.028, at ViT $D = 128$, in favour of the joint arm). The endpoint test loss of the joint CNN arm diverges with width, reaching 2.334 ± 0.169 at $D = 1024$ against 1.119 ± 0.054 frozen, while its training loss falls to 0.029; the ViT runs are in a memorization regime at every width, with training losses of 0.015–0.14 and test losses above 3, so gradient-ratio observables are uninformative there.

D.4 EXPERIMENT 1.2 (V5): RESIDUAL OVERLAP WITH THE FAST KERNEL SUBSPACE

Model and protocol. This experiment combines the model of the verified shallow instance with the training protocol of Appendix D.3. The model is the linear per-pixel map $\mathbb{R}^{64} \rightarrow \mathbb{R}^{16}$ followed by a two-layer ReLU head of width D under theorem initialization with $\sigma_b = 0.5$; that head ini-

Table 3: Experiment 1.2 (v4): endpoint and peak test accuracy and endpoint losses, mean $\pm$ sd over three seeds, computed from the per-run rows of `results/exp1_2v4_summary.csv`. *Peak* is the maximum over the run’s per-epoch test evaluations and is therefore selection-optimistic. C_g is the spatial module’s parameter count; $C_f = 1024$ in every row.

Head	Arm	D	C_g	Final test acc.	Peak test acc.	Final test loss	Final train loss
CNN	joint	16	2,610	0.618 ± 0.061	0.682 ± 0.055	0.650 ± 0.037	0.453 ± 0.002
CNN	frozen	16	2,610	0.644 ± 0.044	0.688 ± 0.054	0.628 ± 0.038	0.465 ± 0.004
CNN	joint	64	10,434	0.574 ± 0.038	0.703 ± 0.021	0.698 ± 0.035	0.421 ± 0.004
CNN	frozen	64	10,434	0.611 ± 0.042	0.688 ± 0.054	0.658 ± 0.036	0.437 ± 0.004
CNN	joint	256	41,730	0.540 ± 0.023	0.712 ± 0.006	1.070 ± 0.112	0.290 ± 0.009
CNN	frozen	256	41,730	0.574 ± 0.026	0.690 ± 0.053	0.787 ± 0.029	0.345 ± 0.008
CNN	joint	1024	166,914	0.534 ± 0.013	0.711 ± 0.007	2.334 ± 0.169	0.029 ± 0.004
CNN	frozen	1024	166,914	0.560 ± 0.024	0.690 ± 0.051	1.119 ± 0.054	0.153 ± 0.019
ViT	joint	64	84,546	0.533 ± 0.021	0.707 ± 0.008	3.121 ± 0.131	0.136 ± 0.030
ViT	frozen	64	84,546	0.547 ± 0.050	0.702 ± 0.037	3.387 ± 0.233	0.132 ± 0.024
ViT	joint	128	300,162	0.546 ± 0.047	0.706 ± 0.019	3.149 ± 0.936	0.039 ± 0.016
ViT	frozen	128	300,162	0.567 ± 0.038	0.679 ± 0.057	3.302 ± 0.451	0.025 ± 0.010
ViT	joint	256	1,124,610	0.503 ± 0.050	0.707 ± 0.023	3.779 ± 0.547	0.031 ± 0.020
ViT	frozen	256	1,124,610	0.539 ± 0.007	0.700 ± 0.035	3.522 ± 0.160	0.015 ± 0.001

tialization is required by the verified instance and is not PyTorch’s default. Optimizer, data, loss and schedule are exactly those of Appendix D.3, with $D \in \{128, 512, 2048\}$, seeds 42–44 and both a joint and a frozen arm. The frozen arm here keeps `requires_grad` true on the spectral parameters but excludes them from the optimizer’s parameter list; the trajectory is identical to a hard freeze, and the logged spectral gradient is the counterfactual signal a frozen encoder would have received. Artifacts: `results/exp1_2v5_residual_export.csv` (198 rows) and `results/exp1_2v5_residual_export_REPORT.md`.

Probe and normalization. All kernel quantities are computed on a fixed probe of $N = 256$ supervised sites with $C = 2$ classes, so the probe logit space is $\mathbb{R}^{512}$. The probe is drawn once per run with a fixed seed from the first 16 training images and held fixed for the whole trajectory; non-probe pixels are set to the ignore index, so the GGN machinery sees the same site set. With $z = N^{-1/2} \text{stack}(\hat{y})$ and averaged cross-entropy, $r = N^{-1/2}(\text{softmax}(z) - \text{onehot}(y))$, $J_b = N^{-1/2} \partial \hat{y} / \partial b$ and $K_b = J_b J_b^\top$, so $\nabla_b \mathcal{L} = J_b^\top r$ holds exactly. We write $a_{\text{eff}} = r^\top K_\theta r / \|r\|^2$ and $\kappa_{\text{eff}} = r^\top K_\phi r / \|r\|^2$ for the realised Rayleigh quotients, f_k for the fraction of $\|r\|^2$ inside the top- k eigenspace of K_ϕ , and $\text{share} = a_{\text{eff}} / (a_{\text{eff}} + \kappa_{\text{eff}})$, which coincides by construction with the instantaneous gradient-flow attribution bound. Table 4 gives the three identities the runner recomputes at each of the 198 measurements.

Table 4: Experiment 1.2 (v5): validation identities, maximum relative error over all 198 measurements, from `results/exp1_2v5_residual_export_REPORT.md` §2. All three are at float64 round-off, so the kernels here are the same objects the GGN machinery measures, in the same normalization.

Identity	Max. relative error
$a_{\text{eff}} \ r\ ^2 = \ \nabla_\theta \mathcal{L}\ ^2$ (autograd)	1.30×10^{-15}
$\kappa_{\text{eff}} \ r\ ^2 = \ \nabla_\phi \mathcal{L}\ ^2$ (autograd)	1.62×10^{-14}
$\lambda_{\text{max}}(H^{1/2} K_\phi H^{1/2}) = \lambda_{\text{max}}(G_{\phi\phi})$ (power iteration)	2.47×10^{-15}

Headline table. Table 5 reproduces the joint arm of §3 of the report at four log-spaced checkpoints for every width and seed. The worst-case-coercivity proxy $\lambda_{\text{max}}(K_\theta) / (\lambda_{\text{max}}(K_\theta) + \lambda_{\text{min}}(K_\phi))$ is between 0.9998 and 1.0000 at every one of the 198 rows and is omitted from the table; $\lambda_{\text{min}}(K_\phi)$ ranges from 2.0×10^{-6} to 1.8×10^{-4} and is likewise omitted. Full trajectories at all eleven checkpoints for all eighteen runs, and the frozen arm, are in the report.

Table 5: Experiment 1.2 (v5), joint arm: probe measurements at four checkpoints, every width and seed, from `results/expl_2v5_residual_export_REPORT.md` §3. *share* is $a_{\text{eff}}/(a_{\text{eff}} + \kappa_{\text{eff}})$; *proxy* is $\lambda_{\text{max}}(K_{\theta})/(\lambda_{\text{max}}(K_{\theta}) + \lambda_{\text{max}}(K_{\phi}))$. These are probe quantities at eleven checkpoints, not a certificate about the intervening steps.

D	seed	step	a_{eff}	κ_{eff}	$\lambda_{\text{max}}(K_{\theta})$	$\lambda_{\text{max}}(K_{\phi})$	f_1	f_{20}	f_{100}	share	proxy
128	42	0	0.704	0.858	2.393	19.839	$1.80 \cdot 10^{-2}$	0.3441	0.4356	0.4510	0.1076
128	42	32	0.017	0.011	4.887	21.590	$1.48 \cdot 10^{-5}$	0.0390	0.1223	0.6022	0.1846
128	42	426	0.012	0.022	3.764	21.884	$9.52 \cdot 10^{-6}$	0.0594	0.1296	0.3513	0.1468
128	42	2400	0.025	0.043	6.965	22.678	$3.68 \cdot 10^{-6}$	0.0288	0.1778	0.3688	0.2350
128	43	0	0.299	4.219	1.532	21.735	$1.34 \cdot 10^{-1}$	0.4465	0.5173	0.0662	0.0659
128	43	32	0.013	0.060	3.275	23.963	$1.10 \cdot 10^{-3}$	0.0375	0.1516	0.1755	0.1202
128	43	426	0.009	0.018	2.229	24.267	$1.81 \cdot 10^{-4}$	0.0298	0.1404	0.3405	0.0841
128	43	2400	0.036	0.037	8.669	26.355	$2.49 \cdot 10^{-5}$	0.0525	0.1913	0.4927	0.2475
128	44	0	0.711	0.948	3.793	16.447	$5.30 \cdot 10^{-3}$	0.3073	0.3813	0.4285	0.1874
128	44	32	0.049	0.078	4.917	17.693	$1.53 \cdot 10^{-4}$	0.0400	0.1711	0.3826	0.2175
128	44	426	0.022	0.026	5.010	17.847	$8.86 \cdot 10^{-6}$	0.0307	0.1702	0.4611	0.2192
128	44	2400	0.029	0.042	6.428	18.976	$1.04 \cdot 10^{-4}$	0.0309	0.2020	0.4073	0.2530
512	42	0	1.098	4.107	2.208	70.380	$1.28 \cdot 10^{-2}$	0.5162	0.5947	0.2109	0.0304
512	42	32	0.027	0.071	8.021	72.767	$1.22 \cdot 10^{-4}$	0.0513	0.1544	0.2763	0.0993
512	42	426	0.021	0.041	7.199	72.745	$2.62 \cdot 10^{-5}$	0.0402	0.1718	0.3393	0.0901
512	42	2400	0.065	0.094	24.622	72.428	$4.01 \cdot 10^{-4}$	0.0154	0.1741	0.4101	0.2537
512	43	0	0.464	7.095	1.717	67.270	$3.67 \cdot 10^{-2}$	0.4456	0.5469	0.0614	0.0249
512	43	32	0.023	0.052	4.829	69.680	$1.14 \cdot 10^{-6}$	0.0334	0.1328	0.3050	0.0648
512	43	426	0.013	0.307	3.550	70.279	$2.90 \cdot 10^{-3}$	0.0369	0.1388	0.0413	0.0481
512	43	2400	0.077	0.085	8.980	70.889	$9.68 \cdot 10^{-6}$	0.0374	0.2226	0.4732	0.1124
512	44	0	0.218	14.482	1.749	61.928	$2.33 \cdot 10^{-2}$	0.4084	0.4749	0.0148	0.0275
512	44	32	0.050	0.236	5.380	64.643	$2.66 \cdot 10^{-4}$	0.0460	0.1826	0.1734	0.0768
512	44	426	0.019	0.064	4.804	65.177	$2.64 \cdot 10^{-6}$	0.0253	0.1555	0.2328	0.0686
512	44	2400	0.073	0.113	9.265	68.294	$1.12 \cdot 10^{-5}$	0.0144	0.1853	0.3935	0.1195
2048	42	0	0.525	6.973	1.805	253.786	$8.38 \cdot 10^{-6}$	0.3850	0.4677	0.0701	0.0071
2048	42	32	0.044	0.324	10.807	254.624	$9.03 \cdot 10^{-4}$	0.0373	0.1713	0.1197	0.0407
2048	42	426	0.026	0.361	11.223	255.856	$9.58 \cdot 10^{-4}$	0.0385	0.1612	0.0663	0.0420
2048	42	2400	0.196	0.320	41.018	252.457	$2.56 \cdot 10^{-7}$	0.0206	0.1848	0.3796	0.1398
2048	43	0	0.179	148.674	1.579	276.775	$2.17 \cdot 10^{-1}$	0.6836	0.7327	0.0012	0.0057
2048	43	32	0.049	0.354	9.486	274.651	$4.62 \cdot 10^{-4}$	0.0193	0.1151	0.1217	0.0334
2048	43	426	0.022	1.213	5.681	276.344	$3.30 \cdot 10^{-3}$	0.0270	0.1585	0.0175	0.0201
2048	43	2400	0.153	0.179	17.424	281.103	$8.36 \cdot 10^{-6}$	0.0299	0.1326	0.4617	0.0584
2048	44	0	0.495	63.759	2.389	253.499	$4.96 \cdot 10^{-2}$	0.4682	0.5447	0.0077	0.0093
2048	44	32	0.018	0.257	8.093	252.495	$1.78 \cdot 10^{-4}$	0.0237	0.1879	0.0648	0.0311
2048	44	426	0.034	1.025	8.764	257.769	$1.40 \cdot 10^{-3}$	0.0264	0.1860	0.0323	0.0329
2048	44	2400	0.131	0.576	30.982	257.813	$1.10 \cdot 10^{-3}$	0.0263	0.1584	0.1848	0.1073

Cumulative gradient-energy fractions. Table 6 gives $\sum_t \|\nabla_{\theta} \mathcal{L}\|^2 / \sum_t (\|\nabla_{\theta} \mathcal{L}\|^2 + \|\nabla_{\phi} \mathcal{L}\|^2)$ over the whole 2400-step run, from the per-step logs the runner writes.

Table 6: Experiment 1.2 (v5): cumulative gradient-energy fraction over the full run, from results/exp1_2v5_residual_export_REPORT.md §6. These are mini-batch gradient norms over the 32-image training batches (8192 sites), not probe quantities, and the optimizer is Adam, so this is a gradient-energy attribution and not a decomposition of the realised loss decrease. On the frozen arm the spectral parameters never move, so that column is a counterfactual ratio.

D	seed	joint	frozen
128	42	0.2896	0.7256
128	43	0.0806	0.1883
128	44	0.3432	0.6091
512	42	0.1303	0.3308
512	43	0.0618	0.1230
512	44	0.0415	0.1336
2048	42	0.0173	0.0642
2048	43	0.0042	0.0059
2048	44	0.0124	0.0310

What the numbers say. The following are the report’s own summaries, over the joint arm’s 99 rows unless stated otherwise. The spatial kernel is extremely ill-conditioned on the probe logit space: $\lambda_{\max}/\lambda_{\min}$ has median 1.28×10^6 (range 2.42×10^5 to 1.06×10^7), so a coercivity hypothesis $K_{\phi} \succeq \kappa I$ taken on the whole output space is available only with $\kappa = \lambda_{\min}$ and yields nothing. The visited residual is not in that near-null part — κ_{eff} exceeds $\lambda_{\min}$ by between $225\times$ and a median $5.0 \times 10^3\times$ — and it is not in the coercive top either: κ_{eff} is a median 0.0025 of $\lambda_{\max}$ (range 2.5×10^{-4} to 0.537), and after step 75 the top eigendirection carries a median 8.6×10^{-5} of $\|r\|^2$ (down from a median 0.0233 at $t = 0$), the top-20 eigenspace a median 0.0298 (down from 0.446) and the top-100 of 512 a median 0.1612 . The spectral side collapses the same way: the top left singular direction of J_{θ} carries a median 6.2×10^{-4} of $\|r\|^2$ after step 75, down from 0.1513 at $t = 0$. Measured against the realised share, the top-eigenvalue proxy understates the encoder’s share in 85% of joint rows, by a median factor 2.72 and up to $11.6\times$, and overstates it by up to $11.1\times$ in the remaining 15%; it errs in both directions and bounds the share in neither. This is how we read those rows: the diagnostic limitation illustrated by the counterexample to top-eigenvalue certificates is observed on this instance.

Scope of these measurements. Four restrictions apply and are not optional caveats. (i) The cumulative column is a *gradient-energy fraction* on the same sequence, not an attribution of Adam’s loss decrease; the algebraic weighted-sum identity holds for any optimizer on one sequence, but the realised displacement under Adam is not proportional to the gradient. (ii) Quantities (i)–(v) of the report are computed on the fixed 256-site probe while the cumulative column uses training mini-batches: these are the same functional on different sequences, and we report probe results as probe results. Eleven checkpoints cannot certify an almost-everywhere trajectory condition or the intervening steps. (iii) The appropriate isotropic reference for the subspace fractions lives in the per-site class-contrast space: with $\Pi = I_N \otimes (I_C - \mathbf{1}\mathbf{1}^T/C)$, a random unit residual there has expected projected mass $\text{tr}(P\Pi)/[N(C-1)]$ rather than $\text{rank}(P)/(NC)$, and for $C = 2$ the difference is consequential. That reference has not been recomputed for the projectors actually used, so we make no claim that any measured mass falls below an isotropic level; the measured masses themselves stand. (iv) The cross-entropy Hessian weighting is a different quantity, not a correction to a squared-loss approximation: for cross-entropy $r^{\top} K_b r = \|\nabla_b \mathcal{L}\|^2$ already holds exactly. For completeness, §5 of the report records that the weighted share moves by at most a factor 2.721 and at least 0.348 relative to the unweighted one on this instance. Finally, a cumulative $O(1/D)$ law does not follow from these three widths: under the phase-restricted attribution result of the mathematics appendix, a total share of order $1/D$ requires the complementary phase to carry an $O(1/D)$ fraction of the energy, and that fraction has not been measured here.

D.5 EXPERIMENT 1.3: THE EQUAL-INFORMATION CALIBRATION

Calibration. “Equal information” between the spectral pathway (per-pixel content of X along a direction u) and the spatial pathway (content along an orthogonal direction $v \perp u$, modulated by a smooth per-position function of the class) requires an operational definition. Three were compared: *bayes*, which sets α, β so that a single-modality logistic regression on α -only data and a position-majority classifier on β -only data both reach a target accuracy of 0.75; *ntk*, which equalizes the leading Gram-matrix eigenvalues of the two modalities’ per-pixel features; and *margin*, which equalizes the L_2 margins of the two single-modality logistic regressions.

Setup and results. For each mode the CNN composition is trained at $D = 256$ for 150 epochs on $N_{\text{train}} = 512$ samples over three seeds, in four conditions: joint, frozen at random spectral initialization, spectral-only ($\beta = 0$) and spatial-only ($\alpha = 0$). Table 7 compares the three modes.

Table 7: Experiment 1.3: calibration-mode comparison (CNN, $D = 256$, three seeds). *spec/spat* are the achievable single-modality accuracies; *gap* is $a_{\text{frozen}} - a_{\text{joint}}$.

Mode	α, β	spec/spat	joint	frozen	gap
bayes	0.82, 18.75	0.75 / 0.75	0.48	0.61	+0.13
ntk	0.99, 4.91	0.78 / 0.58	0.53	0.57	+0.04
margin	5.00, 5.00	0.95 / 0.59	0.60	0.69	+0.09

Under *bayes* calibration the single-modality baselines are equal by construction ($\alpha = 0.82$, $\beta = 18.75$ give spectral-only 0.746 and spatial-only 0.754 under the logistic-regression and position-majority bounds). On the full CNN model, spectral-only training reaches 0.53 and spatial-only training reaches 0.53; frozen-spectral training on the full data reaches 0.61; joint training reaches 0.48.

Two caveats bound this result. First, the effect size is modest in absolute terms. The quoted per-condition standard error of roughly 0.02 is a pooled figure over three seeds and 128 test samples and does not include between-seed variance, so it understates the uncertainty; at three seeds the intervals are loose, and 0.48 is not distinguishable from the 0.50 chance floor. Second, the CNN reaches only 0.53 on single-modality data against a 0.75 calibration target, so the network is operating far from the Bayes bound and part of the joint-training deficit could in principle reflect generic overfitting with more trainable parameters. What the direction of the result establishes is that joint training here undershoots baselines that demonstrably use only one modality, which is not what a reading in which the spectral features are simply irrelevant would produce.

D.6 EXPERIMENT 1.6: COMPARISON WITH A LOGIT-MAGNITUDE PENALTY

Design. 120 runs at width 256 for 100 epochs with weight decay 0, sweeping the conditions *joint*, *frozen* and *sd* (spectral decoupling, $(\lambda/2) \|\text{logits}\|^2$ on the pre-softmax logits, summed over classes and averaged over examples, following Pezeshki et al. (2021)) with $\lambda \in \{10^{-3}, 10^{-2}, 10^{-1}, 1\}$, over noise levels $\{0.05, 0.10, 0.15, 0.20\}$ and five seeds. Artifacts: `results/exp1_6_summary.csv` (120 rows), `results/exp1_6_notes.md` and per-run logs under `results/exp1_6/`.

Results. Table 8 aggregates over the five seeds and four noise levels; the means were computed by us from the rows of `results/exp1_6_summary.csv`, with λ selected per noise level by highest mean final test accuracy. Paired Wilcoxon tests matched on (noise, seed), $n = 20$, recomputed from the same rows, give $T = 88$, $p = 0.55$ for joint against the selected- λ penalty, and $T = 6$, $p = 2.7 \times 10^{-5}$ for joint against frozen.

Caveats. This experiment was asserted in an earlier draft of the work without being presented; it is set out here in full, and the following qualifications belong with it. (i) The frozen condition trades peak accuracy for a smaller peak-to-final difference: its mean peak is 0.583 against 0.660, because a random spectral projection sometimes destroys signal. (ii) Its seed-to-seed variability is large — the

Table 8: Experiment 1.6: peak and final test accuracy and their difference, averaged over five seeds and four noise levels, computed from `results/exp1_6_summary.csv`. The penalty row uses the λ with the highest mean final accuracy at each noise level (10^{-2} at noise 0.05, 10^{-1} at the other three).

Condition	Peak acc.	Final acc.	Peak – final
joint	0.658	0.548	0.111
logit penalty, selected λ	0.660	0.550	0.110
frozen	0.583	0.517	0.066

standard deviation of final accuracy across seeds is 0.085–0.141 for frozen against 0.018–0.040 for the other two conditions — and one seed at noise 0.20 ends at final accuracy 0.372, below chance. (iii) At width 256 the joint condition does not collapse dramatically (0.66 peak to 0.55 final); larger drops appear at $D = 2048$ in Experiment 1.2, so this is a moderate-capacity comparison. (iv) The best penalty strength depends on the noise level, so a single λ across noise levels is not the best available setting for that intervention. We report the comparison descriptively, as a contrast between two interventions on one synthetic instance; nothing here is evaluated as a recommendation for practice.

D.7 SOLVABLE SERIAL INSTANCE: NUMERICAL VERIFICATION

Zero-context instance. Table 9 verifies the closed forms and the bounds of the solvable serial cross-entropy instance at $a_0 = 0$, $v_0 = 1$ and margin $m = \log 19 = 2.944$, from `results/toy_serial_ce.csv`. The solver column is a stiff ODE integration of the full $(2 + M)$ -dimensional state; a separate explicit-Euler gradient-descent path, run at four step sizes, has observed convergence order 1.0000004 and agrees with an independent numpy analytic-gradient implementation to 0 relative error.

Table 9: Solvable serial instance ($a_0 = 0$, $v_0 = 1$, $m = \log 19$): closed form against ODE solver, from `results/toy_serial_ce.csv`. The last block is the two-timescale displacement bound against the realised encoder displacement. Every relative discrepancy between closed form and solver in this table is at most 1.2×10^{-13} .

Quantity	$M = 16$	$M = 64$	$M = 256$	$M = 1024$
$\lambda_{\theta}(0)$	0.25	0.25	0.25	0.25
$\lambda_{\phi}(0)$	4.0	16.0	64.0	256.0
$\mathcal{D}_{\text{curv}}(0)$	16.0	64.0	256.0	1024.0
$a(T_m)$	0.142325	0.042094	0.011216	0.002857
suppression m/a_M	20.688	69.949	262.53	1030.7
$M a_M$ (limit m)	2.2772	2.6940	2.8712	2.9253
T_m	0.90784	0.28680	0.078763	0.020252
T_m lower bound $\Psi(m)/(M+1+2m^2)$	0.60992	0.25437	0.076345	0.020094
T_m upper bound $\Psi(m)/(M+1)$	1.2320	0.32222	0.081496	0.020434
$\sup_t (q_J - q_F)$	0.38232	0.13813	0.039802	0.010376
uniform gap bound	11.297	2.9547	0.74729	0.18737
$(M+1) \times \text{gap}$	6.4994	8.9787	10.229	10.635
$\ W(T_m) - W(0)\ _2$	0.219021	0.071050	0.019658	0.005064
displacement bound	1.370601	0.358465	0.090662	0.022732
bound / displacement	6.258	5.045	4.612	4.489
normalized-readout control: $a(T_m)$	1.026947	1.026947	1.026947	1.026947
normalized-readout control: $\ W(T_m) - W(0)\ $	1.177117	1.177117	1.177117	1.177117
residual envelope excess on $[0, 20T_m]$	0	0	0	0
readout spread $\max_j \beta_j - \min_j \beta_j$ at T_m	0	0	$1.4 \cdot 10^{-17}$	0
invariant $q = Q_M(a)$, max rel. error	$2.0 \cdot 10^{-11}$	$6.7 \cdot 10^{-13}$	$1.5 \cdot 10^{-13}$	$6.8 \cdot 10^{-14}$

The bound stays within 4.5 to 6.3 times the realised displacement across this width range while both fall by more than a factor of 40. The normalized-readout control, in which the readout is $M^{-1/2}Uh$ with $O(1)$ entries trained at unit rate, removes the width dependence exactly: the encoder’s movement is the $M = 1$ value at every width, to 1.7×10^{-14} relative error. The eigenvalue constants $\lambda_\theta(0) = 1/4$ and $\lambda_\phi(0) = M/4$ hold for the mean-over-examples binary cross-entropy with a unit-scaled scalar logit; summing instead of averaging, or rescaling the logit, multiplies both blocks equally and leaves $\mathcal{D}_{\text{curv}}(0) = M$ unchanged.

Isotropic extension. Table 10 collects the verification of the general-initialization version, in which $a(0) = a_0$, $v(0) = v_0 \neq 0$ and the readouts start unequal but sum to exactly zero. The grid is $M \in \{1, 4, 16, 64, 256, 1024, 4096\}$, $a_0 \in \{-1.5, -0.3, 0, 0.4, 1.7\}$, $v_0 \in \{-1.3, -0.2, 0.05, 0.2, 1.0\}$ and $m \in \{2.0, 2.9, 4.0\}$: 525 non-trivial cases plus 54 immediate-stop cases with $a_0 \geq m$, integrated with DOP853 at $\text{rtol } 10^{-10}$ and $\text{atol } 10^{-12}$ with a terminal event at $q = m$.

Table 10: Isotropic serial instance: verification summary and worst-case bound slacks over the whole grid, from `results/toy_serial_ce_isotropic_REPORT.md` §1–§2. Slacks are bound – value for upper bounds and value – bound for lower bounds; a negative entry would be a violation, and there are none. $\Delta = m - a_0$.

Quantity	Value
Max closed-form vs ODE relative error over $u, v(T_m), b(T_m), T_m$ (525 cases)	1.260×10^{-11}
Max relative error of the conserved quantity along the whole trajectory	1.518×10^{-10}
Max root accuracy $ Q_{\text{iso}}(u_M) - m $	3.553×10^{-15}
Max $ \sum_j \beta_j(0) $ (zero-sum initialization is bitwise exact)	0
Max spread of $\beta_j(T_m) - \beta_j(0)$	7.772×10^{-16}
Max frozen-comparator ODE vs closed-form relative error	5.993×10^{-11}
Max $M^{-1/2}$ -readout control vs its $M = 1$ value	1.335×10^{-12}
Bound violations, all cases and all bounds	0
Min slack, $u_M \leq \Delta/(1 + Mv_0^2)$	2.221×10^{-10}
Min slack, $u_M > 0$	4.333×10^{-5}
Min slack, $ v(T_m) - v_0 \leq \Delta^2/(2M v_0 ^3)$	1.493×10^{-9}
Min slack, encoder displacement bound	3.918×10^{-10}
Min slack, $T_m \leq \Psi_{a_0}(m)/(1 + Mv_0^2)$	1.748×10^{-9}
Min slack, $T_m \geq \Psi_{a_0}(m)/(1 + Mv_0^2 + 2\Delta^2/v_0^2)$	3.175×10^{-9}
Min slack, $q_J(t) \geq q_F(t)$	0
Min slack, $\sup_t(q_J - q_F) \leq C_0/(p_0 M v_0^2)$	1.495×10^{-4}
Reversal error at $\sigma = 1, m = 2, M = 64 / 1024 / 16384$	0.8207 / 0.8380 / 0.8397
Monte-Carlo limit $\Phi(m/2\sigma)$ on the same 4000 draws	0.8413
Draws disagreeing with the limit that satisfy $Mv_0^2 > m/(m - 2a_0)$	0 of 48000
Spectral-only comparator reversal error, every case	0

The update fraction $(a(T_m) - a_0)/(m - a_0)$ falls towards zero in M at every (a_0, v_0) on the grid — for instance 0.288, 0.0423, 0.000964 at $M = 1, 16, 1024$ with $a_0 = -1.5, v_0 = 1, m = 2.9$ — while $a(T_m)/m$ and $a(T_m)/a_0$ converge to a_0/m and to 1 instead, so those two ratios are not the quantities that vanish. Under the $M^{-1/2}$ readout the suppression is 0.336399 at every width, against 0.336, 0.0472 and 0.000969 for the unnormalized readout at $M = 1, 16$ and 1024. Every Monte-Carlo draw that falls short of the reversal limit fails the sufficient finite-width condition, either because $|v_0|$ is small or because a_0 sits just below $m/2$, where the threshold diverges.

E PRODUCTION MODEL DETAILS

E.1 PIPELINE, CAPACITIES AND DATA

The data are 336×336 pixel tissue-microarray cores collected by quantum-cascade-laser infrared spectroscopy over the $1000\text{--}1800 \text{ cm}^{-1}$ fingerprint region, with per-pixel four-class tissue labels. Each pixel carries an $S = 942$ dimensional vector, obtained by stacking baseline-corrected ab-

sorbance with its first two derivatives. The compositional model is the Block Vision Transformer of O’Leary et al. (2026): a linear per-pixel $f_\theta : \mathbb{R}^{942} \rightarrow \mathbb{R}^K$, then 16×16 patch tokenization, a 12-layer, 12-head vision transformer of embedding width M , a transposed-convolution decoder and a convolutional segmentation head. Parameter counts in Table 11 are read from the instantiated model.

Table 11: Module capacities of the production pipeline, counted from the instantiated model. K is the spectral bottleneck width. The last row is the single-channel field-standard input convention, for comparison.

Configuration	C_f	C_g	C_g/C_f
$K = 64$, three-channel input	61,108	18,271,540	299
$K = 128$, three-channel input	121,588	21,472,564	177
$K = 64$, single-channel input	20,916	18,271,540	874

Channel convention. The published architecture operates on the second-derivative spectrum alone, the field-standard preprocessing in infrared histopathology. The models analysed here instead stack raw absorbance with its first two derivatives, following a companion tokenization study (anonymous, under review). The choice affects only C_f , and the three-channel convention is the conservative one for the present purpose: under the field standard the capacity asymmetry is roughly three times larger. Within the spatial module the dominant blocks are the transformer (5.34×10^6 parameters) and the transposed convolution (9.44×10^6), both $\Theta(M^2)$ in the embedding width.

The head hypothesis is structurally unmet. The segmentation head is the chain $\text{Conv}_{3 \times 3}(M+K \rightarrow 96) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Conv}_{3 \times 3}(96 \rightarrow 48) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Conv}_{3 \times 3}(48 \rightarrow 4)$, so the final affine classifier sees $48 \times 3 \times 3 = 432$ input features at every width we build ($M \in \{48, 192, 384\}$, $K \in \{64, 128\}$): the $\Theta(M)$ feature map is separated from the logits by two nonlinearities. The curvature-disparity theorem requires a dense classifier over penultimate features of dimension $\Theta(M)$ with no downstream nonlinearity, and that hypothesis is not satisfied here. Failing a hypothesis removes a guarantee; it does not reverse the conclusion.

E.2 GEOMETRY AT INITIALIZATION

Gauss–Newton blocks of the instantiated pipeline were measured on real training batches at random initialization, by Lanczos with full reorthogonalization on matrix-free GGN block operators, validated to 10^{-12} against dense computation, sweeping $M \in \{48, 96, 192, 384\}$ with three seeds and four batches per width. Table 12 collects the results.

Two controls bracket the denominator. Removing the spectral normalization deflates $\lambda_{\max}(G_{\theta\theta})$ by $2.4\times$ ($471 \rightarrow 194$ at $M = 192$) and pushes the ratio across parity (1.18 at $M = 192$, 1.31 at $M = 384$). Input *centering* changes nothing when the architecture contains internal batch normalization: $\mathcal{D}_{\text{curv}}$ is identical to three decimals with and without centering, because a linear projection followed by normalization absorbs constant input shifts exactly. The near-rank-one input statistics make the cap side of the curvature-disparity bound uninformative at buildable widths independently of the head-hypothesis failure: the data concentrate almost all their second-moment mass in one direction, and that direction is the mean spectrum. The directional measurements establish an anisotropy, not its consequence for learning; whether it starves learning is a hypothesis, not a measurement.

E.3 WHAT CARRIES THE WIDTH TREND

The measurements in this subsection are made on the production model at random initialization, at three widths $M \in \{48, 192, 384\}$ and three seeds, on one core (the densest of the first 24 breast fold-0 training cores, with 29,664 labelled pixels of 112,896), with cuDNN TF32 disabled. Four normalization modes are compared: `train` (both head batch normalizations using batch statistics), `eval_bn1` and `eval_bn2` (only the first or only the second head normalization at its initial running statistics) and `eval_head` (both). Artifacts: `results/exp1_8c_interior_witness.csv` and `results/exp1_8d_REPORT.md`.

Table 12: Production geometry at initialization. Ratios are mean $\pm$ sd over three seeds and four batches per width. Slopes are log-log fits over the width sweep. Effective ranks are out of $S = 942$. Directional entries are measured at $M = 192$ over the nine measurements whose batch contains both classes (three seeds $\times$ three batches).

Quantity	Breast	Prostate
$\mathcal{D}_{\text{curv}}$ at $M = 48$	0.33 ± 0.12	—
$\mathcal{D}_{\text{curv}}$ at $M = 192$	0.46 ± 0.21	—
$\mathcal{D}_{\text{curv}}$ at $M = 384$	0.70 ± 0.19	1.09 ± 0.52
log-log slope of $\lambda_{\max}(G_{\theta\theta})$ vs M	-0.003	-0.012
log-log slope of $\lambda_{\max}(G_{\phi\phi})$ vs M	$0.38 (115 \rightarrow 258)$	0.45
Effective rank of Σ_X , raw inputs	1.07	1.02
Effective rank of Σ_X , post-normalization	1.21	—
$ \cos $ between v_1 and the mean spectrum (post-norm./raw)	$> 0.995 / > 0.9999$	—
θ -block top eigenvector’s norm fraction in the v_1 -induced subspace	0.954 ± 0.017	—
Trace fraction in the top 40 of 61,108 spectral directions	0.83 ± 0.12	—
Curvature ratio, v_1 vs full class contrast ($ \cos(d, v_1) = 0.45$)	$3.7 \times (2.6-4.8)$	—
Curvature ratio, v_1 vs v_1 -orthogonalized contrast	$12-28 \times$	—
Squared-norm fraction of the contrast retained after orthogonalization	$76-84\%$	—
Rank at which $\lambda_{\max}(G_{\phi\phi})$ overtakes the θ spectrum	$2-5$	—

The incoming features are flat in width. The only downstream layer whose input dimension grows with M is the first head convolution. The top eigenvalue of the patch-unfolded second-moment matrix of the features entering it is unchanged to four significant figures across the three widths, at 356.3, 403.7 and 332.4 for seeds 0, 1 and 2 respectively; the leading direction sits inside the width-independent $K = 64$ skip channels. Table 13 also records a discrepancy that matters for any normalization argument: the same Gram computed over the full batch-normalization group (all positions, ignored pixels included) rather than over labelled pixels is substantially smaller, and the cosine between the two centered top eigenvectors is far from one.

Table 13: Incoming 3×3 -patch Gram of the first head convolution, top eigenvalue, from `results/exp1_8d_REPORT.md`. Values are identical across $M \in \{48, 192, 384\}$ and across all four normalization modes, so a single column suffices. *labelled* uses the labelled-pixel mask; *full group* uses the batch-normalization group. The last column is the cosine between the two centered top eigenvectors.

seed	labelled	labelled, centered	full group	cos(labelled, full)
0	356.3	68.19	263.3	0.2793
1	403.7	54.60	306.8	0.5502
2	332.4	67.14	253.0	0.2176

Normalization quantities. Table 14 gives the per-channel batch statistics of the two head normalizations, measured over the group they actually normalize, together with the activation energy entering and leaving the first one. Under fan-in initialization the pre-normalization variance of the first head normalization falls roughly as $1/(M + K)$, and its inverse-standard-deviation gain grows correspondingly. The stabilizer $\varepsilon = 10^{-5}$ is never within three orders of magnitude of the variance: the fraction of channels with variance below 10ε is 0 in every row of both normalizations at every width, seed and mode.

Witness Jacobian-vector products. A unit perturbation of the first head convolution is propagated by forward-mode differentiation through the actual head in the actual normalization mode, so that the train-mode derivative includes the dependence on the batch statistics. The final scalar $e_{\text{final}} = N^{-1} \sum_n d_n^\top (\text{diag}(p_n) - p_n p_n^\top) d_n$ is a lower-bound witness for $\lambda_{\max}(G_{\phi\phi})$ and is verified against the same matvec the eigensolver uses; all 216 rows pass at relative discrepancy below 10^{-3} , with a maximum of 1.269×10^{-5} over the 198 rows with nonzero curvature. Table 15 gives the stage energies for two directions in two modes.

Table 14: Head normalization quantities at initialization, train mode, from results/exp1_8d_REPORT.md Table 1. Per-seed values are per-channel medians (variance, gain) or labelled-pixel energies; the *mean* column is the mean of the three seed values, computed by us. $\text{gain}_i = \gamma_i / \sqrt{\text{var}_i + \varepsilon}$. The last row is the product of the mean variance with $M + K$, $K = 64$.

Quantity	seed 0	seed 1	seed 2	mean	M
First head BN, median variance	0.1470	0.1467	0.1587	0.1508	48
	0.0678	0.0589	0.0656	0.0641	192
	0.0346	0.0310	0.0398	0.0351	384
First head BN, median gain	2.609	2.611	2.511	2.577	48
	3.842	4.120	3.905	3.956	192
	5.375	5.682	5.010	5.356	384
Second head BN, median gain	2.892	3.122	3.039	3.018	48
	3.110	3.177	3.195	3.161	192
	2.966	3.060	3.121	3.049	384
Pre-BN energy, labelled pixels	25.64	26.28	23.91	25.28	48
	11.07	10.67	11.19	10.98	192
	5.765	5.543	6.015	5.774	384
Post-BN energy, labelled pixels	132.1	130.8	133.2	132.0	48
	130.4	131.5	132.5	131.5	192
	130.7	128.5	130.8	130.0	384
Mean variance $\times (M+K)$	—	—	—	16.9 / 16.4 / 15.7	48 / 192 / 384

Table 15: Witness JVP stage energies, mean over three seeds, from results/exp1_8d_REPORT.md Table 2a. ggn_top is the full top GGN eigenvector of the convolution block; s9cen is the centered leading direction of the incoming patch Gram. e_{pre} is the perturbation energy before the first head normalization, e_{logit} is measured on labelled pixels.

Mode	Direction	M	e_{pre}	e_{bn1}	e_{bn2}	e_{logit}	e_{final}
train	ggn_top	48	260.6	2875	4453	261.7	60.73
train	ggn_top	192	259.4	5905	7553	505.2	118.5
train	ggn_top	384	256.8	9168	13810	1018	227.6
train	s9cen	48	76.53	1538	1224	29.23	5.761
train	s9cen	192	76.53	2305	1914	50.83	9.862
train	s9cen	384	76.53	3997	3906	124.3	21.95
eval_head	ggn_top	48	261.9	261.9	42.42	9.607	2.366
eval_head	ggn_top	192	262.8	262.8	42.18	10.35	2.487
eval_head	ggn_top	384	262.5	262.5	46.53	10.45	2.372
eval_head	s9cen	48	76.53	76.53	9.103	0.4667	0.1049
eval_head	s9cen	192	76.53	76.53	8.460	0.4613	0.1013
eval_head	s9cen	384	76.53	76.53	9.762	0.4450	0.1011

A spatially constant bias perturbation $\Delta b = e_c$ is annihilated exactly by the immediately following train-mode normalization: in `train` and `eval_bn2` modes, at every width and seed, all stage energies after the centering step are exactly zero and the quadratic form agrees with the matvec to at most 1.14×10^{-17} in absolute value. In `eval_head` the same perturbation gives e_{final} between 7.8×10^{-4} and 2.0×10^{-3} , and in `eval_bn1` between 0.0154 and 0.0692.

Slopes and attribution. Table 16 gives the log-log slopes against M per seed and pooled, together with the mode-to-mode ratios of levels.

Table 16: Log-log slopes against M over the three widths, per seed and pooled, and mode-to-mode ratios of the levels (each mode’s value divided by its train-mode value at the same width and seed, then averaged over seeds and over the three widths), from `results/exp1_8d_REPORT.md`. λ_{WW} is the top curvature of the first head convolution’s weight block; λ_ϕ is the whole spatial block. Three-width slopes are descriptive, not asymptotic exponents.

Mode	slope of λ_{WW}				λ_ϕ	level ratio vs train	
	seed 0	seed 1	seed 2	pooled	pooled slope	λ_{WW}	λ_ϕ
train	0.509	0.785	0.350	+0.613	+0.398	1	1
eval_bn1	0.429	0.634	0.542	+0.563	+0.281	0.903	1.03
eval_bn2	0.544	0.632	0.508	+0.571	+0.253	0.156	0.227
eval_head	-0.112	0.181	-0.084	+0.006	-0.331	0.026	0.037

Three readings, and one non-reading. The positive width trend in the spatial block’s curvature at initialization is sensitive to head batch normalization: holding both head normalizations at their initial running statistics removes it ($+0.613 \rightarrow +0.006$ for the convolution block, $+0.398 \rightarrow -0.331$ for the whole spatial block) while leaving the incoming features unchanged. Either normalization alone preserves the trend ($+0.563$ and $+0.571$), which is evidence against attributing it uniquely to the first one, even though most of the drop in the *level* attaches to the second. The mechanism this supports is a finite-range normalization gain acting on width-diluted activations, over the range of widths measured, and not the width-linear feature-Gram mechanism of the shallow theory. The non-reading: the `eval_head` intervention changes the normalization derivative, the forward logits, the downstream gates and the cross-entropy metric at the same time, so a difference between modes establishes sensitivity to the intervention, not a unique cause; and the observation that the variance never approaches the stabilizer is a heuristic about the regime, not a guarantee.

Parameterization control. Table 17 scales the weights and bias of the first head convolution by c and, in the exact invariance, the following stabilizer by c^2 . The logits are unchanged and the block’s top curvature scales by exactly c^{-2} : the raw Euclidean curvature of a layer feeding a normalization is a parameterization quantity, not an architectural one (van Laarhoven, 2017).

Table 17: Function-preserving scale control at seed 0, from `results/exp1_8d_REPORT.md` Table 3. *ratio* is $c^2 \lambda_{WW}(c) / \lambda_{WW}(1)$, which is 1 under exact invariance. *logit dev.* is $\max |\text{logits}(c) - \text{logits}(1)|$, reported both with the stabilizer co-scaled by c^2 and with it held fixed (the implementation’s actual behaviour); λ_{WW} , *ratio* and $\|W\|_F^2$ are taken from the fixed-stabilizer variant, and the co-scaled variant differs from it by at most one unit in the fourth significant figure.

M	c	λ_{WW}	ratio	logit dev. (co-scaled)	logit dev. (fixed ε)	$\ W\ _F^2$
48	1/4	719.4	1.000	0	$9.40 \cdot 10^{-4}$	2.003
48	1/2	179.8	1.000	0	$1.89 \cdot 10^{-4}$	8.014
48	1	44.96	1.000	0	0	32.06
48	2	11.24	1.000	0	$4.63 \cdot 10^{-5}$	128.2
48	4	2.810	1.000	0	$5.84 \cdot 10^{-5}$	512.9
192	1/4	1935	0.9998	0	$3.00 \cdot 10^{-3}$	2.002
192	4	7.560	1.000	0	$1.89 \cdot 10^{-4}$	512.6
384	1/4	1927	0.9991	0	$4.69 \cdot 10^{-3}$	1.999
384	4	7.535	1.000	0	$2.95 \cdot 10^{-4}$	511.8

Eigensolver accuracy. Every row of the width sweep terminates on the combined criterion (relative change of the Rayleigh quotient below tolerance *and* residual below 10^{-3}), never on the iteration or wall-clock budget. Over all 36 (mode, width, seed) rows the relative residual $\|Gv - \lambda v\| / \lambda$ is at most 3.81×10^{-4} for the convolution weight block, at most 3.81×10^{-4} for the weight-bias block and at most 9.97×10^{-4} for the whole spatial block, and the Rayleigh quotient of the returned iterate agrees with the reported eigenvalue to the printed precision in every row.

E.4 CONTROLLED RETRAINING: ORIGINAL PROTOCOL

The pipeline was retrained on breast fold 0 at reduced depth (six transformer layers) with full instrumentation. Two protocols were run and both are reported: the *original* protocol (39 runs), which is the pre-registered one and whose joint-versus-frozen contrast an audit of the training code later showed to be confounded, and the *matched* protocol (21 runs), which removes the confounds, saves the selected checkpoint and re-runs the pre-registered comparisons. Sixty logged runs in total. Per-step residuals, block gradient norms, per-epoch parameter displacements and input-Jacobian norms are recorded in both; the matched protocol additionally logs the per-step clipping coefficient and the per-block applied update norms.

The original protocol comprises arms {joint-linear, frozen-random, frozen-PCA, joint-MLP} $\times$ widths $M \in \{48, 192\} \times$ three seeds under AdamW for 60 epochs, plus a {joint-linear, frozen-random} $\times$ three-seed pair under momentum SGD at both widths and a frozen-PCA SGD triple at $M = 48$ as a positive control. The spectral parameters sit in a zero-weight-decay group in every arm, and frozen arms log counterfactual spectral gradients. Table 18 gives the AdamW arms.

Table 18: Controlled study, original protocol, AdamW arms: validation macro-F1 (mean $\pm$ sd over three seeds) under two checkpoint policies, and relative spectral displacement at end of training. *Final-5* is the pre-registered metric. *Best-val* is the maximum over the 60 per-epoch evaluations on the same 28 validation cores; no checkpoint was saved at that maximum and this study has no test set, so that column is exploratory and selection-optimistic. The two policies order the arms oppositely. These arms also differ in clip scope and in normalization-affine treatment, so this contrast is confounded.

Arm	Final-5 (pre-registered)		Best-val (exploratory)		$\ \Delta\theta\ / \ \theta_0\ $
	$M=48$	$M=192$	$M=48$	$M=192$	
Joint (linear)	0.538 ± 0.082	0.646 ± 0.050	0.873 ± 0.030	0.858 ± 0.056	0.54–0.64
Frozen random	0.648 ± 0.063	0.731 ± 0.023	0.840 ± 0.025	0.817 ± 0.017	0
Frozen PCA	0.731 ± 0.003	0.726 ± 0.012	0.769 ± 0.021	0.862 ± 0.068	0
Joint (MLP)	0.566 ± 0.070	0.575 ± 0.184	0.900 ± 0.000	0.897 ± 0.001	0.85–0.96

On the pre-registered final-window metric the joint-linear minus frozen-random gaps are -0.110 at $M = 48$ (90% paired- t interval $[-0.333, +0.114]$) and -0.085 at $M = 192$ ($[-0.185, +0.015]$), neither resolvable at $n = 3$; against frozen-PCA they are -0.193 ($[-0.334, -0.052]$) and -0.081 ($[-0.175, +0.014]$). Taking each run’s maximum over its 60 evaluations reverses the ordering: joint minus frozen-random becomes $+0.033$ and $+0.041$, joint minus frozen-PCA $+0.104$ and -0.004 . The switch is an arithmetic identity — the change in every gap equals the difference in peak-to-final degradation between the two arms — and averaged over seeds that degradation is 0.335 ($M = 48$) and 0.212 ($M = 192$) for joint-linear against 0.192 and 0.086 for frozen-random.

Three asymmetries. First, gradient-norm clipping at 1.0 was computed over the optimizer’s parameters, that is over $\theta + \phi$ in the joint arms but over ϕ alone in the frozen arms, so at identical gradients the frozen arm takes the larger ϕ step: the median extra clip factor paid by the joint arm is 1.10–1.12 under SGD at $M = 48$ and 3.0–3.6 under AdamW at $M = 192$. Second, the normalization following the spectral projection contributes 128 affine parameters that are trained in joint-linear, frozen in frozen-random and structurally absent in frozen-PCA. Third, no checkpoint was saved during training, so the best-validation column is a post-hoc curve maximum that no saved model realizes, taken over 60 evaluations of the same 28 validation cores that define the comparison.

E.5 MOMENTUM-SGD CONTROLS AND NORMALIZATION RECALIBRATION

The two optimizer families constitute an optimizer-family control; the SGD arm is not an instantiation of the two-timescale theorem’s gradient flow. It uses momentum 0.9; weight decay 0.01 on ϕ and 0 on θ ; mini-batches of four cores with reshuffling; and global gradient-norm clipping at 1.0, which binds on 73–89% of steps. Logged gradient norms are recorded before clipping, so the plotted theory quantities are not the quantities the trajectory applied. Under momentum SGD the paired joint minus frozen-random gap is +0.002 at $M = 48$ and +0.011 at $M = 192$ on best-validation, with 90% paired- t intervals $[-0.008, +0.011]$ and $[-0.007, +0.030]$; on the final-window metric the same pairs give +0.028 and -0.057 , with intervals $[-0.173, +0.228]$ and $[-0.124, +0.010]$ — gaps that differ in sign between the widths, with intervals an order of magnitude wider than the effect they bracket. Both arms lose 0.12–0.22 macro-F1 between their peak and their final window. Descriptively, the spectral share of the summed squared gradient norms drops from about 0.94 (AdamW joint) to about 0.26 (SGD joint); momentum, clipping, weight decay and adaptivity all differ between the two families, so no single factor is isolated.

Recalibration of normalization statistics. On disposable copies of the final checkpoints, with all weights fixed, the normalization running statistics were re-estimated on training data only (29 batches) and the model re-evaluated. For this intervention and these checkpoints, the recovered fraction of the joint arms’ peak-to-final drop is 0.17 at $M = 48$ and 0.07 at $M = 192$: recalibration recovers a nonzero part of the degradation but not most of it. What it does undo is the instability of the endpoint number: it repairs the two collapsed joint checkpoints (+0.24 and +0.15 macro-F1) and shrinks the seed standard deviation by roughly $3.7\times$, while leaving the frozen arms essentially unchanged ($|\Delta| \leq 0.02$, and ≤ 0.006 for frozen-PCA, whose spectral stage carries no normalization).

E.6 MATCHED PROTOCOL

The matched protocol equalizes the three asymmetries and re-runs the pre-registered comparisons at $M = 48$: the reduction-stage normalization affine parameters are placed in ϕ and trained wherever that normalization exists; the clip norm at 1.0 is computed over ϕ only under the same clipping rule and scope in every arm; and the best-validation checkpoint is saved. AdamW at learning rate 10^{-4} , 60 epochs, three seeds, flip and 90° -rotation augmentation on in all arms. The logged per-step clip coefficient averages 0.33–0.57 across arms — the coefficients and gradients differ by arm, which is an outcome of the matched rule — and the logged applied update norms confirm that the spectral share of the applied update tracks the learning-rate multiplier (0.02, 0.17 and 1.36 times the ϕ update norm for $\times 0.1$, $\times 1$ and $\times 10$). Table 19 gives all seven arms.

Table 19: Controlled study, matched protocol ($M = 48$, AdamW, three seeds, mean $\pm$ sd). Identical normalization-affine treatment and the same ϕ -only clipping rule and scope in every arm; best-validation checkpoints saved. *Final-5* is the pre-registered metric; *best-val* is the maximum over 60 per-epoch evaluations on the 28 fold-0 validation cores and is exploratory and selection-optimistic, since this study has no test set. The drift column is the relative spectral-parameter displacement at end of training; its standard deviations were computed by us from `results/e3c_analysis_runs.csv`. Rows 1–2 and rows 3–4 are the two matched pairs; see the text.

Arm	Final-5 (pre-registered)	Best-val (exploratory)	$\ \Delta\theta\ / \ \theta_0\ $
Frozen random	0.684 ± 0.037	0.814 ± 0.040	0
Joint (linear)	0.694 ± 0.056	0.837 ± 0.029	0.566 ± 0.026
Frozen pretrained	0.690 ± 0.033	0.861 ± 0.022	0
Fine-tuned pretrained	0.710 ± 0.052	0.857 ± 0.009	0.020 ± 0.005
Joint, spectral lr $\times 0.1$	0.626 ± 0.055	0.841 ± 0.007	0.083 ± 0.006
Joint, spectral lr $\times 10$	0.561 ± 0.114	0.886 ± 0.037	4.910 ± 0.402
Joint, cosine schedule	0.660 ± 0.020	0.825 ± 0.033	0.359 ± 0.013

Two matched pairs, not a single-factor design. The four core arms do not all differ in one factor. Joint-linear and frozen-random use a linear reduction stage with normalization; frozen-pretrained

and fine-tuned-pretrained use a pretrained multilayer spectral encoder without reduction-stage normalization. The clean freezing comparisons are therefore the two matched pairs, joint-linear against frozen-random and fine-tuned against frozen-pretrained. Comparing across the pairs — frozen-pretrained against frozen-random, for instance — changes architecture, initialization and training history, and validation selection, not only informativeness.

Verdicts. All comparisons in Table 20 are paired by seed with 90% t -intervals ($t = 2.920$, $n = 3$), conditional on one validation split, and are descriptive seed summaries rather than a population or multi-factor causal analysis.

Table 20: Matched protocol: paired differences with 90% paired- t intervals ($n = 3$). The last two rows compare the two protocols and are reported as descriptive seed summaries; saving a checkpoint changes what is available to evaluate, not the final-window trajectory.

Contrast	Final-5	Best-val
Fine-tuned – frozen-pretrained	+0.021 [−0.022, +0.063]	−0.004 [−0.048, +0.039]
Joint-linear – frozen-random	+0.011 [−0.142, +0.163]	+0.023 [−0.066, +0.112]
Frozen-pretrained – frozen-random	—	+0.047 [−0.042, +0.136]
Spectral lr $\times 10 - \times 1$	−0.133 [−0.419, +0.152]	+0.048 [−0.016, +0.112]
Spectral lr $\times 0.1 - \times 1$	−0.068 [−0.173, +0.037]	+0.004 [−0.039, +0.046]
Joint/frozen gap: matched – original	+0.120 [−0.230, +0.470]	−0.010 [−0.105, +0.085]

The pre-registered prediction that fine-tuning a pretrained spectral encoder falls below leaving it frozen is not supported: neither difference is resolvable at $n = 3$, and the fine-tuned arm moved θ by only 2% of its initial norm. The logged applied update norms verify nonzero applied encoder updates under AdamW in that arm; this is a statement about the applied updates, not an attribution of the loss decrease. The pre-registered harmful-drift ordering $\times 0.1 \geq \times 1 \geq \times 10$ is not observed: on best-validation the $\times 10$ arm is above $\times 1$ in 3/3 seeds while its final-window endpoint is the lowest and most variable of any arm at 0.561. Three multiplier means do not establish a systematic relation between displacement and peak score, and the intervals are wide. The cosine schedule improved neither mean endpoint (0.825 best-val and 0.660 final-5 against 0.837 and 0.694 for the constant learning rate); one schedule that did not help does not rule out an excessive late learning rate.

Disclosures. Width 48 only; $n = 3$; a single validation fold of 28 fold-0 cores; no test set. The best-validation column is exploratory and the final-window column is the pre-specified metric. The pretrained encoders were selected by pixel validation macro-F1 on the *same* fold-0 validation cores the runner later scores, which is a selection advantage for both pretrained arms, and were pretrained with weight decay 0.01 on θ . Pixel pretraining used the unaugmented centre-crop protocol while subsequent training used random crop plus augmentation; this affects comparisons across the two pairs, not the frozen-versus-fine-tuned pair that starts from the same checkpoint. Augmentation is on in all arms. Finally, the summary artifact `results/e3c_analysis_runs.csv` does not expose per-run configuration hashes or data-order pairing, so before the pairs are described as completely matched a compact provenance manifest should accompany it, recording architecture, initial checkpoint hash, parameter-group membership, clipping and weight-decay settings, normalization mode and state, augmentation and data-order seed, scheduler and selection rule; identical seed labels alone do not establish bit-identical mini-batch order across different model constructions.

E.7 WHAT IS NOT CERTIFIED ON THIS PIPELINE

The spatial module’s input-Jacobian operator norm grows over training, from 3.2 to 157 on average for joint-linear (up to 414) and from 1.7 to 1.2×10^4 for joint-MLP, which rejects an initialization-scale stability approximation for the containment constant of the displacement theorem, though not the existence of a finite-horizon constant over the training window. The fitted per-step residual-envelope rate for joint-linear grows from $\hat{\mu} = 6.0 \times 10^{-4}$ ($M = 48$) to 1.9×10^{-3} ($M = 192$), a factor 3.2 for a $4\times$ width increase; the ratio is invariant to the step-clock-to-flow-time conversion, the absolute values are not, and two widths do not establish a trend.

The earlier proxy instantiation of the displacement bound is withdrawn. Even after correcting the time units and using the supremum gain, the proxy exceeds the measured displacement by roughly $17\text{--}47\times$ under AdamW and roughly $450\text{--}1900\times$ under momentum SGD; the raw per-step residual violates the fitted envelope on $19\text{--}28\%$ of joint-linear steps, so the envelope assumption is verified for the smoothed curve rather than for the logged quantity. Momentum, gradient clipping and adaptive preconditioning all lie outside the theorem’s hypotheses, and under AdamW the measured displacement exceeds $\eta \sum_k \|\nabla_{\theta} \mathcal{L}_k\|$ by more than an order of magnitude, so no step-size-rescaled flow bound describes that trajectory in either direction. No certified instantiation of the displacement bound exists for this pipeline at present.

Finally, the scalar gradient-ratio diagnostic is blind here: it *rises* during joint AdamW training (joint-linear epoch-1 median $0.6\text{--}0.8$, rising to $3\text{--}5$ by epoch 60), including in runs whose validation score is falling. A single global ratio cannot see a direction-wise anisotropy, and the initialization geometry of Appendix E.2 is what such a ratio would have to resolve.

F VERIFICATION RECORD

Every proof reproduced or stated in this appendix was independently re-derived line by line, and every displayed inequality was checked numerically at the parameter values recorded below. The scripts live under `code/audits/` in the supplementary repository; each is a standalone program that reads and writes no repository file, and each records its own seed and command in its header. Table 21 lists them; a seed is a NumPy generator seed unless marked `torch`, in which case it is the global torch seed, and `--` marks a deterministic script. The recorded environment is Python 3.12.13, numpy 2.0.2, scipy 1.16.3, torch 2.9.1, CPU only.

Two entries are not clean, and both are artefacts of the audit code rather than of the mathematics. In `audit_a1.py`, the single failing line is the sub-check “ $\text{share} \leq 1$ ” of the affine bridge, whose integral the script evaluates with a rectangle rule that over-estimates a decaying integrand (16 of 200 trials exceed one, maximum 1.73); recomputing the same integral in closed form gives 0 of 300 trials above one with maximum 0.919, and the identical claim passes exactly, over 800 non-commuting-kernel instances, in `astra_math_02_checks.py`. In `audit_a2.py`, every check passes until the script aborts inside a functional Jacobian transform applied to a normalization *module*, because the installed torch version forbids the in-place update of that module’s batch counter under the transform; the aborted and subsequent checks are duplicated in `audit_a2_bn.py`, which constructs the same layer without running statistics and passes them all, so nothing is left unverified. Both scripts are retained unmodified so that the record matches the run.

Script	Verifies	Seed	Result
audit_A1.py	Discrete displacement and clipping bounds; monotone comparison and its Grönwall form; affine bridge, its counterexample and the attribution share; the top-eigenvalue counterexample; the constants of the initialization certificate	np 0	PASS (18/19)
audit_a2.py	Interior curvature witness identity and the ReLU interior theorem’s probability and curvature-floor events; the normalization counterexamples; the cross-entropy gradient-gain counterexample	np 0	partial
audit_a2_bn.py	The normalization annihilation counterexample against a real BatchNorm1d in both stabilizer regimes; the width-proportional weighted feature Gram; the mean-direction witness with non-constant features	torch 1	PASS
check_p5.py	Solvable serial instance: invariants, fitting-time bracket, residual envelope, Gauss–Newton constants, gradient-norm bound, joint-versus-frozen gap, displacement table, normalized-readout control	—	PASS
audit_blocks.py	In the paper’s own normalization: the gradient identity $\nabla_b \mathcal{L} = J_b^\top r$, the head-weight block identity for a shared-weight 1×1 convolutional head, and the degenerate stationary-point counterexample to Corollary 1	torch 0	PASS
check_theoremN.py	Scoped nonlinear existence result: the containment inequality, the three-term Jacobian decomposition, gate-flip containment, perturbation and moment bounds, boundary mass, and the attribution share along the nonlinear flow	np 0	PASS
astra_math_02_checks.py	Finite-horizon attribution with non-commuting kernels by exact spectral integral; the isotropic serial instance with signed context and unequal zero-sum readouts; the normalization derivative and co-scaled curvature law; ReLU feature and gate tube bounds	np 902	PASS
<i>Subdirectory math02_sections_2_4_6/</i>			
a1_finite_horizon.py	Finite-horizon attribution over 24 nonlinear runs, sharpness at equality, and the old-versus-new bound	20260910	PASS
a2_chernoff.py	Truncated matrix Chernoff certificate: truncation radius, minimum eigenvalue floor, exponent constant, width threshold	424242	PASS
a2_chernoff_biting.py	The same certificate in a second parameterization, with empirical failure frequencies across widths	909090	PASS
a2_chernoff_constant.py	The constant in the certificate, stressed on the extremal balls-in-bins family	5150	PASS
c23_bullets.py	Reduction of the full flow to its three-variable closure, residual identities, and the clipping equivalence against a real optimizer	246810	PASS
i_toy_ode.py	The serial instance’s update suppression, displacement, fitting time and frozen-comparator bracket over 24 high-accuracy solves	31337	PASS
i6_reversal_mc.py	The contextual-reversal error limit, its per-draw form, the sufficient condition, and the tail bound	77007	PASS
b_bn_torch.py	Normalization Jacobian and spectrum against a real BatchNorm2d; the width-constant curvature ratio; the function-preserving rescaling law; uniform versus channelwise rescaling	13579	PASS
b_bn_crosschannel.py	Exact block-diagonalization of the normalization derivative across channels	13579	PASS

Table 21: Numerical verification record for the results of Appendices A and B. Both non-clean